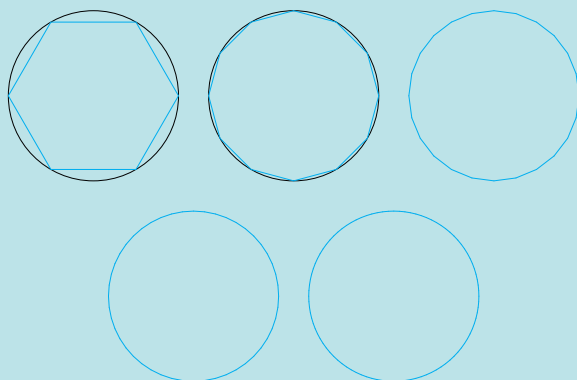


First Principles of Math



Sindre Sogge Heggen

*"Wahrlich es ist nicht das Wissen, sondern das Lernen,
nicht das Besitzen, sondern das Erwerben,
nicht das Da-Seyn, sondern das Hinkommen,
was den grössten Genuss gewährt"*

*"It is not knowing, rather learning,
not possessing, rather obtaining,
not being present, rather reaching there,
which serves the greatest joy."*

— Carl Friedrich Gauss

This document is created by Sindre Sogge Heggen. The text is written in L^AT_EX and the figures are made using Asymptote.

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13.09.2026

Foreword

The extent and applications of mathematics are enormous, but a fair share of it is founded on a manageable amount of principles: I wish to present these in this book. I have chosen to call a principle in summarized form a *rule*. You will find the rules in blue text boxes, usually followed by an example of its usage. One of the main targets of this book is presenting the logical justification for the rules. In Chapter 1-5 you will find explanations¹ preceding every rule, while in chapter 6 some explanations are found directly after stating the rules (and eventual examples). As of chapter 7, some explanations are found in a concluding section named *Explanations*. This indicates that they are rather intricate or are so intuitively true that many will find the explanation superfluous.

The structure of the book

The book consists of a *Part I* and a *Part II*. *Part I* focuses on the basic understanding of the numbers and operations of calculation. *Part II* introduces the concept of algebra and the closely related topics of powers, equations, and functions. In addition, both *Part I* and *Part II* end with a chapter on geometry.

Notice! You will not find practice problems and applications of mathematics in real life in this book. These are two very important elements to come, either integrated in this book or as an independent document.

A note on convention

Although I am very much aware of the convention of writing commas and dots in center-aligned equations, I opted against this². In this way, a center-aligned equation is a grammatical hybrid; it can end with both an invisible comma or dot, or nothing at all.

¹To *explain* the rules rather than *proving* them is a deliberate decision. A proof demands mathematical rigor that often forces a lot of assumptions and definitions along the way. This can make the main insight disappear in the crowd of details. However, some of the explanations are valid as proofs.

²I've never liked the looks of it.

I want to thank:

Anne Jordal Myrset

Charlotte Merete Dahl

Brenden Callahan

For brilliant suggestions and comments.

Symbols

$=$	"equals"
$<$	"less than"
$>$	"greater than"
$\leq$	"less than or equal to"
$\geq$	"greater than or equal to"
$\in$	"included in"
$\vee$	"or"
$\wedge$	"and"
$[a, b]$	"closed interval from a to b "
$ a $	"length/absolute value of a "
$\perp$	"perpendicular to"
$\parallel$	"parallel with"
$\triangle$	"triangle"
$\square$	"quadrilateral"

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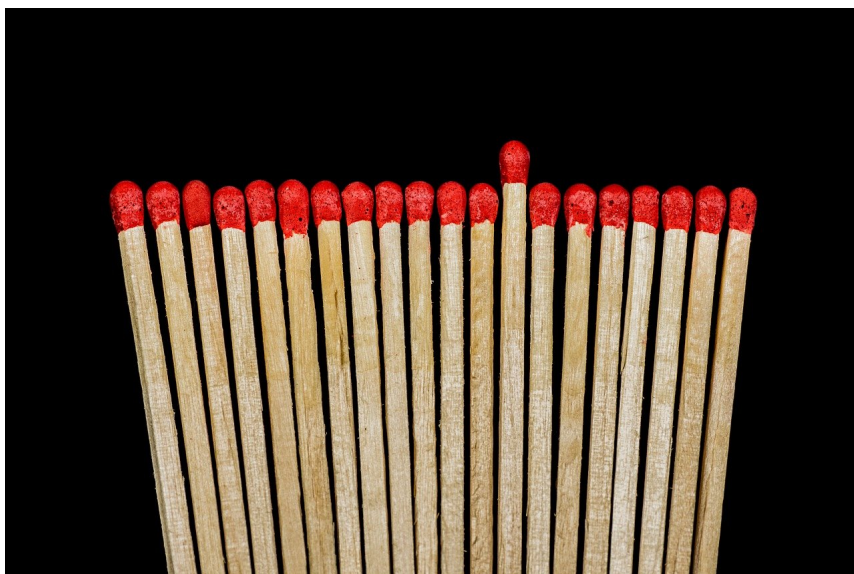
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Part I

Numbers, calculations, and geometry

Chapter 1

The numbers



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1.1 The Equality Sign, Sets, and Number Lines

The Equality Sign

As the name suggests, the **equality sign** $=$ indicates that something is equal. To what extent and when something can be considered equal is a philosophical discussion, so initially, we are left with this: *The kind of equality $=$ referred to must be understood from the context in which the sign is used.* This way, we can study some basic properties of our numbers and later return to more precise meanings of the sign.

The language box

Common ways to say $=$ are:

- "is equal to"
- "is the same as"
- "corresponds to"

Sets and Number Lines

In this book, we will use two ways to represent numbers: numbers as a *set* and numbers as a *position on a line*. All representations of numbers are based on the understanding of the numbers 0 and 1.

Numbers as Quantities

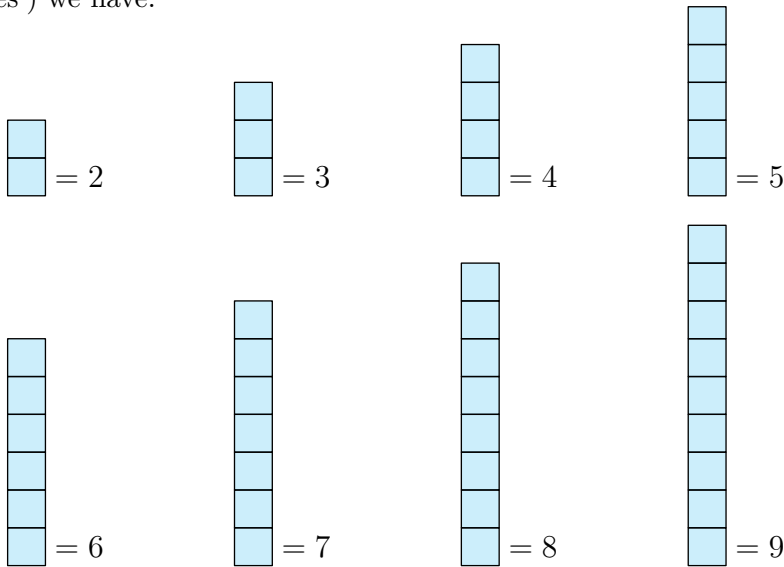
When we talk about a set, the number 0 means ¹ "nothing". A figure where nothing is present will thus be the same as 0:

$$= 0$$

1 will be represented as a single square:

 = 1

Other numbers will then be defined based on how many unit squares ('ones') we have:

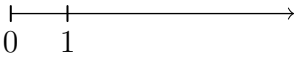


Numbers as Positions on a Line

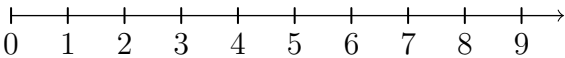
When we place numbers on a line, 0 will be our starting point:



Then we place 1 a certain distance to the right of 0:



Other numbers will then be defined based on how many unit lengths ('ones') we are away from 0:



¹In [Chapter ??](#) we will see that there are other interpretations of 0.

Positive Integers

We will soon see that numbers do not necessarily have to be *whole* numbers of ones, but numbers that *are* have a special name:

1.1 Positive Integers

Numbers that are a whole number of ones are called **positive¹integers**. The positive integers are

1, 2, 3, 4, 5 and so on.

Positive integers are also called **natural numbers**.

What about 0?

Some authors also include 0 in the concept of natural numbers. In some contexts, this makes sense, while in others it does not.

Inequality Signs

We also have symbols that indicate that values are *not* equal. Sometimes it is enough to say that they are not equal, while other times there is also a need to specify which value is larger (and which is smaller).

The language box

$\neq$	"is not equal to"
$<$	"is less than"
$>$	"is greater than"
$\leq$	"is less than or equal to"
$\geq$	"is greater than or equal to"

¹What the word *positive* means will be discussed in [Chapter ??](#).

Example

$$0 \neq 1$$

$$7 < 9$$

$$5 > 2$$

$$\text{positive even numbers} \geq 2$$

$$3 \leq \text{positive odd numbers}$$

Reading Direction

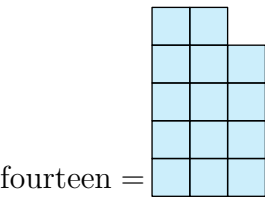
Strictly speaking, we only have three symbols for inequality, namely $\neq$, $<$ and $\leq$. The last two symbols require a reading direction from left to right. The inequality $3 < 5$ is read as "3 is less than 5". But we can also say that $<$ is a directional symbol, where the pointed end should point toward the smaller value. This means that we can read $3 < 5$ from the right as "5 is greater than 3". The fact that we also use the symbols $>$ and $\geq$ is a result of sometimes wanting to say "greater than" when reading from the left.

1.2 Numbers, Digits, and Value

Our numbers are built from the **digits** 0, 1, 2, 3, 4, 5, 6, 7, 8, and 9, and their *position*. The digits and their positions define ¹ the **value** of the number.

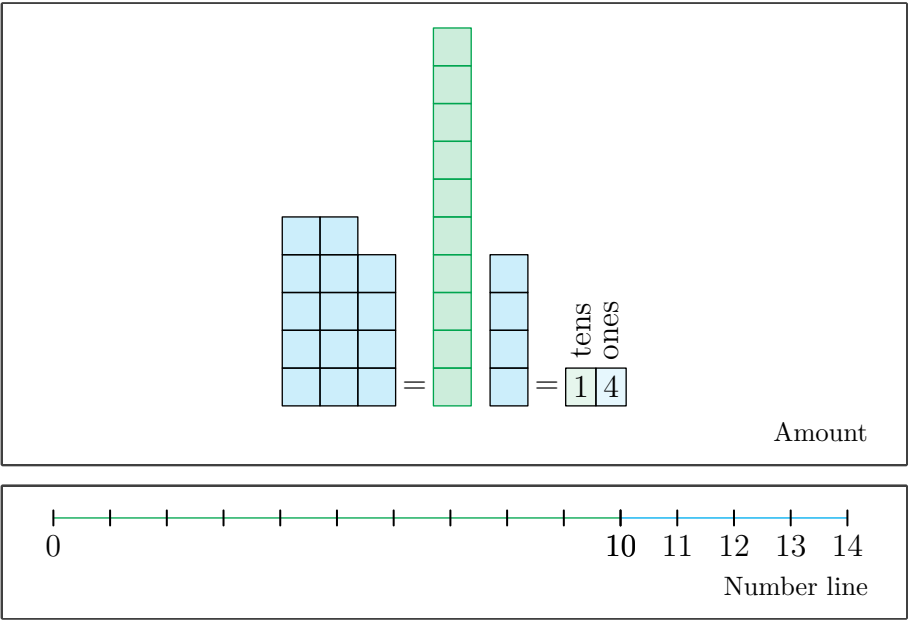
Integers Greater Than 9

Let’s take the number ‘fourteen’ as an example and write it using our digits.



A group of ten ‘ones’ we call a ‘ten’. From ‘fourteen’, we can make 1 ‘ten’, and in addition, we have 4 ‘ones’. So we write ‘fourteen’ like this:

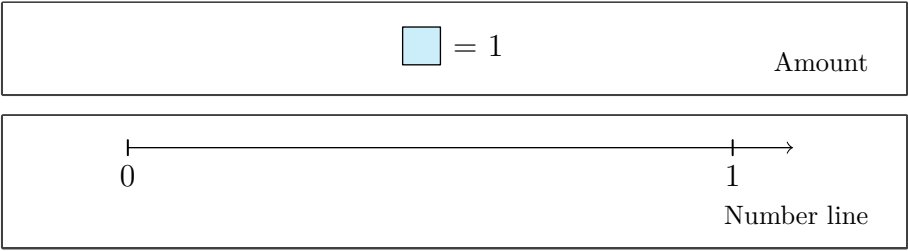
fourteen = 14



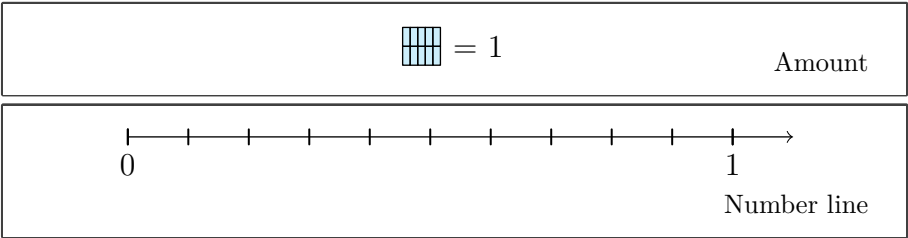
¹Eventually, we will also see that *sign* helps define the value of the number (see [Chapter ??](#)).

Decimal Numbers

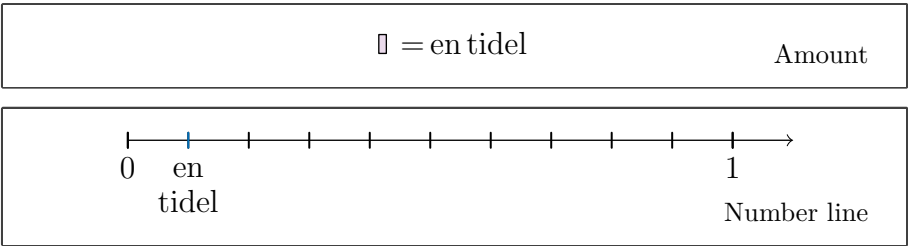
In many cases, we do not have a whole number of ones, and then we will need to divide 1 into smaller parts. Let's start by drawing a ¹ :



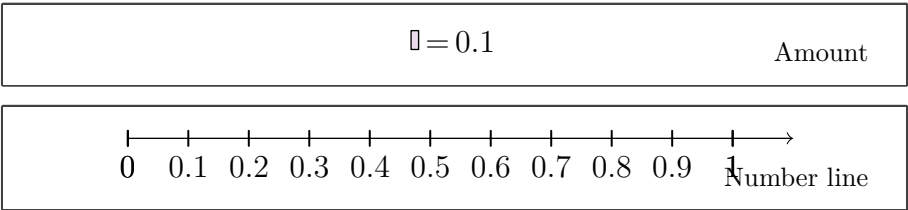
Then we divide our unit into 10 smaller parts:



Since we have divided 1 into 10 parts, we call such a part a 'tenth':

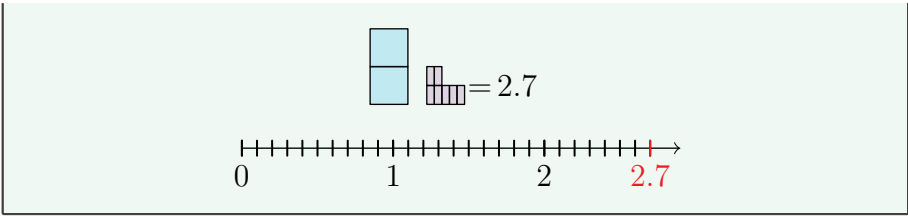


Tenths are written using the **decimal point** :



Example

¹Here we define the unit length as larger than before.



The language box

In many countries, a comma , is used as the decimal separator:

3,5 (*Norwegian, German and ‘others’*)
3.5 (*English*)

When we combine mathematical symbols ¹, we get what we call a mathematical **expression**.

¹Digits are considered mathematical symbols.

The Decimal System

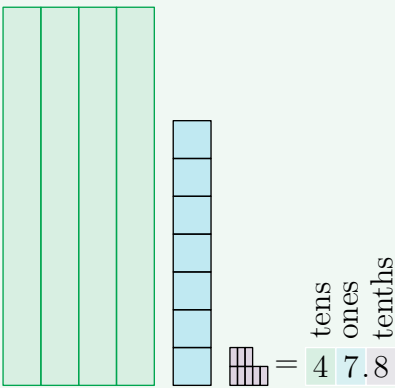
We have now seen how we can express the value of numbers by placing digits according to the number of tens, ones, and tenths, and it doesn't stop there:

1.2 The Decimal System

The value of a number is given by the digits 0, 1, 2, 3, 4, 5, 6, 7, 8, and 9, and their position. With the digit representing ones as the starting point:

- digits to the left (in order) represent the number of tens, hundreds, thousands, and so on.
- digits to the right (in order) represent the number of tenths, hundredths, thousandths, and so on.

Example 1



Example 2

thousands
hundreds
tens
ones
tenths
hundredths
3805.72

1.3 Even and Odd Numbers

Integers that have 0, 2, 4, 6, or 8 in the ones place are called **even numbers**.

Integers that have 1, 3, 5, 7, or 9 in the ones place are called **odd numbers**.

Example

The first ten (positive) even numbers are

0, 2, 4, 6, 8, 10, 12, 14, 16, and 18

The first ten (positive) odd numbers are

1, 3, 5, 7, 9, 11, 13, 15, 17, and 19

1.3 Coordinate System

In many cases, it is useful to use two number lines simultaneously. This is called a **coordinate system**¹. We then place one number line (an axis) that runs *horizontally* and one that runs *vertically*. A position in a coordinate system is called a **point**.

1.4 Coordinate System

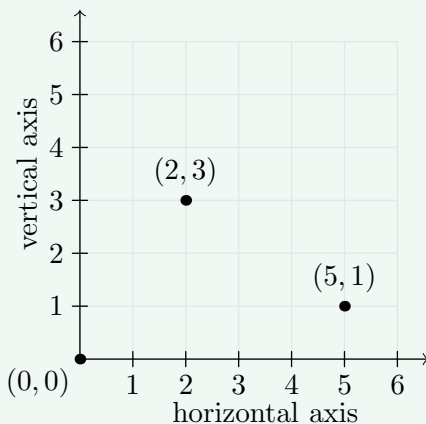
A point is written as two numbers inside a parenthesis. The two numbers are called the **first coordinate** and the **second coordinate** of the point.

The first coordinate shows the point's position along the horizontal axis.

The second coordinate shows the point's position along the vertical axis.

Example 1

In the figure, we see the points $(2, 3)$, $(5, 1)$ and $(0, 0)$.



The language box

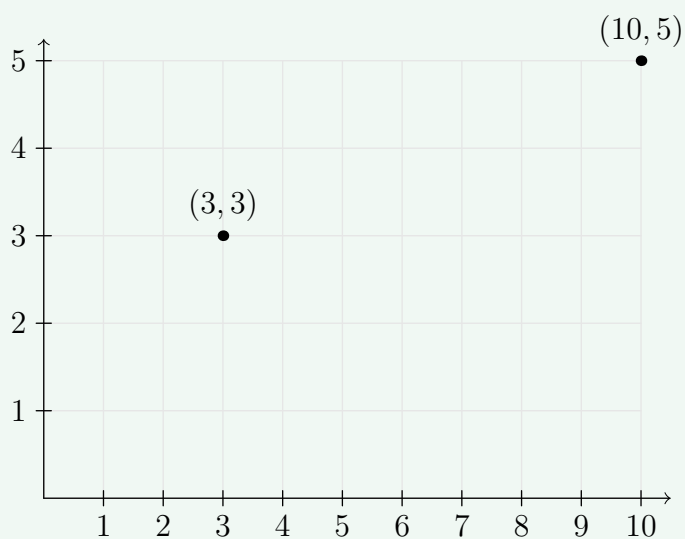
The point where the axes meet, that is, $(0, 0)$, is called the **origin**.

¹Strictly speaking, there are many types of coordinate systems, but in this book, we use the term for only one type, namely the **Cartesian coordinate system**. It is named after the French philosopher and mathematician René Descartes.

Note

The size of a unit length can be different on the two axes.

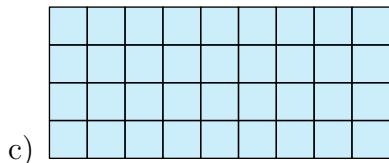
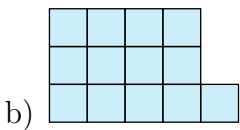
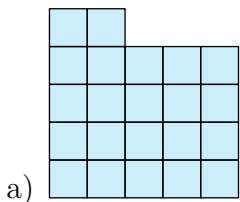
Example 2



Exercises for Chapter 1

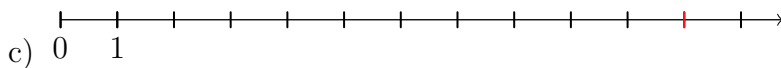
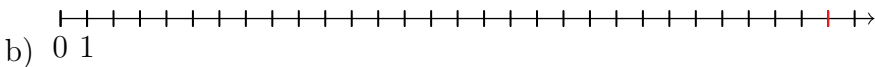
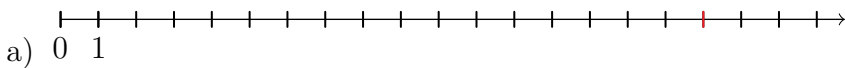
1.1.1

Write the value of the number.



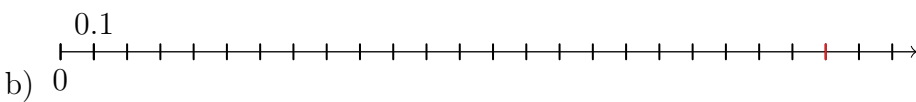
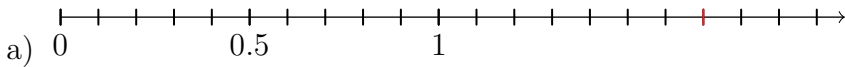
1.1.2

Write the value of the number marked red.



1.1.3

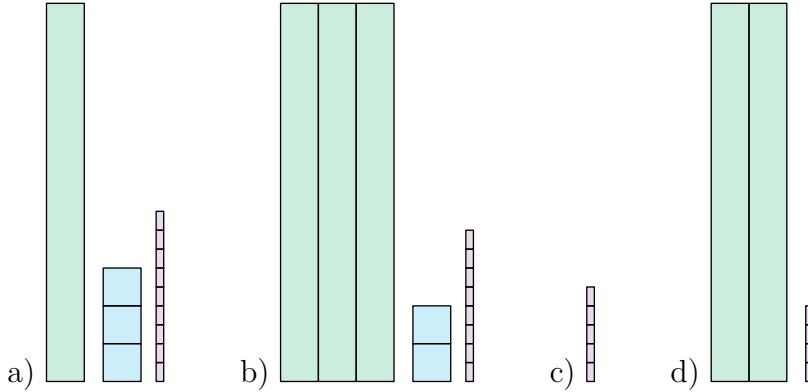
Write the value of the number marked red.



1.1.4

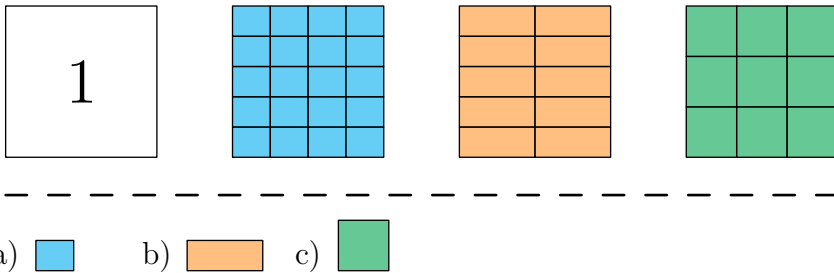
 = 10  = 1  = 0,1
--

Write the value of the number.



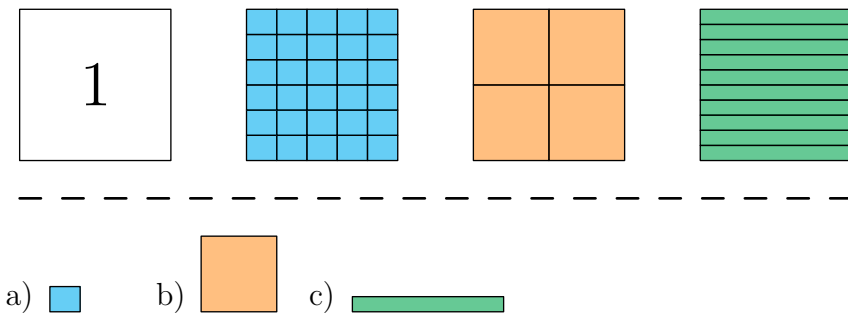
1.1.5

Choose which one of a), b) and c) have the value 0.1.



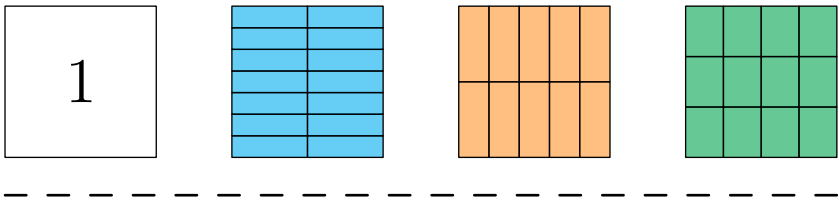
1.1.6

Choose which one of a), b) and c) have the value 0.1.



1.1.7

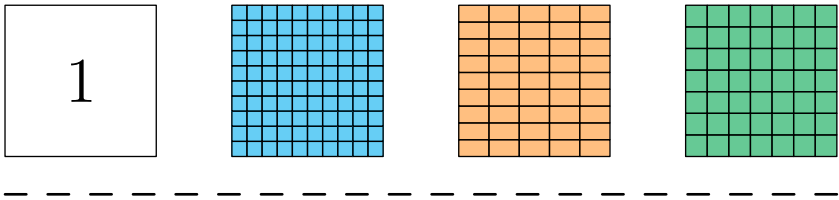
Choose which one of a), b) and c) have the value 0.1.



- a)  b)  c) 

1.1.8

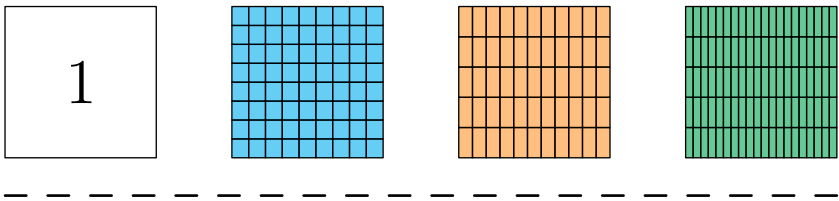
Choose which one of a), b) and c) have the value 0.01.



- a)  b)  c) 

1.1.9

Choose which one of a), b) and c) have the value 0.01.



- a)  b)  c) 

Chapter 2

The four elementary operations



Bilde fra pxhere.com

2.1 Mathematical Operations

When we combine two or more numbers to create a new number, we perform a **mathematical operation**. **Addition**, **subtraction**, **multiplication**, and **division** are our fundamental mathematical operations, and we call them the four basic **arithmetic operations**. When we use these operations, we create **calculations**. The value we get from performing a mathematical operation is often called the **answer** to the calculation.

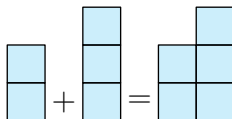
2.2 Addition

Addition with Quantities: Adding

When we have a quantity and want to add more, we use the **plus sign**

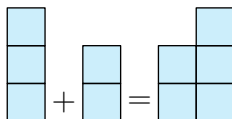
+. If we have 2 and want to add 3, we write

$$2 + 3 = 5$$



The order in which we add the numbers does not matter; starting with 2 and then adding 3 is the same as starting with 3 and then adding 2:

$$3 + 2 = 5$$



The language box

An addition problem consists of two or more **terms** and one **sum**. In the addition problem

$$2 + 3 = 5$$

both 2 and 3 are terms, while 5 is the sum.

Common ways to say $2 + 3$ are:

- "2 plus 3"
- "2 added to 3"
- "2 and 3 added together"

Adding numbers is also called **summing**.

2.1 Addition is Commutative

The sum is the same regardless of the order of the terms.

Example

$$2 + 5 = 7 = 5 + 2$$

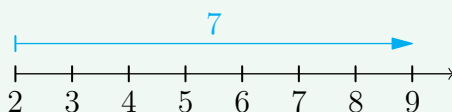
$$6 + 3 = 9 = 3 + 6$$

Addition on the Number Line: Moving to the Right

On a number line, addition with positive numbers means moving *to the right*:

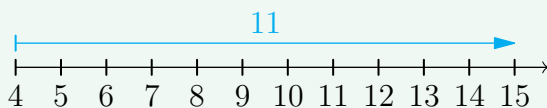
Example 1

$$2 + 7 = 9$$



Example 2

$$4 + 11 = 15$$



A New Meaning of =

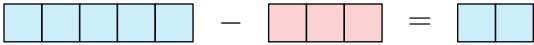
+ gives us the ability to express numbers in many different ways. For example, $5 = 2 + 3$ and $5 = 1 + 4$. In this context, = means "has the same value as". This also applies to subtraction, multiplication, and division, which we will cover in the next three sections.

2.3 Subtraction

Subtraction with Quantities: Taking Away

When we have a quantity and remove part of it, we use the **minus sign** $-$. If we have 5 and want to remove 3, we write

$$5 - 3 = 2$$



The language box

A subtraction problem consists of two or more **terms** and one **difference**. In the subtraction problem

$$5 - 3 = 2$$

both 5 and 3 are terms, while 2 is the difference.

Common ways to say $5 - 3$ are:

- "5 minus 3"
- "5 subtracted by 3"
- "3 taken from 5"
- "3 subtracted from 5"

A New Interpretation of 0

In chapter [Chapter 1](#), we mentioned that 0 can be interpreted as "nothing". Subtraction gives us the ability to express 0 using other numbers. For example, $7 - 7 = 0$ and $19 - 19 = 0$.

Subtraction on the Number Line: Moving to the Left

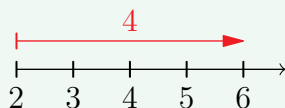
In [section 6.2](#), we saw that $+$ (with positive numbers) means moving *to the right* along the number line. With $-$ we do the opposite; we move *to the left*!! ¹ :

Note

In *Example 1* and *Example 2* below, we move in the opposite direction of the arrow. This means we start at the arrowhead and move to the end. This may seem a bit odd at first, but it will be useful, especially in [Chapter 5](#), to think this way.

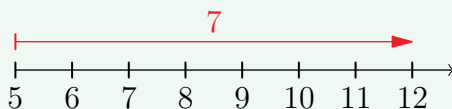
Example 1

$$6 - 4 = 2$$



Example 2

$$12 - 7 = 5$$



¹In figures with number lines, red arrows indicate that you start at the arrowhead and move to the other end.

Inverse Operations

The figures showing addition and subtraction on the number line illustrate why we say that addition and subtraction are **inverse operations**.

For example: Since $2 + 4 = 6$, we know that $6 - 4 = 2$ and that $6 - 2 = 4$.

Canceling Terms

If we have an expression where numbers with the same value are added as many times as they are subtracted, we say they **cancel** each other out. In the expression

$$4 + 3 - 4 + 4 + 7 - 4 + 1$$

we have as many 4s added as we have 4s subtracted. The total sum and difference of the 4s is equal to 0. Thus, we can simplify our expression to

$$3 + 7 + 1$$

Note! In [Chapter ??](#), the word *cancellation* will have a different meaning.

2.4 Multiplication

Multiplication with Whole Numbers: Introductory Definition

When we add the same numbers together, we can use the **multiplication sign** $\cdot$ to write our calculations more concisely:

Example

$$4 + 4 + 4 = 4 \cdot 3$$

$$8 + 8 = 8 \cdot 2$$

$$1 + 1 + 1 + 1 + 1 = 1 \cdot 5$$

The language box

A multiplication problem consists of two or more **factors** and one **product**. In the multiplication problem

$$4 \cdot 3 = 12$$

4 and 3 are factors, while 12 is the product.

Common ways to say $4 \cdot 3$ are:

- "4 times 3"
- "4 multiplied by 3"
- "4 multiplied with 3"

Many websites and books in English use the symbol $\times$ as the multiplication sign. In most programming languages, the symbol $*$ is used.

Multiplication of Quantities

Let’s use a figure to visualize the multiplication problem $2 \cdot 3$:

$$2 \cdot 3 = \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} + \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} + \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} = \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \square & \square & \square \\ \hline \end{array}$$

And then we can observe the product of $3 \cdot 2$:

$$3 \cdot 2 = \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \square \\ \hline \end{array} + \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \square \\ \hline \end{array} = \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \square & \square \\ \hline \end{array}$$

2.2 Multiplication is Commutative

The product is the same regardless of the order of the factors.

Example

$$3 \cdot 4 = 12 = 4 \cdot 3$$

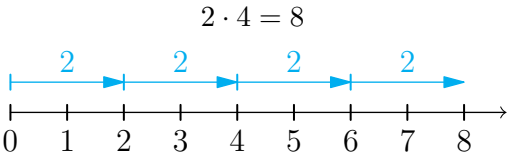
$$6 \cdot 7 = 42 = 7 \cdot 6$$

$$8 \cdot 9 = 72 = 9 \cdot 8$$

Multiplication on the Number Line

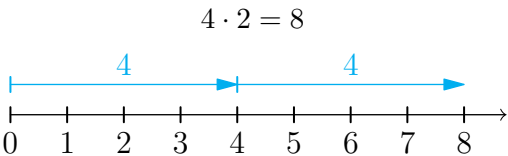
We can also use the number line to solve multiplication problems. For example, we can find what $2 \cdot 4$ is by thinking like this:

” $2 \cdot 4$ means moving 2 spaces *to the right*, 4 times.”



The number line can also be used to see that the order of the factors does not matter:

” $4 \cdot 2$ means moving 4 spaces *to the right*, 2 times.”



Final Definition of Multiplication with Positive Whole Numbers

It might seem most natural to interpret "2 times 3" as "3, 2 times". Then

$$\text{"2 times 3"} = 3 + 3$$

On page 111, we presented $2 \cdot 3$, i.e., "2 times 3", as $2 + 2 + 2$!. With this interpretation, $3 + 3$ corresponds to $3 \cdot 2$, but the fact that multiplication is a commutative operation ([Rule 6.2](#)) means that one interpretation does not exclude the other; $2 \cdot 3 = 2 + 2 + 2$ and $2 \cdot 3 = 3 + 3$ are two expressions with the same value. Thus, we can also write $2 \cdot 3 = 3 + 3$ and $3 \cdot 2 = 2 + 2 + 2$.

2.3 Multiplication as Repeated Addition

Multiplication with a positive whole number can be expressed as repeated addition.

Example 1

$$4 + 4 + 4 = 4 \cdot 3 = 3 + 3 + 3 + 3$$

$$8 + 8 = 8 \cdot 2 = 2 + 2 + 2 + 2 + 2 + 2 + 2 + 2$$

$$1 + 1 + 1 + 1 + 1 = 1 \cdot 5 = 5$$

Note

The fact that multiplication with positive whole numbers can be expressed as repeated addition does not exclude other expressions. It is not wrong to write that $2 \cdot 3 = 1 + 5$.

2.5 Division

`:` is the **division sign**. Division has three different meanings, here shown with $12 : 3$:

2.4 The Three Meanings of Division

- **Equal Distribution of Quantities**
 $12 : 3 =$ "The number in each group when 12 is divided into 3 equal groups"
- **Number of Times One Number Fits into Another**
 $12 : 3 =$ "How many times 3 fits into 12"
- **Inverse Operation of Multiplication**
 $12 : 3 =$ "The number you multiply 3 by to get 12"

The language box

A division problem consists of a **dividend**, a **divisor**, and a **quotient**. In the division problem

$$12 : 3 = 4$$

12 is the dividend, 3 is the divisor, and 4 is the quotient.

Common ways to say $12 : 3$ are:

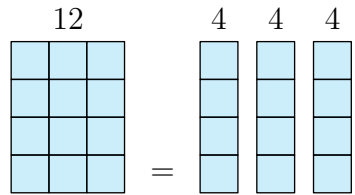
- "12 divided by 3"
- "12 divided into 3"
- "12 over 3"

In some contexts, $12 : 3$ is called the "**ratio** between 12 and 3". Then, 4 is the **ratio value**.

Often, `/` is used instead of `:`, especially in programming languages.

Equal Distribution of Quantities

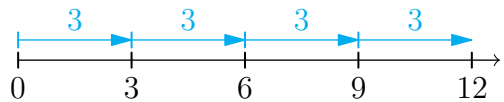
The calculation $12 : 3$ tells us to distribute 12 into 3 equal groups:



We see that each group contains 4 squares, which means that

$$12 : 3 = 4$$

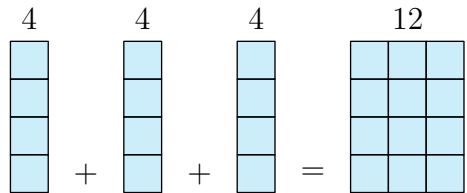
Number of Times One Number Fits into Another



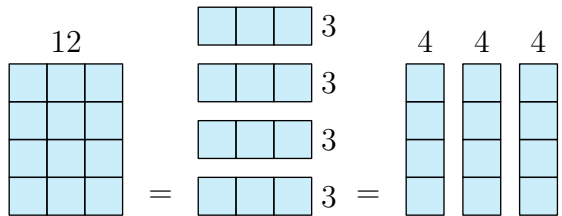
3 fits 4 times into 12, so $12 : 3 = 4$.

Inverse Operation of Multiplication

We have seen that if we divide 12 into 3 equal groups, we get 4 in each group. So, $12 : 3 = 4$. If we add these groups together again, we naturally get 12:



But this is the same as multiplying 4 by 3. So, if we know that $4 \cdot 3 = 12$, we also know that $12 : 3 = 4$. Additionally, we then know that $12 : 4 = 3$.



Example 1

Since $6 \cdot 3 = 18$, then

$$18 : 6 = 3$$

$$18 : 3 = 6$$

Example 2

Since $5 \cdot 7 = 35$, then

$$35 : 5 = 7$$

$$35 : 7 = 5$$

Exercises for Chapter 2

2.1.1

Write the numbers as the sum of two numbers.

Example

We can write 3 as $1 + 2$

- a) 4 b) 5 c) 6 d) 7 e) 8 f) 9

2.1.2

Write the numbers as the sum of three numbers.

Example

We can write 4 as $1 + 2 + 1$

- a) 5 b) 6 c) 7 d) 8 e) 9 f) 10

2.1.3

When the sum of two numbers equals 10, these numbers are called **friends of ten**. For example are 1 and 9 friends of ten because $1 + 9 = 10$.

1) Find the ten-friend of

- a) 2 b) 3 c) 4 d) 5

2) When exercise 1) is completed, why is it "unnecessary" to find the ten-friends of 6, 7 og 8?

2.1.4

Note: You can allow yourself answering the questions by simply checking a couple of examples. For proof, see [Exercise 12](#).

Choose the correct alternative of 1), 2) and 3).

- a) The sum of two even numbers is
 - 1) an even number.
 - 2) an odd number.
 - 3) sometimes an even number and sometimes an odd number.
- b) The sum of two odd numbers is
 - 1) an even number.
 - 2) an odd number.
 - 3) sometimes an even number and sometimes an odd number.
- c) The sum of an even number and an odd number is
 - 1) an even number.
 - 2) an odd number.
 - 3) sometimes an even number and sometimes an odd number.

2.2.1

Write the number as the difference of two numbers.

Example

We can write 1 as $8 - 7$.

- a) 2 b) 3 c) 4 d) 5 e) 6 f) 7 g) 8

2.2.2

Note: You can allow yourself answering the questions by simply checking a couple of examples. For proof, see [Exercise 12](#).

Choose the correct alternative of 1), 2) and 3).

- a) The difference of two even numbers is
- 1) an even number.
 - 2) an odd number.
 - 3) sometimes an even number and sometimes an odd number.
- b) The difference of two odd numbers is
- 1) an even number.
 - 2) an odd number.
 - 3) sometimes an even number and sometimes an odd number.
- c) The difference of an even number and an odd number is
- 1) an even number.
 - 2) an odd number.
 - 3) sometimes an even number and sometimes an odd number.

2.3.1

Write as a multiplication, and write the alternative sum.

Example

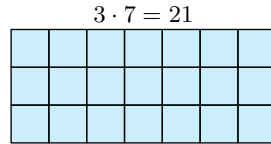
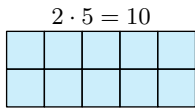
$$3 + 3 + 3 + 3 + 3 = 3 \cdot 5 = 5 + 5 + 5$$

- a) $2 + 2 + 2$
- b) $3 + 3 + 3 + 3 + 3 + 3$
- c) $4 + 4$
- d) $5 + 5 + 5 + 5 + 5 + 5 + 5 + 5 + 5 + 5$
- e) $6 + 6 + 6 + 6$
- f) $7 + 7 + 7 + 7$

2.3.2

Draw boxes to find the answer to the multiplications.

Example



- a) $4 \cdot 5$
- b) $8 \cdot 3$
- c) $2 \cdot 9$
- d) $5 \cdot 6$
- e) $7 \cdot 8$

2.3.3

- a) Will an integer multiplied by 2 always result in an even number or an odd number?
- b) Will an even number multiplied by 5 always result in an even number or an odd number? What digit will always be positioned at the one-place?
- c) Will an odd number multiplied by 5 always result in an even number or an odd number? What digit will always be positioned at the one-place?

Chapter 3

Factorization and order of operations

3.1 Order of operations

Priority of the operations

Look at the following calculation:

$$2 + 3 \cdot 4$$

This *could* have been interpreted in two ways:

1. "2 plus 3 equals 5. 5 times 4 equals 20. The answer is 20."
2. "3 times 4 equals 12. 2 plus 12 equals 14. The answer is."

But the answers are not the same! This points out the need to have rules for what to calculate first. One of these rules is that multiplication and division is to be calculated *before* addition or subtraction, which means that

$$\begin{aligned} 2 + 3 \cdot 4 &= \text{"Calculate } 3 \cdot 4, \text{ then add } 2\text{"} \\ &= 2 + 12 \\ &= 14 \end{aligned}$$

But what if we wanted to calculate $2 + 3$ first, then multiply the sum by 4? We use parentheses to tell that something is to be calculated first:

$$\begin{aligned} (2 + 3) \cdot 4 &= \text{"Calculate } 2 + 3, \text{ multiply by } 4 \text{ afterwards"} \\ &= 5 \cdot 4 \\ &= 20 \end{aligned}$$

3.1 Order of operations

1. Expressions with parentheses
2. Multiplication or division
3. Addition or subtraction

Example 1

Calculate

$$23 - (3 + 9) + 4 \cdot 7$$

Answer

$$\begin{aligned} 23 - (3 + 9) + 4 \cdot 7 &= 23 - 12 + 4 \cdot 7 && \text{Parentheses} \\ &= 23 - 12 + 28 && \text{Multiplication} \\ &= 39 && \text{Addition and subtraction} \end{aligned}$$

Example 2

Calculate

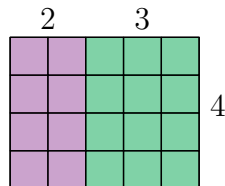
$$18 : (7 - 5) - 3$$

Answer

$$\begin{aligned} 18 : (7 - 5) - 3 &= 18 : 2 - 3 && \text{Parentheses} \\ &= 9 - 3 && \text{Division} \\ &= 6 && \text{Addition and subtraction} \end{aligned}$$

Multiplication involving parentheses

How many boxes are present in this figure?



Two correct interpretations include:

1. It is $2 \cdot 4 = 8$ purple boxes and $3 \cdot 4 = 12$ green boxes. In total there are $8 + 12 = 20$ boxes. This we can write as

$$2 \cdot 4 + 3 \cdot 4 = 20$$

2. It is $2 + 3 = 5$ boxes horizontally and 4 boxes vertically, so there are $5 \cdot 4 = 20$ boxes in total. This we can write as

$$(2 + 3) \cdot 4 = 20$$

From these two calculations it follows that

$$(2 + 3) \cdot 4 = 2 \cdot 4 + 3 \cdot 4$$

3.2 Distributive law

When an expression enclosed by parentheses is a factor, we can multiply the other factors with each term inside the parentheses.

Example 1

$$(4 + 7) \cdot 8 = 4 \cdot 8 + 7 \cdot 8$$

Example 2

$$\begin{aligned}(10 - 7) \cdot 2 &= 10 \cdot 2 - 7 \cdot 2 \\ &= 20 - 14 \\ &= 6\end{aligned}$$

Note: Obviously, it would be easier to calculate like this:

$$(10 - 7) \cdot 2 = 3 \cdot 2 = 6$$

Example 2

Calculate $12 \cdot 3$.

Answer

$$\begin{aligned}12 \cdot 3 &= (10 + 2) \cdot 3 \\ &= 10 \cdot 3 + 2 \cdot 3 \\ &= 30 + 6 \\ &= 36\end{aligned}$$

Note

We introduced parentheses as an indicator of what to calculate first, but [Rule 3.2](#) gives an alternative and equivalent interpretation of parentheses. The rule is especially useful when working with algebra (see [Part II](#)).

Multiplying by 0

Earlier we have seen that 0 can be expressed as the difference between two numbers, and this can help us calculate when multiplying by 0. Let's look at the calculation

$$(2 - 2) \cdot 3$$

By [Rule 3.2](#), we get

$$\begin{aligned}(2 - 2) \cdot 3 &= 2 \cdot 3 - 2 \cdot 3 \\ &= 6 - 6 \\ &= 0\end{aligned}$$

Since $0 = 2 - 2$, this means that

$$0 \cdot 3 = 0$$

3.3 Multiplication by 0

If 0 is a factor, the product equals 0.

Example 1

$$7 \cdot 0 = 0$$

$$0 \cdot 219 = 0$$

Associative laws

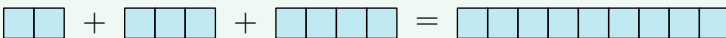
3.4 Associative law for addition

The placement of parentheses between terms has no impact on the sum.

Example

$$(2 + 3) + 4 = 5 + 4 = 9$$

$$2 + (3 + 4) = 2 + 7 = 9$$


$$\begin{array}{|c|c|c|} \hline & & \\ \hline \end{array} + \begin{array}{|c|c|c|} \hline & & \\ \hline \end{array} + \begin{array}{|c|c|c|} \hline & & \\ \hline \end{array} = \begin{array}{|c|c|c|c|c|c|c|} \hline & & & & & & \\ \hline \end{array}$$

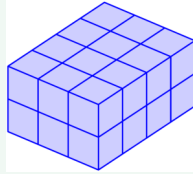
3.5 Associative law for multiplication

The placement of parentheses between factors has no impact on the product.

Example

$$(2 \cdot 3) \cdot 4 = 6 \cdot 4 = 24$$

$$2 \cdot (3 \cdot 4) = 2 \cdot 12 = 24$$



Opposite to addition and multiplication, neither subtraction nor division is associative:

$$(12 - 5) - 4 = 7 - 4 = 3$$

$$12 - (5 - 4) = 12 - 1 = 11$$

$$(80 : 10) : 2 = 8 : 2 = 4$$

$$80 : (10 : 2) = 80 : 5 = 16$$

We have seen how parentheses helps indicating the *priority* of operations, but the fact that subtraction and division are non-associative brings the need of having a rule of in which *direction* to calculate.

3.6 Direction of calculations

Operations which by [Rule 3.1](#) have equal priority, are to be calculated from left to right.

Example 1

$$\begin{aligned} 12 - 5 - 4 &= (12 - 5) - 4 \\ &= 7 - 4 \\ &= 3 \end{aligned}$$

Example 2

$$\begin{aligned} 80 : 10 : 2 &= (80 : 10) : 2 \\ &= 8 : 2 \\ &= 4 \end{aligned}$$

Example 3

$$\begin{aligned} 6 : 3 \cdot 4 &= (6 : 3) \cdot 4 \\ &= 2 \cdot 4 \\ &= 8 \end{aligned}$$

3.2 Factorization

If an integer dividend and an integer divisor results in an integer quotient, we say that the dividend is *divisible* by the divisor. For example is 6 divisible with 3 because $6 : 3 = 2$, and 40 is divisible with 10 because $40 : 10 = 4$. The concept of divisibility contributes to the definition of *prime numbers*:

3.7 Primal

A natural number larger than 1, and only divisible by itself and 1, is a prime number.

Example

The first five prime numbers are 2, 3, 5, 7 and 11.

3.8 Factorization

Factorization involves writing a number as the product of other numbers.

Example

Factorize 24 in three different ways.

Answer

$$24 = 2 \cdot 12$$

$$24 = 3 \cdot 8$$

$$24 = 2 \cdot 3 \cdot 4$$

The language box

Since 12 is divisible by 4, we say that 4 is a factor of 12.

3.9 Prime factorization

Factorization involving prime factors only is called prime factorization.

Example

Prime factorize 12.

Answer

$$12 = 2 \cdot 2 \cdot 3$$

The prime numbers from 1-100

91	92	93	94	95	96	97	98	99	100
81	82	83	84	85	86	87	88	89	90
71	72	73	74	75	76	77	78	79	80
61	62	63	64	65	66	67	68	69	70
51	52	53	54	55	56	57	58	59	60
41	42	43	44	45	46	47	48	49	50
31	32	33	34	35	36	37	38	39	40
21	22	23	24	25	26	27	28	29	30
11	12	13	14	15	16	17	18	19	20
1	2	3	4	5	6	7	8	9	10

Exercises for Chapter 3

3.1.1

Calculate by writing the numbers as sums of ones, tens and hundreds, and applying the distributive law.

Example

$$15 \cdot 3 = (10 + 5) \cdot 3 = 10 \cdot 3 + 5 \cdot 3 = 30 + 15 = 45$$

$$147 \cdot 2 = (100 + 40 + 7) \cdot 2 = 100 \cdot 2 + 40 \cdot 2 + 7 \cdot 2 = 200 + 80 + 14 = 294$$

- a) $17 \cdot 2$ b) $59 \cdot 3$ c) $25 \cdot 4$ d) $582 \cdot 2$ e) $981 \cdot 3$

3.2.1

Write the numbers as a multiplication between two factors.

- a) 100 b) 30 c) 40 d) 70
e) 42 f) 32 g) 84 h) 90

3.2.2

Prime factorize the numbers from [Exercise 3.2.1](#).

Note: It is recommended that the reader find it's own ways of prime factorization, but those who insist on following a schematic method can look up [Exercise 3.2.5](#).

3.2.3

Factorize the numbers from [Exercise 3.2.1](#) in three different ways.

3.2.4

6 is called a **perfect number** because the sum of it's factors 6 (including 1, but excluding 6) equals 6: $1 + 2 + 3 = 6$. Find the next perfect number (it lies between 15 og 30).

3.2.5

The examples below shows a method for prime factorization of numbers. Explain the method.

Example 1

$$84 = 2 \cdot 2 \cdot 3 \cdot 7$$

	:	
84	2	42
42	2	21
21	3	7

Example 2

$$595 = 5 \cdot 7 \cdot 17$$

	:	
595	5	119
119	7	17

Ponder 1

Explain why the product of two odd numbers always equals an odd number.

Ponder 2

Explain a method for checking if a number is a prime number.

Chapter 4

Fractions

4.1 Introduction

4.1 Fractions as rewriting of division

A fraction is a different way of writing a division. In a fraction the dividend is called the *numerator* and the divisor the denominator.

$$1 : 4 = \frac{1}{4} \begin{array}{l} \longleftarrow \text{Numerator} \\ \longleftarrow \text{Denominator} \end{array}$$

The language box

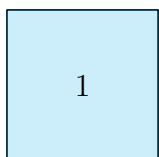
Common ways of saying $\frac{1}{4}$ are¹

- "one fourth"
- "1 of 4"
- "1 over 4"

¹We also have the expressions from the language box on page 113.

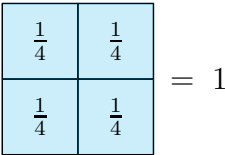
Fractions as amounts

Let us present $\frac{1}{4}$ as an amount. We then think of the number 1 as a box¹:

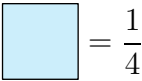

$$= 1$$

¹For practical reasons, we choose a unit box larger than the one used in [Chapter 1](#).

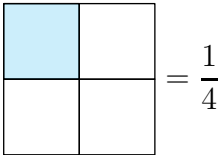
Further, we divide this box into four smaller, equal-sized boxes. The sum of these boxes equals 1.



One such box equals $\frac{1}{4}$:



However, if you from a figure only are to see how large a fraction is, the size of 1 must be known, and to make this more apparent we'll also include the "empty" boxes:



In this way, the blue and the empty boxes tell us how many pieces 1 is divided into, while the blue boxes alone tells how many of these boxes are *actually* present. In other words,

number of blue boxes = numerator

number of blue boxes + number of empty boxes = denominator

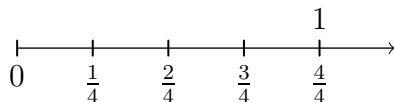
$= \frac{2}{3}$

$= \frac{7}{10}$

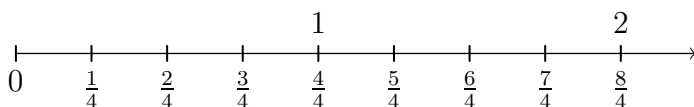
$= \frac{19}{20}$

Fractions on the number line

On the number line, we divide the length between 0 and 1 into as many pieces as the denominator indicates. In the case of a fraction with denominator 4, we divide the length between 0 and 1 into 4 equal lengths:



Moreover, fractions larger than 1 are easily presented on the number line:



Numerator and denominator summarized

Although already mentioned, the interpretations of the numerator and the denominator are of such importance that we shortly summarize them:

- The denominator tells how many pieces 1 is divided into.
- The numerator tells how many of these pieces are present.

4.2 Values, expanding and simplifying

4.2 The value of a fraction

The value of a fraction is given by dividing the numerator by the denominator.

Example

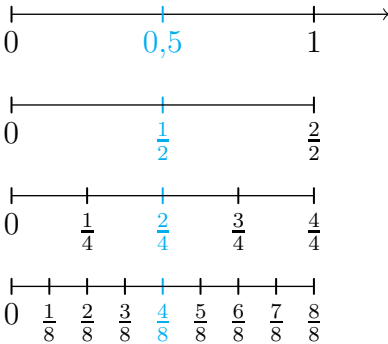
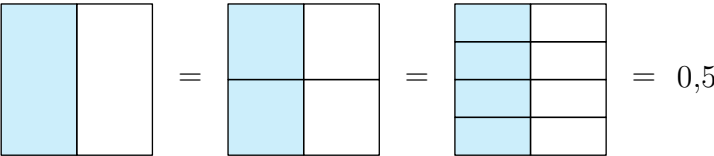
Find the value of $\frac{1}{4}$.

Answer $\frac{1}{4} = 0.25$

Fractions with equal value

Fractions can have the same value even though they look different. If you calculate $1 : 2$, $2 : 4$ and $4 : 8$, you will in every case end up with 0.5 as the answer. This means that

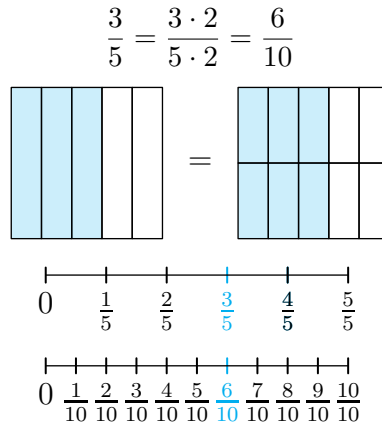
$$\frac{1}{2} = \frac{2}{4} = \frac{4}{8} = 0,5$$



Expanding

The fact that fractions can look different but have the same value, implies that we can change a fraction's look without changing its value. Let's, as an example, change $\frac{3}{5}$ into a fraction of equal value but with denominator 10:

- We can make $\frac{3}{5}$ into a fraction with denominator 10 if we divide each fifth into 2 equal pieces. In that case, 1 is divided into $5 \cdot 2 = 10$ pieces in total.
- The numerator of $\frac{3}{5}$ indicates that there are 3 fifths. When these are divided by 2, they make up $3 \cdot 2 = 6$ tenths. Hence $\frac{3}{5}$ equals $\frac{6}{10}$.



Simplifying

Notice that we can also go "the opposite way". We can change $\frac{6}{10}$ into a fraction with denominator 5 by dividing both the numerator and the denominator by 2:

$$\frac{6}{10} = \frac{6 : 2}{10 : 2} = \frac{3}{5}$$

4.3 Expanding of fractions

We can either multiply or divide both the numerator and the denominator by the same number without alternating the fractions value.

Multiplying by a number larger than 1 is called *expanding* the fraction. Dividing by a number larger than 1 is called *simplifying* the fraction.

Example 1

Expand $\frac{3}{5}$ into a fraction with denominator 20.

Answer

Since $5 \cdot 4 = 20$, we multiply both the numerator and the denominator by 4:

$$\begin{aligned}\frac{3}{5} &= \frac{3 \cdot 4}{5 \cdot 4} \\ &= \frac{12}{20}\end{aligned}$$

Example 2

Expand $\frac{150}{50}$ into a fraction with denominator 100.

Answer

Since $50 \cdot 2 = 100$, we multiply both the numerator and the denominator by 2:

$$\begin{aligned}\frac{150}{50} &= \frac{150 \cdot 2}{50 \cdot 2} \\ &= \frac{300}{100}\end{aligned}$$

Example 3

Simplify $\frac{18}{30}$ into a fraction with denominator 5.

Answer

Since $30 : 6 = 5$, we divide both the numerator and the denominator by 6:

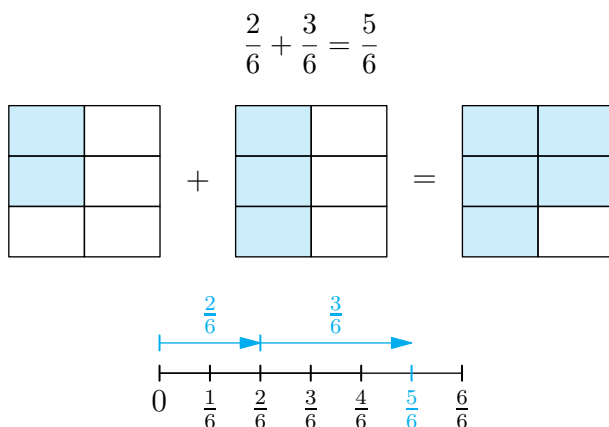
$$\begin{aligned}\frac{18}{30} &= \frac{18 : 6}{30 : 6} \\ &= \frac{3}{5}\end{aligned}$$

4.3 Addition and subtraction

Addition and subtraction of fractions are in large parts focused around the denominators. Recall that the denominators indicate the partitioning of 1. If fractions have equal denominators, they represent amounts of equal-sized pieces. In this case it makes sense calculating addition or subtraction of the numerators. However, if fractions have unequal denominators, they represent amounts of different-sized pieces, and hence addition and subtraction of the numerators makes no sense directly.

Equal denominators

If we, for example, have 2 sixths and add 3 sixths, the sum is 5 sixths:



4.4 Addition/subtraction of fractions with equal denominators

When adding/subtracting fractions with equal denominators, we find the sum/difference of the numerators and keep the denominator.

Example 1

$$\begin{aligned}\frac{2}{7} + \frac{8}{7} &= \frac{2+8}{7} \\ &= \frac{10}{7}\end{aligned}$$

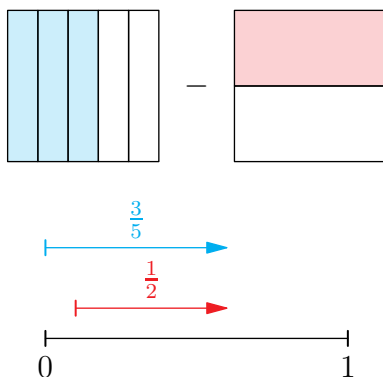
Example 2

$$\begin{aligned}\frac{7}{9} - \frac{5}{9} &= \frac{7-5}{9} \\ &= \frac{2}{9}\end{aligned}$$

Unequal denominators

Let's examine the calculation¹

$$\frac{3}{5} - \frac{1}{2}$$

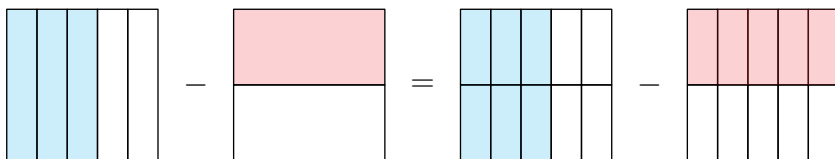


To write the difference as a single fraction, the two terms need to have denominators of equal value. Both of the fractions can have denominator 10:

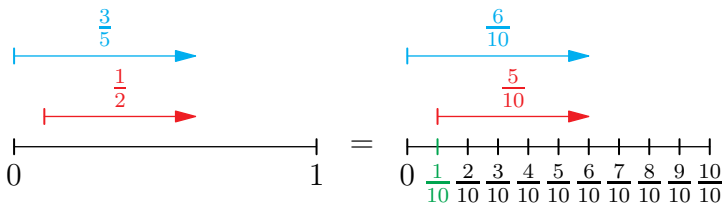
$$\frac{3}{5} = \frac{3 \cdot 2}{5 \cdot 2} = \frac{6}{10} \qquad \frac{1}{2} = \frac{1 \cdot 5}{2 \cdot 5} = \frac{5}{10}$$

Hence

$$\frac{3}{5} - \frac{1}{2} = \frac{6}{10} - \frac{5}{10}$$



¹Recall that the red-colored arrow indicates that you shall start at the arrowhead and then move to the other end.



Summarized, we have expanded the fractions such that they have denominators of equal value, that is 10. When the denominators are equal, we can calculate the difference of the numerators:

$$\begin{aligned}\frac{3}{5} - \frac{1}{2} &= \frac{6}{10} - \frac{5}{10} \\ &= \frac{1}{10}\end{aligned}$$

4.5 Addition/subtraction of fractions with unequal denominators

When calculating addition/subtraction of fractions, we must expand the fractions such that they have denominators of equal value, and then apply [Rule 4.4](#).

Example 1

Calculate

$$\frac{2}{9} + \frac{6}{7}$$

Both denominators can be transformed into 63 if multiplied by a fitting integer. Therefore, we expand the fractions as follows:

$$\begin{aligned}\frac{2 \cdot 7}{9 \cdot 7} + \frac{6 \cdot 9}{7 \cdot 9} &= \frac{14}{63} + \frac{54}{63} \\ &= \frac{68}{63}\end{aligned}$$

Common denominator

In *Example 1* above, 63 is called a *common denominator* because there exists integers which, when multiplied by the original denominators, results in 63:

$$9 \cdot 7 = 63$$

$$7 \cdot 9 = 63$$

Multiplying together the original denominators always results in a common denominator but one can avoid large numbers by finding the *smallest* common denominator. Take, for example,

$$\frac{7}{6} + \frac{5}{3}$$

$6 \cdot 3 = 18$ is a common denominator, but it's worth noticing that $6 \cdot 1 = 3 \cdot 2 = 6$ is too. Hence,

$$\begin{aligned}\frac{7}{6} + \frac{5}{3} &= \frac{7}{6} + \frac{5 \cdot 2}{3 \cdot 2} \\ &= \frac{7}{6} + \frac{10}{6} \\ &= \frac{17}{6}\end{aligned}$$

Example 2

Calculate

$$\frac{3}{2} - \frac{5}{8} + \frac{10}{4}$$

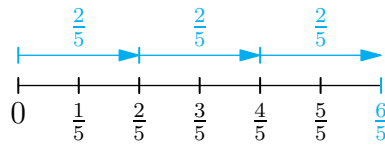
Answer

All denominators can be transformed into 8 if multiplied by a fitting integer. Therefore, we expand the fractions as follows:

$$\begin{aligned}\frac{3}{2} - \frac{5}{8} + \frac{10}{4} &= \frac{3 \cdot 4}{2 \cdot 4} - \frac{5}{8} + \frac{10 \cdot 2}{4 \cdot 2} \\ &= \frac{12}{8} - \frac{5}{8} + \frac{20}{8} \\ &= \frac{27}{8}\end{aligned}$$

4.4 Fractions multiplied by integers

In [Section 6.4](#) we observed that multiplying by an integer corresponds to repeated addition. Hence, if we are to calculate $\frac{2}{5} \cdot 3$, we can write

$$\begin{aligned}\frac{2}{5} \cdot 3 &= \frac{2}{5} + \frac{2}{5} + \frac{2}{5} \\ &= \frac{2+2+2}{5} \\ &= \frac{6}{5}\end{aligned}$$


Noticing that $2 + 2 + 2 = 2 \cdot 3$, we get

$$\begin{aligned}\frac{2}{5} \cdot 3 &= \frac{2 \cdot 3}{5} \\ &= \frac{6}{5}\end{aligned}$$

Multiplication of integers and fractions are also commutative¹:

$$\begin{aligned}3 \cdot \frac{2}{5} &= 3 \cdot 2 : 5 \\ &= 6 : 5 \\ &= \frac{6}{5}\end{aligned}$$

4.6 Fractions multiplied by integers

When multiplying a fraction by an integer, we multiply the numerator by the integer.

¹Recall that $\frac{2}{5}$ corresponds to $2 : 5$.

Example 1

$$\begin{aligned}\frac{1}{3} \cdot 4 &= \frac{1 \cdot 4}{3} \\ &= \frac{4}{3}\end{aligned}$$

Example 2

$$\begin{aligned}3 \cdot \frac{2}{5} &= \frac{3 \cdot 2}{5} \\ &= \frac{6}{5}\end{aligned}$$

An interpretation of multiplying by a fraction

By [Rule 4.6](#) we can make an interpretation of multiplying by a fraction. For example, multiplying 3 by $\frac{2}{5}$ can be interpreted in these two following ways:

- We multiply 3 by 2 and divide by 5:

$$(3 \cdot 2) : 5 = \frac{3 \cdot 2}{5} = \frac{6}{5}$$

- We divide 3 by 5 and multiply the quotient by 2:

$$3 : 5 = \frac{3}{5} \quad , \quad \frac{3}{5} \cdot 2 = \frac{3 \cdot 2}{5} = \frac{6}{5}$$

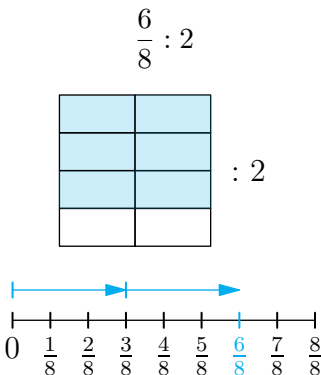
4.5 Fractions divided by integers

It is now important to recall two things:

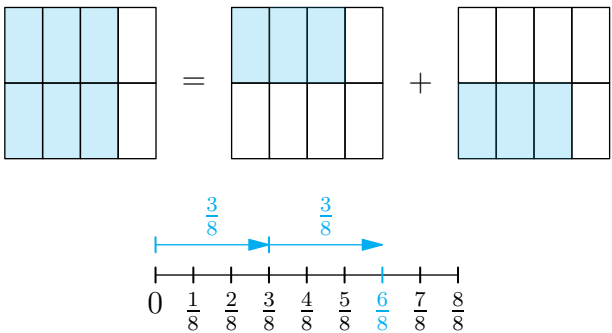
- Division can be interpreted as an equal distribution of amounts
- In a fraction, it is the numerator which indicates the amount (the denominator indicates the partitioning of 1)

When the numerator is divisible by the divisor

Let's calculate



We have 6 eights which are to be equally distributed into 2 groups.
This results in $6 : 2 = 3$ eights.



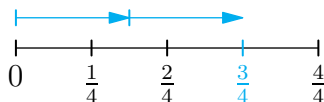
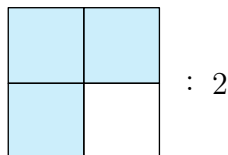
Thus

$$\frac{6}{8} : 2 = \frac{3}{8}$$

When the numerator is not divisible by the denominator

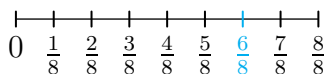
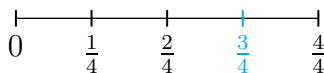
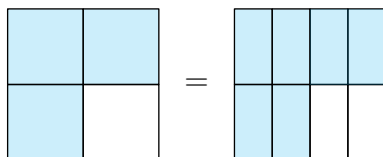
What if we are to divide $\frac{3}{4}$ by 2?

$$\frac{3}{4} : 2$$

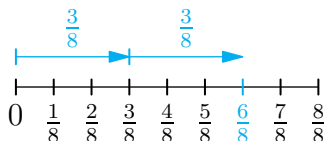
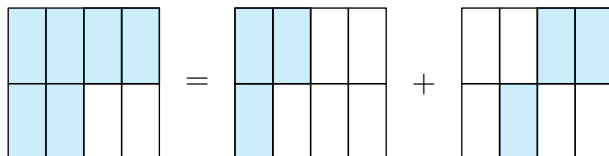


Thing is, we can always expand the fraction such that the numerator becomes divisible by the divisor. Since 2 is the divisor, we expand the fraction by 2:

$$\frac{3}{4} = \frac{3 \cdot 2}{4 \cdot 2} = \frac{6}{8}$$



Now we have 6 eights. 6 eights divided by 2 equals 3 eights:



Hence

$$\frac{3}{4} : 2 = \frac{3}{8}$$

In effect, we have multiplied $\frac{3}{4}$ by 2:

$$\begin{aligned}\frac{3}{4} : 2 &= \frac{3}{4 \cdot 2} \\ &= \frac{3}{8}\end{aligned}$$

4.7 Fractions divided by integers

When dividing a fraction by an integer, we multiply the denominator by the integer.

Example 1

$$\begin{aligned}\frac{5}{3} : 6 &= \frac{5}{3 \cdot 6} \\ &= \frac{5}{18}\end{aligned}$$

Note

At the start of this section we found that

$$\frac{4}{8} : 2 = \frac{2}{8}$$

In that case, there was no need to multiply the denominator by 2, such as [Rule 4.7](#) implies. However, if we do, we have

$$\frac{4}{8} : 2 = \frac{4}{8 \cdot 2} = \frac{4}{16}$$

Now,

$$\frac{2}{8} = \frac{2 \cdot 2}{8 \cdot 2} = \frac{4}{16}$$

Hence, unsurprisingly, the two answers are of equal value.

4.6 Fractions multiplied by fractions

We have seen that¹ multiplying a number by a fraction involves multiplying the number by the numerator and then dividing the product by the denominator. Let us apply this to calculate

$$\frac{5}{4} \cdot \frac{3}{2}$$

Firstly, we multiply $\frac{5}{4}$ by 3, then we divide the resulting product by 2. By [Rule 4.6](#), we have

$$\frac{5}{4} \cdot 3 = \frac{5 \cdot 3}{4}$$

And by [Rule 4.7](#), we get

$$\frac{5 \cdot 3}{4} : 2 = \frac{5 \cdot 3}{4 \cdot 2}$$

Hence

$$\frac{5}{4} \cdot \frac{3}{2} = \frac{5 \cdot 3}{4 \cdot 2}$$

4.8 Fractions multiplied by fractions

When multiplying a fraction by a fraction, we multiply numerator by numerator and denominator by denominator.

Example 1

$$\begin{aligned}\frac{4}{7} \cdot \frac{6}{9} &= \frac{4 \cdot 6}{7 \cdot 9} \\ &= \frac{24}{63}\end{aligned}$$

Example 2

$$\begin{aligned}\frac{1}{2} \cdot \frac{9}{10} &= \frac{1 \cdot 9}{2 \cdot 10} \\ &= \frac{9}{20}\end{aligned}$$

¹Look at the text box titled "An interpretation of multiplying by a fraction" on page 69.

4.7 Cancellation of fractions

When the numerator and the denominator are of equal value, the fractions value always equals 1. For example, $\frac{3}{3} = 1$, $\frac{25}{25} = 1$ etc. We can exploit this fact to simplify expressions involving fractions.

Let us simplify the expression

$$\frac{8 \cdot 5}{9 \cdot 8}$$

Since $8 \cdot 5 = 5 \cdot 8$, we can write

$$\frac{8 \cdot 5}{9 \cdot 8} = \frac{5 \cdot 8}{9 \cdot 8}$$

And, as recently seen ([Rule 4.8](#)), we have

$$\frac{5 \cdot 8}{9 \cdot 8} = \frac{5}{9} \cdot \frac{8}{8}$$

Since $\frac{8}{8} = 1$,

$$\begin{aligned} \frac{5}{9} \cdot \frac{8}{8} &= \frac{5}{9} \cdot 1 \\ &= \frac{5}{9} \end{aligned}$$

When multiplication is exclusively present in a fraction, you can always shuffle the way we did in the above expressions. However, when you have understood the outcome of the shuffling, it is better to apply *cancellation*. You then draw a line across two and two equal factors, thus indicating that they constitute a fraction which equals 1. Hence, our most recent example can be simplified to

$$\frac{\cancel{8} \cdot 5}{9 \cdot \cancel{8}} = \frac{5}{9}$$

4.9 Cancellation of factors

When multiplication is exclusively present in a fraction, we can cancel pair of equal factors in numerator and denominator.

Example 1

Cancel as many factors as possible in the fraction.

$$\frac{3 \cdot 12 \cdot 7}{7 \cdot 4 \cdot 12}$$

Answer

$$\frac{3 \cdot \cancel{12} \cdot \cancel{7}}{\cancel{7} \cdot 4 \cdot \cancel{12}} = \frac{3}{4}$$

Example 2

Simplify the fraction $\frac{12}{42}$.

Answer

We notice that 6 is a factor of both 12 and 42, and thus

$$\begin{aligned}\frac{12}{42} &= \frac{\cancel{6} \cdot 2}{\cancel{6} \cdot 7} \\ &= \frac{2}{7}\end{aligned}$$

Example 3

Simplify the fraction $\frac{48}{16}$.

Answer

We notice that 16 is a factor of 48, and thus

$$\begin{aligned}\frac{48}{16} &= \frac{3 \cdot \cancel{16}}{\cancel{16}} \\ &= \frac{3}{1} \\ &= 3\end{aligned}$$

Note: If all factors are canceled in the numerator or the denominator, 1 takes their place.

Simplifying by prime factorization

It's not necessarily easy to notice common factors, as we did in *Example 2* and *Example 3* above. To be sure you have not overlooked any common factors, you can always prime factorize (see page ??) both the numerator and the denominator. For example,

$$\begin{aligned}\frac{12}{42} &= \frac{\cancel{2} \cdot 2 \cdot \cancel{3}}{\cancel{2} \cdot \cancel{3} \cdot 7} \\ &= \frac{2}{7}\end{aligned}$$

Fractions simplify calculations

The decimal number 0.125 can be written as the fraction $\frac{1}{8}$. The calculation

$$0.125 \cdot 16$$

is, for the most of us, rather strenuous to carry out. However, exploiting the nature of fractions, we have

$$\begin{aligned}0.125 \cdot 16 &= \frac{1}{8} \cdot 16 \\ &= \frac{2 \cdot \cancel{8}}{\cancel{8}} \\ &= 2\end{aligned}$$

"Cancelling zeros"

A number such as 3000 equals $3 \cdot 10 \cdot 10 \cdot 10$, while 700 equals $7 \cdot 10 \cdot 10$. Hence, we can simplify $\frac{3000}{700}$ like this:

$$\begin{aligned}\frac{3000}{700} &= \frac{3 \cdot \cancel{10} \cdot \cancel{10} \cdot 10}{7 \cdot \cancel{10} \cdot \cancel{10}} \\ &= \frac{3 \cdot 10}{7} \\ &= \frac{30}{7}\end{aligned}$$

In practice, this is the same as "cancelling zeros":

$$\frac{300\cancel{0}}{70\cancel{0}} = \frac{30}{7}$$

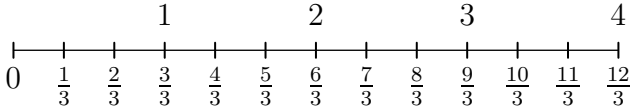
Aware! Zeros are the only digits we can "cancel" this way. For example, $\frac{123}{13}$ cannot be simplified in any way. Also, we can only "cancel" zeros which are right-most situated, e.g. we cannot "cancel" zeros in the fraction $\frac{101}{10}$.

4.8 Division by fractions

Division by studying the number line

Let's calculate $4 : \frac{2}{3}$. Since the fraction have denominator 3, it could be wise to transform also 4 into a fraction with denominator 3.

$$4 = \frac{12}{3}$$

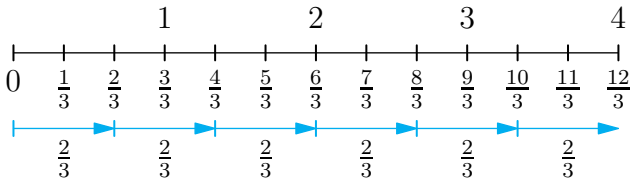


Recall that one of the interpretations of $4 : \frac{2}{3}$ is

"The number of $\frac{2}{3}$'s added to make 4."

By studying a number line, we find that 6 instances of $\frac{2}{3}$ added together equals 4. Hence

$$4 : \frac{2}{3} = 6$$



A general method

We can't study the number line every time we are to divide by a fraction, so here we shall find a general method, again with $4 : \frac{2}{3}$ as our example. In this case, we apply the following interpretation of division:

$$4 : \frac{2}{3} = \text{"The number to multiply } \frac{2}{3} \text{ by to make 4."}$$

We begin the search of this number by multiplying $\frac{2}{3}$ by the number which results in the product equal to 1. This number is the *inverted fraction* of $\frac{2}{3}$, namely $\frac{3}{2}$:

$$\frac{2}{3} \cdot \frac{3}{2} = 1$$

Now we only have to multiply by 4 to make 4:

$$\frac{2}{3} \cdot \frac{3}{2} \cdot 4 = 4$$

Therefore, to make 4 we must multiply $\frac{2}{3}$ by $\frac{3}{2} \cdot 4$. Consequently,

$$\begin{aligned} 4 : \frac{2}{3} &= \frac{3}{2} \cdot 4 \\ &= 6 \end{aligned}$$

4.10 Fractions divided by fractions

When dividing a number by a fraction, we multiply the number by the inverted fraction.

Example 1

$$\begin{aligned} 3 : \frac{2}{9} &= 3 \cdot \frac{9}{2} \\ &= \frac{27}{2} \end{aligned}$$

Example 2

$$\begin{aligned} \frac{4}{3} : \frac{5}{8} &= \frac{4}{3} \cdot \frac{8}{5} \\ &= \frac{32}{15} \end{aligned}$$

Example 3

$$\begin{aligned}\frac{3}{5} \div \frac{3}{10} &= \frac{3}{5} \cdot \frac{10}{3} \\ &= \frac{30}{15}\end{aligned}$$

In this case we should also observe that the fraction can be simplified:

$$\begin{aligned}\frac{30}{15} &= \frac{2 \cdot \cancel{15}}{\cancel{15}} \\ &= 2\end{aligned}$$

Note: Canceling factors along the way saves the labor of working with large numbers:

$$\begin{aligned}\frac{3}{5} \cdot \frac{10}{3} &= \frac{\cancel{3} \cdot 2 \cdot \cancel{5}}{\cancel{5} \cdot \cancel{3}} \\ &= 2\end{aligned}$$

4.9 Rational numbers

4.11 Rational numbers

Any number which can be expressed as a fraction is a *rational number*.

Note

Rational numbers is a collective name of

- **Integers**

For example $4 = \frac{4}{1}$.

- **Decimal numbers with a finite number of digits**

For example $0,2 = \frac{1}{5}$.

- **Decimal numbers with infinite digits in a repeating manner**

For example ¹ $0.08\bar{3} = \frac{1}{12}$.

¹ $\bar{3}$ indicates that 3 repeats infinite. Another way of expressing this is by using $\dots$. That is, $0.08\bar{3} = 0.08333333\dots$

4.12 Mixed numbers

If we add an integer with a fraction where the numerator has a lower value than the denominator, we have a **mixed number**

Example 1

Three different mixed numbers:

$$2 + \frac{5}{7} \qquad 8 + \frac{2}{7} \qquad \frac{1}{10} + 4$$

Note

In a lot of texts you will find numbers like those from *Example 1* written like this:

$$2\frac{5}{7} \qquad 8\frac{2}{7} \qquad 4\frac{1}{10}$$

Example 2

Write the fraction $\frac{17}{3}$ as a mixed number.

Answer

We observe that 17 is the numerator and 3 the denominator. The largest number we can multiply by 3 without getting a product larger than 17 is 5, so

$$\begin{aligned}\frac{17}{3} &= \frac{5 \cdot 3 + 2}{3} \\ &= \frac{5 \cdot \cancel{3} + 2}{\cancel{3}} \\ &= 5 + \frac{2}{3}\end{aligned}$$

Example 3

Write the mixed number $3 + \frac{4}{5}$ as a fraction.

Answer

We have $3 = \frac{3}{1}$, so

$$3 + \frac{4}{5} = \frac{3}{1} + \frac{4}{5}$$

Moreover¹,

$$\begin{aligned}\frac{3}{1} + \frac{4}{5} &= \frac{3 \cdot 5}{1 \cdot 5} + \frac{4}{5} \\ &= \frac{15}{5} + \frac{4}{5} \\ &= \frac{19}{5}\end{aligned}$$

¹See [Rule ??](#).

Exercises for Chapter 4

4.1.1

Find the value of the fraction.

a) $\frac{18}{3}$ b) $\frac{20}{4}$ c) $\frac{10}{5}$ d) $\frac{42}{6}$ e) $\frac{63}{7}$ f) $\frac{32}{8}$

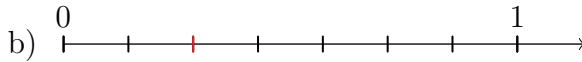
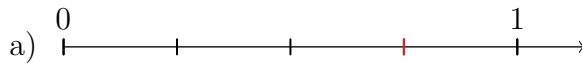
4.1.2

Find the value of the fraction. Use a calculator if necessary.

a) $\frac{1}{2}$ b) $\frac{1}{4}$ c) $\frac{1}{5}$ d) $\frac{3}{4}$ e) $\frac{2}{5}$ f) $\frac{3}{5}$ g) $\frac{4}{5}$
f) $\frac{3}{2}$ g) $\frac{1}{3}$ h) $\frac{5}{2}$ i) $\frac{5}{6}$ j) $\frac{7}{5}$ k) $\frac{11}{4}$ l) $\frac{7}{10}$

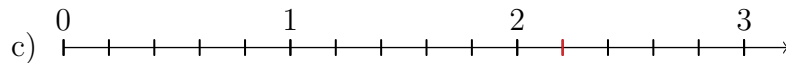
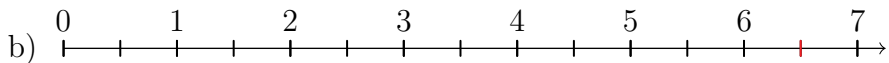
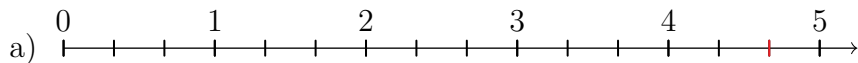
4.1.3

Write the fraction positioned at the red mark.



4.1.4

Write the fraction positioned at the red mark.



4.2.1

Example

$$\frac{9}{8} \text{ expanded by } 3 = \frac{9 \cdot 3}{8 \cdot 3} = \frac{27}{24}$$

Expand

a) $\frac{10}{3}$ by 2. b) $\frac{3}{4}$ by 3. c) $\frac{3}{7}$ by 4.

c) $\frac{9}{8}$ by 5. d) $\frac{9}{5}$ by 6. e) $\frac{11}{4}$ by 7.

4.2.2

Expand

a) $\frac{7}{3}$ to a fraction with denominator 15.

b) $\frac{3}{4}$ to a fraction with denominator 32.

c) $\frac{10}{9}$ to a fraction with denominator 63.

4.2.3

Example

$$\frac{10}{8} \text{ simplified by } 2 = \frac{10 : 2}{8 : 2} = \frac{5}{4}$$

Simplify

a) $\frac{14}{26}$ by 2. b) $\frac{15}{12}$ by 3. c) $\frac{20}{16}$ by 4.

c) $\frac{35}{50}$ by 5. d) $\frac{54}{18}$ by 6. e) $\frac{49}{63}$ by 7.

4.2.4

Simplify

a) $\frac{27}{12}$ to a fraction with denominator 4.

b) $\frac{36}{20}$ to a fraction with denominator 5.

c) $\frac{18}{63}$ to a fraction with denominator 7.

4.3.1

Calculate.

a) $\frac{4}{3} + \frac{6}{3}$ b) $\frac{5}{4} + \frac{9}{4}$ c) $\frac{1}{6} + \frac{10}{6}$ d) $\frac{8}{7} + \frac{2}{7}$ e) $\frac{1}{2} + \frac{1}{2}$

4.3.2

Calculate.

a) $\frac{10}{3} + \frac{4}{3} + \frac{8}{3}$ b) $\frac{4}{5} + \frac{3}{5} + \frac{1}{5}$ c) $\frac{11}{7} + \frac{2}{7} + \frac{4}{7}$

4.3.3

Calculate.

a) $\frac{5}{3} - \frac{4}{3}$ b) $\frac{9}{4} - \frac{7}{4}$ c) $\frac{11}{6} - \frac{1}{6}$ d) $\frac{11}{7} - \frac{4}{7}$ e) $\frac{1}{2} - \frac{1}{2}$

4.3.4

Calculate.

a) $\frac{4}{5} + \frac{3}{5} - \frac{1}{5}$ b) $\frac{11}{7} - \frac{2}{7} - \frac{4}{7}$ c) $\frac{10}{3} - \frac{4}{3} + \frac{8}{3}$

4.3.5

Calculate.

a) $\frac{2}{5} + \frac{3}{6}$ b) $\frac{5}{7} + \frac{4}{9}$ c) $\frac{10}{3} + \frac{7}{8}$ d) $\frac{7}{5} + \frac{9}{4}$ e) $\frac{1}{3} + \frac{1}{2}$

4.3.6

Calculate.

a) $\frac{2}{5} - \frac{3}{10}$ b) $\frac{5}{4} - \frac{4}{9}$ c) $\frac{10}{9} - \frac{1}{8}$ d) $\frac{4}{5} - \frac{1}{4}$ e) $\frac{5}{2} - \frac{5}{3}$

4.3.7

Calculate.

a) $\frac{2}{3} + \frac{1}{2} - \frac{3}{4}$ b) $\frac{10}{2} - \frac{1}{6} + \frac{2}{5}$ c) $\frac{9}{2} - \frac{2}{7} - \frac{1}{8}$

4.4.1

Calculate.

a) $\frac{4}{3} \cdot 5$ b) $\frac{5}{7} \cdot 8$ c) $\frac{9}{10} \cdot 6$ d) $\frac{8}{7} \cdot 10$ e) $\frac{3}{2} \cdot 7$

f) $7 \cdot \frac{4}{3}$ g) $5 \cdot \frac{7}{3}$ h) $3 \cdot \frac{10}{7}$ i) $1 \cdot \frac{5}{11}$ j) $8 \cdot \frac{9}{17}$

4.5.1

Calculate.

a) $\frac{4}{3} : 5$ b) $\frac{5}{7} : 8$ c) $\frac{9}{10} : 6$ d) $\frac{8}{7} : 10$ e) $\frac{3}{2} : 7$

f) $\frac{9}{10} : 11$ g) $\frac{1}{5} : 12$ h) $\frac{9}{10} : 29$ i) $\frac{8}{9} : 51$ j) $\frac{3}{2} : 79$

4.6.1

Calculate.

a) $\frac{4}{3} \cdot \frac{5}{9}$ b) $\frac{7}{8} \cdot \frac{1}{4}$ c) $\frac{2}{7} \cdot \frac{9}{3}$ d) $\frac{10}{3} \cdot \frac{6}{5}$ e) $\frac{3}{2} \cdot \frac{7}{5}$

f) $\frac{2}{3} \cdot \frac{5}{7}$ g) $\frac{8}{9} \cdot \frac{2}{3}$ h) $\frac{10}{3} \cdot \frac{8}{3}$ i) $\frac{4}{5} \cdot \frac{9}{7}$ j) $\frac{7}{2} \cdot \frac{5}{6}$

4.7.1

Calculate.

a) $\frac{3}{10} \cdot \frac{5}{4}$ b) $\frac{17}{8} \cdot \frac{9}{4}$ c) $\frac{23}{8} \cdot \frac{2}{4}$ d) $\frac{7}{81} \cdot \frac{3}{8}$ e) $\frac{7}{8} \cdot \frac{29}{41}$

4.8.1

Cancel as many factors as possible.

a) $\frac{3 \cdot 11 \cdot 8}{4 \cdot 8 \cdot 3}$ b) $\frac{5 \cdot 12 \cdot 7 \cdot 2}{2 \cdot 8 \cdot 12}$ c) $\frac{6 \cdot 10}{6 \cdot 9 \cdot 10}$ d) $\frac{7 \cdot 4 \cdot 3}{7 \cdot 3}$

4.8.2

Simplify the fraction as much as possible.

a) $\frac{28}{16}$ b) $\frac{18}{42}$ c) $\frac{24}{36}$ d) $\frac{56}{49}$ e) $\frac{25}{50}$ f) $\frac{21}{14}$

4.8.3

Example

$$\frac{3}{4} \cdot 20 = \frac{3}{\cancel{4}} \cdot \cancel{4} \cdot 5 = 3 \cdot 5 = 15$$

Exploit that the numerator is a factor in the other factor, and calculate.

a) $\frac{7}{3} \cdot 21$ b) $\frac{9}{5} \cdot 30$ c) $\frac{10}{7} \cdot 49$ d) $\frac{8}{9} \cdot 18$ e) $\frac{5}{4} \cdot 24$
f) $8 \cdot \frac{3}{2}$ g) $35 \cdot \frac{5}{7}$ h) $63 \cdot \frac{2}{9}$ i) $48 \cdot \frac{1}{6}$ j) $27 \cdot \frac{7}{3}$

4.9.1

Calculate.

a) $4 : \frac{9}{8}$ b) $7 : \frac{3}{5}$ c) $10 : \frac{7}{3}$ d) $5 : \frac{4}{5}$ e) $2 : \frac{5}{11}$

4.9.2

Calculate, and simplify the fraction as much as possible.

a) $4 : \frac{8}{9}$ b) $7 : \frac{21}{5}$ c) $10 : \frac{5}{3}$ d) $5 : \frac{5}{4}$ e) $2 : \frac{8}{11}$

4.9.3

Calculate.

a) $\frac{2}{3} : \frac{5}{7}$ b) $\frac{8}{9} : \frac{5}{3}$ c) $\frac{10}{3} : \frac{7}{3}$ d) $\frac{1}{5} : \frac{4}{7}$ e) $\frac{6}{5} : \frac{3}{11}$

4.9.4

Example

$$\frac{3}{4} : \frac{15}{8} = \frac{\cancel{3}}{\cancel{2} \cdot \cancel{2}} \cdot \frac{\cancel{2} \cdot \cancel{2} \cdot 2}{\cancel{3} \cdot 5} = \frac{2}{5}$$

Note: Here, we have chosen to prime factorize the numbers, however, this is not necessary if you detect the common factors of the numerators and the denominators.

Exploit that the numerators and denominators have common factors, and calculate.

a) $\frac{7}{9} : \frac{21}{12}$ b) $\frac{35}{24} : \frac{7}{18}$ c) $\frac{84}{55} : \frac{42}{77}$

Ponder 3

Apply [Rule 4.7](#) and [Rule 4.8](#) to insert the missing integers in place of " _".

- a) Multiplying by $\frac{1}{2}$ is the same as dividing by _ .
- b) Multiplying by $\frac{1}{4}$ is the same as dividing by _ .
- c) Multiplying by $\frac{1}{5}$ is the same as dividing by _ .

Look back at the answers of [Exercise 4.1.2a](#)) - g). Fill inn the inte-
ger missing in place of " _".

- d) Multiplying by 0,5 is the same as dividing by _ .
- e) Multiplying by 0,25 is the same as dividing by _ .
- f) Multiplying by 0,2 is the same as dividing by _ .
- g) Multiplying by 0,75 is the same as multiplying by _ and di-
viding by _ .
- h) Multiplying by 0,4 is the same as multiplying by _ and divid-
ing by _ .
- i) Multiplying by 0,6 is the same as multiplying by _ and divid-
ing by _ .
- j) Multiplying by 0,8 is the same as multiplying by _ and divid-
ing by _ .

Ponder 4

Look up [Rule 4.10](#) and the answers of [Exercise 4.1.2a](#)) - g). Fill in the missing integers in place of " _".

- (a) Dividing by 0,5 is the same as multiplying by _.
- (b) Dividing by 0,25 is the same as multiplying by _.
- (c) Dividing by 0,2 is the same as multiplying by _.
- (d) Dividing by 0,75 is the same as multiplying by _ and dividing by _.
- (e) Dividing by 0,4 is the same as multiplying by _ and dividing by _.
- (f) Dividing by 0,6 is the same as multiplying by _ and dividing by _.
- (g) Dividing by 0,8 is the same as multiplying by _ and dividing by _.

Ponder 5

Calculate.

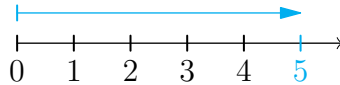
a) $\frac{5}{204} + \frac{7}{198}$ b) $\frac{11}{350} + \frac{17}{315}$

Chapter 5

Negative numbers

5.1 Introduction

Earlier we have seen that e.g. 5 on a number line is placed 5 units to the right of 0.

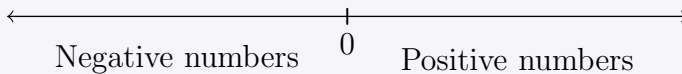


But what if we move in the other direction, that is to the left? The question is answered by introducing *negative numbers*.

5.1 Positive and negative numbers

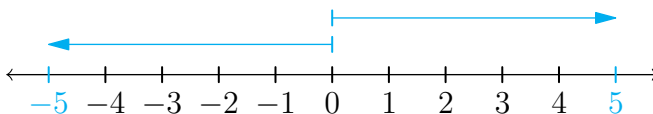
On a number line, the following applies:

- Numbers placed to *the right* of 0 are positive numbers.
- Numbers placed to *the left* of 0 are negative numbers.



However, relying on the number line every time negative numbers are involved would be very inconvenient, therefore we use a symbol to indicate negative numbers. This is $-$, simply the same as the symbol of subtraction. From this it follows that 5 is a positive number, while -5 is a negative number. On the number line,

- 5 is placed 5 units to *the right* of 0.
- -5 is placed 5 units to *the left* of 0.



Hence, the big difference between 5 and -5 is on which side of 0 the numbers are placed. Since 5 and -5 have the same distance from 0, we say that 5 and -5 have equal *length*.

5.2 Length (absolute value/modulus/magnitude)

The length of a number is expressed by the symbol $||$.

The length of a positive number equals the value of the number.

The length of a negative number equals the value of the positive number with corresponding digits.

Example 1

$$|27| = 27$$

Example 2

$$|-27| = 27$$

Sign

Sign is a collective name of $+$ and $-$. $+$ is the sign of 5 and $-$ is the sign of -5 .

5.2 The elementary operations

The introduction of negative numbers brings new aspects to the elementary operations. When adding, subtracting, multiplying or dividing by negative numbers, we'll frequently, for clarity, enclose negative numbers by parentheses. Then we'll write e.g. -4 as (-4) .

Addition

When adding in [Section 6.2](#) $+$ implied moving to *the right*. Negative numbers bring an alternation of the interpretation of $+$:

$+$ "As long and in *the same* direction as"

Let's study the calculation

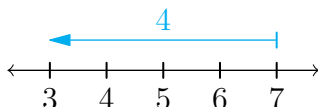
$$7 + (-4)$$

Our altered definition of $+$ implies that

$$7 + (-4) = \text{"}7 \text{ and as long and in the same direction as } (-4)\text{"}$$

(-4) has length 4 and direction to *the left*. Hence, the calculation tells us to start at 7 and then move the length of 4 to *the left*.

$$7 + (-4) = 3$$



5.3 Addition involving negative numbers

Adding a negative number is the same as subtracting the number of equal magnitude.

Example 1

$$4 + (-3) = 4 - 3 = 1$$

Example 2

$$-8 + (-3) = -8 - 3 = -11$$

Notice

[Rule 6.1](#) declares that addition is commutative. This also applies

after introducing negative numbers, for example is

$$7 + (-3) = 4 = -3 + 7$$

Subtraction

In [Section 6.3](#), $-$ implied moving to *the left*. The interpretation of $-$ also needs an alternation when working with negative numbers:

$-$ "As long and in the *opposite* direction as"

Let's study the calculation

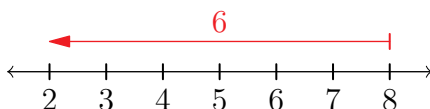
$$2 - (-6)$$

Our altered definition of $-$ implies that

$$2 - (-6) = "2 \text{ and as long and in the } \textit{opposite} \text{ direction as } (-6)"$$

-6 have length 6 and direction to *the left*. When moving an equal length, but in the *opposite* direction, we have to move the length of 6 to *the right*¹. This is equivalent to adding 6:

$$2 - (-6) = 2 + 6 = 8$$



5.4 Subtraction involving negative numbers

Subtracting a negative number is the same as adding the positive number of equal magnitude.

Example 1

$$11 - (-9) = 11 + 9 = 20$$

Example 2

$$-3 - (-7) = -3 + 7 = 4$$

¹Once again, recall that the red colored arrow indicates starting at the arrowhead, then moving to the other end.

Multiplication

In [Section 6.4](#), multiplication by positive integers were introduced as repeated addition. By our altered interpretations of addition and subtraction we can now also alternate the interpretation of multiplication:

5.5 Multiplication by positive and negative integers

- Multiplication by a positive integer corresponds to repeated addition.
- Multiplication by a negative integer corresponds to repeated subtraction.

Example 1

$$\begin{aligned} 2 \cdot 3 &= \text{"As long and in the same direction as 2, 3 times"} \\ &= 2 + 2 + 2 \\ &= 6 \end{aligned}$$

Example 2

$$\begin{aligned} (-2) \cdot 3 &\equiv \text{"As long and in the same direction as } (-2), 3 \text{ times"} \\ &= -6 \end{aligned}$$

Example 3

$$\begin{aligned} 2 \cdot (-3) &\equiv \text{"As long and in the opposite direction as 2, 3 times"} \\ &= -6 \end{aligned}$$

Example 4

$(-3) \cdot (-4) \equiv \underbrace{3 + 3 + 3 + 3}_{\text{As long and in the opposite direction as } -3, 4 \text{ times}}$
 $= 12$

Multiplication is commutative

Example 2 and *Example 3* on page 96 illustrates that [Rule 6.2](#) also implies after introducing negative numbers:

$$(-2) \cdot 3 = 3 \cdot (-2)$$

It would be laborious to calculate multiplication by repeated addition/subtraction every time a negative number were involved, however, as a direct consequence of [Rule 5.5](#) we can make the two following rules:

5.6 Multiplication by negative numbers I

The product of a negative number and a positive number is a negative number.

The magnitude of the factors multiplied together yields the magnitude of the product.

Example 1

Calculate $(-7) \cdot 8$

Answer

Since $7 \cdot 8 = 56$, we have $(-7) \cdot 8 = -56$

Example 2

Calculate $3 \cdot (-9)$.

Answer

Since $3 \cdot 9 = 27$, we have $3 \cdot (-9) = -27$

5.7 Multiplication by negative numbers II

The product of two negative numbers is a positive number.

The magnitude of the factors multiplied together yields the value of the product.

Example 1 $(-5) \cdot (-10) = 5 \cdot 10 = 50$

Example 2 $(-2) \cdot (-8) = 2 \cdot 8 = 16$

Division

From the definition of division (see [Section 6.5](#)), combined with what we now know about multiplication involving negative numbers, it follows that

$-18 : 6 =$ "The number which yields -18 when multiplied by 6 "

$$6 \cdot (-3) = -18, \text{ hence } -18 : 6 = -3$$

$42 : (-7) =$ "The number which yields 42 when multiplied by -7 "

$$(-7) \cdot (-8) = 42, \text{ hence } 42 : (-7) = -8$$

$-45 : (-5) =$ "The number which yields -45 when multiplied by -5 "

$$(-5) \cdot 9 = -45, \text{ hence } -45 : (-5) = 9$$

5.8 Division involving negative numbers

Division between a positive number and a negative number yields a negative number.

Division between two negative numbers yields a positive number.

The magnitude of the dividend divided by the magnitude of the divisor yields the magnitude of the quotient.

Example 1

$$-24 : 6 = -4$$

Example 2

$$24 : (-2) = -12$$

Example 3

$$-24 : (-3) = 8$$

Example 4

$$\frac{2}{-3} = -\frac{2}{3}$$

Example 5

$$\frac{-10}{7} = -\frac{10}{7}$$

5.3 Negative numbers as amounts

Notice: This view of negative numbers will first come into use in [Section 9.2](#), a section a lot of readers can skip without loss of understanding.

So far, we have studied negative number by the aid of number lines. Studying negative numbers as amounts is at first difficult because negative amounts makes no sense! To make an interpretation of negative numbers through the perspective of amounts, we'll use what we shall call the *weight principle*. Then we look upon the numbers as amounts of forces. The positive numbers are amounts of forces acting downwards while the negative numbers are amounts of forces working upwards¹. In this way, the results of calculations involving positive and negative numbers can be looked upon as the result of weighing the amounts. Hence, a positive number and a negative number of equal magnitude will cancel each other.

5.9 Negative numbers as amounts

Negative numbers will be illustrated as a light blue amount:

$$\boxed{} = -1$$

Example

$$1 + (-1) = 0$$

$$\boxed{} + \boxed{} = 0$$

¹From reality one can look upon the positive and the negative numbers as balloons filled with air and helium, respectively. Balloons filled with air acts with a force downwards (they fall), while balloons filled with helium acts with a force upwards (they rise).

Exercises for Chapter 5

5.1.1

Example

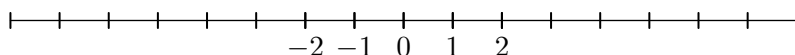
8 is directed to the right, and has length 8.

-7 is directed to the left, and has length 7.

Fill in the missing words.

- a) 9 is directed to the, and has length
- b) 4 is directed to the, and has length
- c) -3 is directed to the, and has length
- d) 12 is directed to the, and has length
- e) -11 is directed to the, and has length
- f) -25 is directed to the, and has length

5.1.2



draw the number line, and place the the numbers on it.

- a) 3 b) -4 c) -8 d) 7 e) -3 f) 5 g) -5

5.1.3

Look back at the numbers from [Exercise 5.1.2](#). Write down which numbers are

- a) positive numbers b) negative numbers

5.2.1

Calculate.

- a) $8 + (-7)$ b) $12 + (-5)$ c) $9 + (-3)$ d) $7 + (-7)$
e) $-5 + 8$ f) $-9 + 10$ g) $-1 + 11$ h) $-4 + 9$

5.2.2

Calculate.

- a) $3 + (-19)$ b) $7 + (-15)$ c) $-20 + (-3)$ d) $7 + (-7)$
e) $-4 + (-19)$ f) $-2 + (-15)$ g) $-8 + 5$ h) $-6 + 6$

5.2.3

Calculate.

- a) $8 - (-7)$ b) $12 - (-5)$ c) $9 - (-3)$ d) $7 - (-7)$
e) $-5 - 8$ f) $-9 - 10$ g) $-1 - 11$ h) $-4 - 9$

5.2.4

Calculate.

- a) $3 - (-19)$ b) $7 - (-15)$ c) $-20 - (-3)$ d) $7 - (-7)$
e) $-4 - (-19)$ f) $-2 - (-15)$ g) $-8 - 5$ h) $-6 - 6$

5.2.5

Calculate.

- a) $3 \cdot (-4)$ b) $5 \cdot (-10)$ c) $7 \cdot (-9)$ d) $4 \cdot (-6)$
e) $(-7) \cdot 8$ f) $(-3) \cdot 9$ g) $(-1) \cdot 12$ h) $(-10) \cdot 4$
i) $(-3) \cdot 7$ j) $(-5) \cdot 5$ k) $(-6) \cdot 2$ l) $(-8) \cdot 9$

5.2.6

Calculate.

- a) $(-32) : 8$ b) $(-42) : 7$ c) $(-30) : 6$ d) $(-20) : 5$
e) $72 : (-9)$ f) $63 : (-7)$ g) $50 : (-10)$ h) $25 : (-5)$
i) $(-72) : (-9)$ j) $(-63) : (-7)$ k) $(-50) : (-10)$

Chapter 6

Calculation methods

6.1 Mathematical Operations

If we combine two or more numbers to create a new number, we perform a **mathematical operation**. **Addition**, **subtraction**, **multiplication**, and **division** are our fundamental mathematical operations, and we call them the four basic **arithmetic operations**. When we use these operations, we create **calculations**. The value we get from performing a mathematical operation is usually called the **answer** to the calculation.

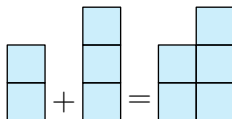
6.2 Addition

Addition with Quantities: Adding

When we have a quantity and want to add more, we use the **plus sign**

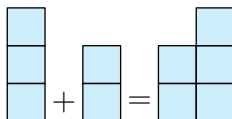
+. If we have 2 and want to add 3, we write

$$2 + 3 = 5$$



The order in which we add the numbers does not matter; starting with 2 and then adding 3 is the same as starting with 3 and then adding 2:

$$3 + 2 = 5$$



The language box

An addition problem consists of two or more **terms** and one **sum**. In the addition problem

$$2 + 3 = 5$$

both 2 and 3 are terms, while 5 is the sum.

Common ways to say $2 + 3$ are

- "2 plus 3"
- "2 added to 3"
- "2 and 3 added together"

Adding numbers is also called **summing**.

6.1 Addition is Commutative

The sum is the same regardless of the order of the terms.

Example

$$2 + 5 = 7 = 5 + 2$$

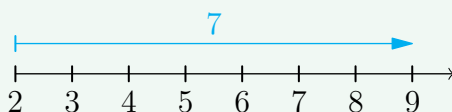
$$6 + 3 = 9 = 3 + 6$$

Addition on the Number Line: Moving to the Right

On a number line, addition with positive numbers means moving *to the right*:

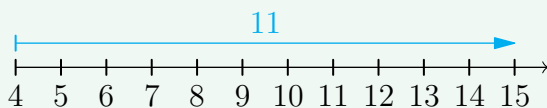
Example 1

$$2 + 7 = 9$$



Example 2

$$4 + 11 = 15$$



A New Meaning of =

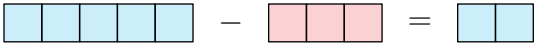
+ gives us the ability to express numbers in many different ways. For example, $5 = 2 + 3$ and $5 = 1 + 4$. In this context, = means "has the same value as." This also applies to subtraction, multiplication, and division, which we will look at in the next three sections.

6.3 Subtraction

Subtraction with Quantities: Taking Away

When we have a quantity and remove part of it, we use the **minus sign** $-$. If we have 5 and want to remove 3, we write

$$5 - 3 = 2$$



The language box

A subtraction problem consists of two or more **terms** and one **difference**. In the subtraction problem

$$5 - 3 = 2$$

both 5 and 3 are terms, while 2 is the difference.

Common ways to say $5 - 3$ are

- "5 minus 3"
- "5 subtracted by 3"
- "3 taken from 5"
- "3 subtracted from 5"

A New Interpretation of 0

In chapter [Chapter 1](#), we mentioned that 0 can be interpreted as "nothing." Subtraction gives us the ability to express 0 using other numbers. For example, $7 - 7 = 0$ and $19 - 19 = 0$.

Subtraction on the Number Line: Moving to the Left

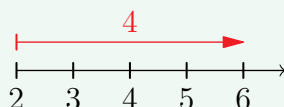
In [section 6.2](#), we saw that $+$ (with positive numbers) means we move *to the right* along the number line. With $-$ we do the opposite; we move *to the left*¹ :

Note

In *Example 1* and *Example 2* below, we move in the opposite direction of the arrow. This means we start at the arrowhead and move to the end. This may seem a bit strange at first, but it will be useful, especially in [Chapter 5](#), to think this way.

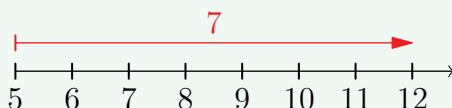
Example 1

$$6 - 4 = 2$$



Example 2

$$12 - 7 = 5$$



¹In figures with number lines, red arrows indicate that you start at the arrowhead and move to the other end.

Inverse Operations

The figures showing addition and subtraction on the number line illustrate why we say that addition and subtraction are **inverse operations**.

For example: Since $2 + 4 = 6$, we know that $6 - 4 = 2$ and $6 - 2 = 4$.

Canceling Terms

If we have an expression where numbers with the same value are added as many times as they are subtracted, we say they **cancel** each other out. In the expression

$$4 + 3 - 4 + 4 + 7 - 4 + 1$$

we have as many 4s added as we have 4s subtracted. The total sum and difference of the 4s is equal to 0. Thus, we can simplify our expression to

$$3 + 7 + 1$$

Note! In [Chapter ??](#), the word *canceling* will have a different meaning.

6.4 Multiplication

Multiplication with Whole Numbers: Introductory Definition

When we add the same numbers together, we can use the **multiplication sign** $\cdot$ to write our calculations more concisely:

Example

$$4 + 4 + 4 = 4 \cdot 3$$

$$8 + 8 = 8 \cdot 2$$

$$1 + 1 + 1 + 1 + 1 = 1 \cdot 5$$

The language box

A multiplication problem consists of two or more **factors** and one **product**. In the multiplication problem

$$4 \cdot 3 = 12$$

4 and 3 are factors, while 12 is the product.

Common ways to say $4 \cdot 3$ are

- "4 times 3"
- "4 multiplied by 3"
- "4 times 3"

Many websites and books in English use the symbol $\times$ as the multiplication sign. In most programming languages, the symbol $*$ is used.

Multiplication of Quantities

Let’s use a figure to visualize the multiplication problem $2 \cdot 3$:

$$2 \cdot 3 = \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} + \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} + \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} = \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \square & \square & \square \\ \hline \end{array}$$

And then we can observe the product of $3 \cdot 2$:

$$3 \cdot 2 = \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \square \\ \hline \end{array} + \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \square \\ \hline \end{array} = \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \square & \square \\ \hline \end{array}$$

6.2 Multiplication is Commutative

The product is the same regardless of the order of the factors.

Example

$$3 \cdot 4 = 12 = 4 \cdot 3$$

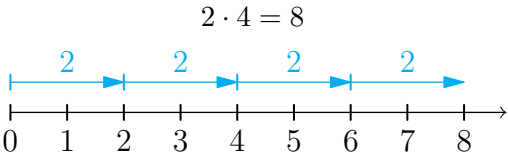
$$6 \cdot 7 = 42 = 7 \cdot 6$$

$$8 \cdot 9 = 72 = 9 \cdot 8$$

Multiplication on the Number Line

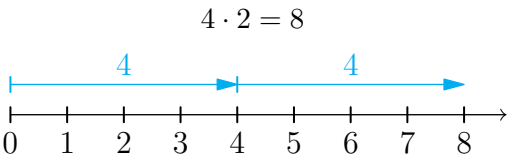
We can also use the number line to solve multiplication problems. For example, we can find what $2 \cdot 4$ is by thinking like this:

” $2 \cdot 4$ means moving 2 spaces *to the right*, 4 times.”



The number line can also be used to see that the order of the factors does not matter:

” $4 \cdot 2$ means moving 4 spaces *to the right*, 2 times.”



Final Definition of Multiplication with Positive Whole Numbers

It might seem most natural to interpret "2 times 3" as "3, 2 times." So,

$$\text{"2 times 3"} = 3 + 3$$

On page 111, we presented $2 \cdot 3$, i.e., "2 times 3," as $2 + 2 + 2$. With this interpretation, $3 + 3$ corresponds to $3 \cdot 2$, but since multiplication is a commutative operation ([Rule 6.2](#)), one interpretation does not exclude the other; $2 \cdot 3 = 2 + 2 + 2$ and $2 \cdot 3 = 3 + 3$ are two expressions with the same value. Thus, we can also write $2 \cdot 3 = 3 + 3$ and $3 \cdot 2 = 2 + 2 + 2$.

6.3 Multiplication as Repeated Addition

Multiplication with a positive whole number can be expressed as repeated addition.

Example 1

$$4 + 4 + 4 = 4 \cdot 3 = 3 + 3 + 3 + 3$$

$$8 + 8 = 8 \cdot 2 = 2 + 2 + 2 + 2 + 2 + 2 + 2$$

$$1 + 1 + 1 + 1 + 1 = 1 \cdot 5 = 5$$

Note

The fact that multiplication with positive whole numbers can be expressed as repeated addition does not exclude other expressions. It is not wrong to write that $2 \cdot 3 = 1 + 5$.

6.5 Division

`:` is the **division sign**. Division has three different meanings, shown here with $12 : 3$:

6.4 The Three Meanings of Division

- **Equal Distribution of Quantities**
 $12 : 3 =$ "The number in each group when 12 is divided into 3 equal large groups "
- **Number of Times One Number Fits into Another**
 $12 : 3 =$ "How many times 3 fits into 12"
- **Inverse Operation of Multiplication**
 $12 : 3 =$ "The number you multiply 3 by to get 12"

The language box

A division problem consists of a **dividend**, a **divisor**, and a **quotient**. In the division problem

$$12 : 3 = 4$$

12 is the dividend, 3 is the divisor, and 4 is the quotient.

Common ways to pronounce $12 : 3$ are

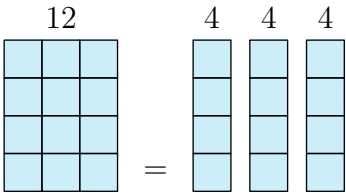
- "12 divided by 3"
- "12 divided into 3"
- "12 over 3"

In some contexts, $12 : 3$ is called the "**ratio** between 12 and 3." In this case, 4 is the **ratio value**.

Often, `/` is used instead of `:`, especially in programming languages.

Equal Distribution of Quantities

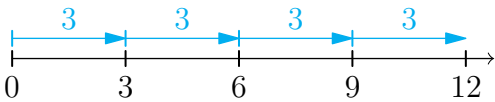
The calculation $12 : 3$ tells us that we should distribute 12 into 3 equal groups:



We see that each group contains 4 squares, which means that

$$12 : 3 = 4$$

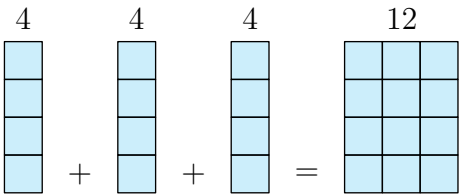
Number of Times One Number Fits into Another



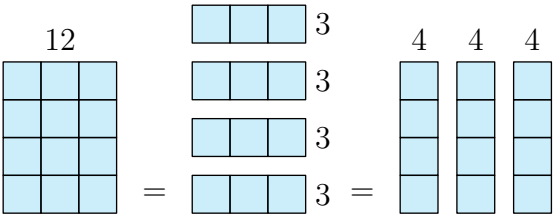
3 fits 4 times into 12, so $12 : 3 = 4$.

Inverse Operation of Multiplication

We have seen that if we divide 12 into 3 equal groups, we get 4 in each group. So, $12 : 3 = 4$. If we add these groups together again, we naturally get 12:



But this is the same as multiplying 4 by 3. So; if we know that $4 \cdot 3 = 12$, then we also know that $12 : 3 = 4$. Additionally, we then know that $12 : 4 = 3$.



Example 1

Since $6 \cdot 3 = 18$, then

$$18 : 6 = 3$$

$$18 : 3 = 6$$

Example 2

Since $5 \cdot 7 = 35$, then

$$35 : 5 = 7$$

$$35 : 7 = 5$$

Exercises for Chapter 6

6.1.1

Calculate.

- a) $12 + 84$ b) $36 + 51$ c) $328 + 571$ d) $242 + 56$

6.1.2

Calculate.

- a) $19 + 84$ b) $86 + 57$ c) $529 + 471$ d) $202 + 808$

6.2.1

Calculate.

- a) $84 - 23$ b) $286 - 52$ c) $529 - 401$ d) $782 - 131$

6.2.2

Calculate.

- a) $78 - 19$ b) $824 - 499$ c) $731 - 208$ d) $1078 - 991$

6.3.1

Calculate.

- a) $12 \cdot 3$ b) $28 \cdot 4$ c) $76 \cdot 5$ d) $43 \cdot 6$
e) $109 \cdot 7$ f) $98 \cdot 8$ g) $213 \cdot 9$

6.3.2

Calculate.

- a) $29 \cdot 12$ b) $83 \cdot 31$ c) $91 \cdot 76$ d) $14 \cdot 83$

6.3.3

Calculate.

- a) $531 \cdot 56$ b) $83 \cdot 701$ c) $91 \cdot 673$ d) $731 \cdot 67$

6.3.4

- a) Use a calculator to calculate $27 \cdot 5$ og $2.7 \cdot 5$.
- b) Use a calculator to calculate $247 \cdot 192$ og $24.7 \cdot 19.2$.
- c) Use a calculator to calculate $928 \cdot 74$ og $9.28 \cdot 7.4$.
- d) Use a calculator to calculate $134 \cdot 4\,249$ og $1.34 \cdot 42.49$.
- e) Compare the pairs of answers from a) - c), and make a rule for how to calculate multiplication involving decimal numbers.

6.3.5

Calculate.

- a) $82.3 \cdot 5$
- b) $9.51 \cdot 7$
- c) $22.4 \cdot 1.7$

6.4.1

Calculate.

- a) $98 : 2$
- b) $87 : 3$
- c) $92 : 4$
- d) $85 : 5$
- e) $72 : 6$

6.4.2

Calculate.

- a) $378 : 2$
- b) $224 : 4$
- c) $495 : 5$
- e) $133 : 7$
- f) $208 : 8$
- g) $873 : 9$

6.5.1

Write the number in standard form.

- a) 98 000
- b) 167 000 000
- c) 4 819
- d) 21
- e) 9 132,27
- f) 893.7
- g) 18 002.1
- h) 302.4

6.5.2

Write the number in standard form.

- a) 0.027 b) 0.0001901 c) 0.32 d) 0.00000020032

6.5.3

Given the calculation

$$900\,000\,000 \cdot 0.00007$$

- a) Explain why the calculation can be written as

$$9 \cdot 10^8 \cdot 7 \cdot 10^{-5}$$

- b) Use power rules (see [Section 8.2](#)) to find the product of the calculation from a).

6.5.4 (T1H21D1)

Calculate, and write answer in standard form.

$$\frac{6.2 \cdot 10^7 + 2.5 \cdot 10^8}{0.000002}$$

Chapter 7

Geometry

7.1 Terms

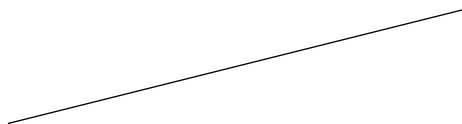
Point

A given position is called a¹ *point*. We mark a point by drawing a dot, which we preferably name by a letter. Below we have drawn the points A and B .

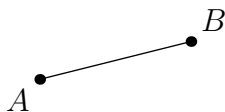


Line and segment

A straight dash with infinite length (!) is called a *line*. The fact that a line has infinite length, makes *drawing* a line impossible, we can only *imagine* a line. Imagining a line can be done by drawing a straight dash and think of its ends as wandering out in each direction.



A straight dash between two points is called a *segment*.



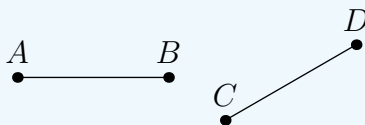
We write the segment between the points A and B as AB .

Notice

A segment is a part of a line, therefore a line and a segment have a lot of attributes in common. When writing about lines, it will be up to the reader to confirm whether the same applies for segments. Hence we avoid the need of writing "lines/segments".

¹See also [Section ??](#).

Segment or length?



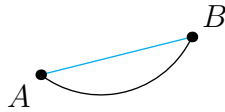
The segments AB and CD have equal length, but they are not the same segment. Still we'll write $AB = CD$. That is, we'll use the same names for the line segments and their lengths (the same applies for angles and their values, see page 123-125).

We'll do this by the following reasons:

- The context will make it clear whether we are talking about a segment or a length.
- Finding it necessary to write e.g. "the length of AB " would make sentences less readable.

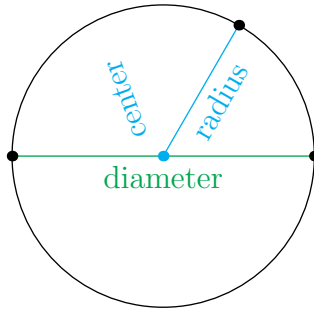
Distance

There are infinite ways one can move from one point to another and some ways will be longer than others. When talking about a distance in geometry, we usually mean the *shortest* distance. For geometries studied in this book the shortest distance between two points will always equal the length of the segment (blue in the below figure) connecting them.



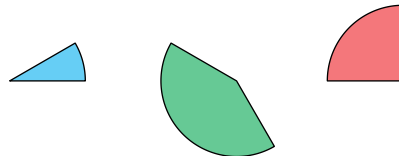
Circle; center, radius and diameter

If we make an enclosed curve where all points on this curve have the same distance to a given point, we have a *circle*. The point which all the points on the curve have an equal distance to is the *center* of the circle. A segment between a point on the curve and the center is called a *radius*. A segment between two points on the curve, passing through the circle center, is called a *diameter*¹.



Arcs and sectors

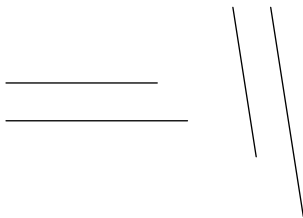
A part of a circular curve is called an *arc*. The shape formed by an arc and two associated radii is called a *sector*. The below figure shows three different sectors.



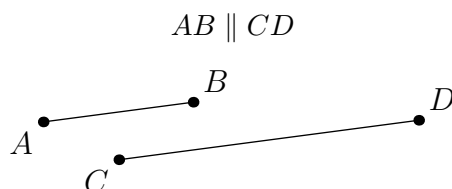
¹As mentioned, *radius* and *diameter* can just as well indicate the length of the segments.

Parallel lines

Lines aligned in the same direction are *parallel*. The below figure shows two pairs of parallel lines.

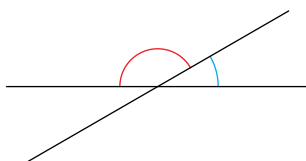


We use the symbol $\parallel$ to indicate that two lines are parallel.



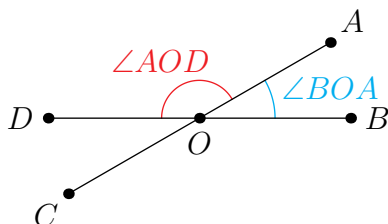
Angles

Non-parallel lines will sooner or later intersect. The gap formed by two non-parallel lines is called an *angle*. We draw angles as small circular curves:



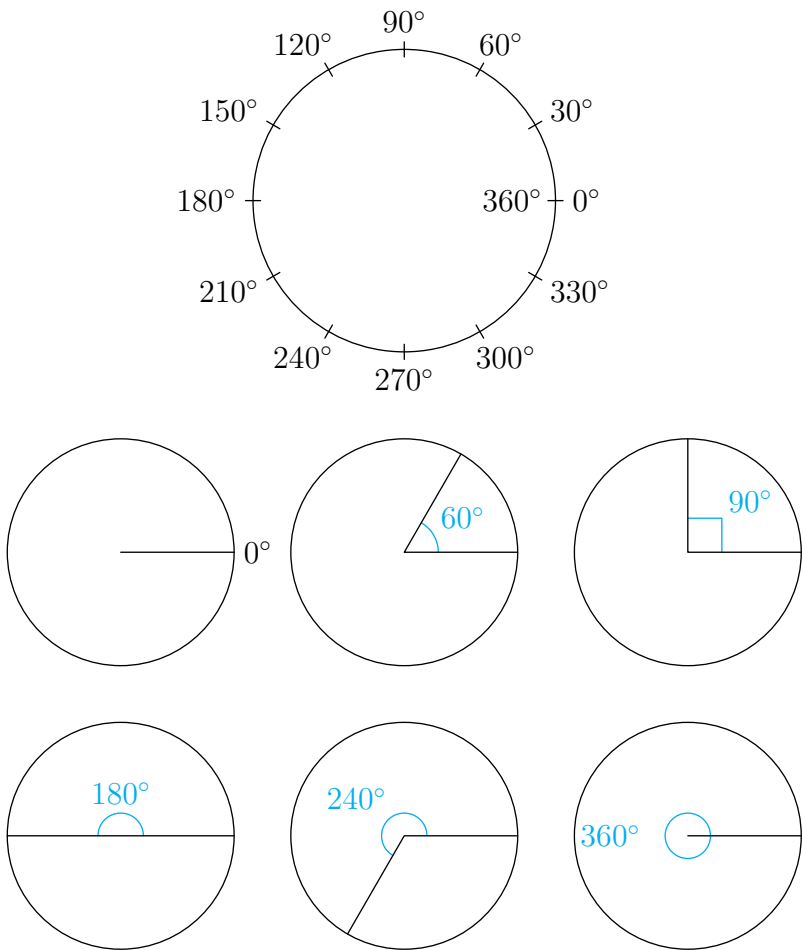
Lines creating an angle are called the *sides* of the angle. The intersection point of the lines are called the *vertex* of the angle. It is common to use the symbol $\angle$ to underline the angle in question. In the below figure we have the following:

- the angle $\angle BOA$ has angle sides OB and OA and vertex O .
- the angle $\angle AOD$ has angle sides OA and OD and vertex O .

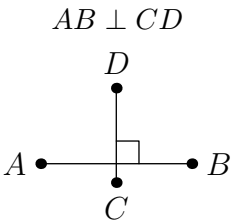


Measure of angles in degrees

When measuring an angle in degrees, we imagine a circular curve divided into 360 equally long pieces. We call one such piece 1 *degree*, indicated by the symbol $^{\circ}$.

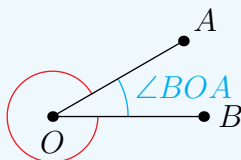


Notice that an angle with measure 90° is indicated by the symbol $\square$. Such an angle is called a *right* angle. Lines which form right angles are said to be *perpendicular* to one another, indicated by the symbol $\perp$.

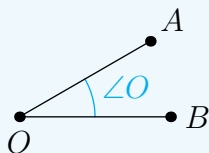


What angle?

Strictly speaking, when two segments (or lines) intersect, they form two angles; the one larger than 180° and the other smaller than 180° . Usually it is the smaller angle we wish to study, therefore it is common to define $\angle AOB$ as the *smaller* angle formed by the segments OA and OB .

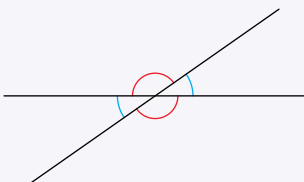


As long as there are only two segments/lines present, it is common using only one letter to indicate the angle:

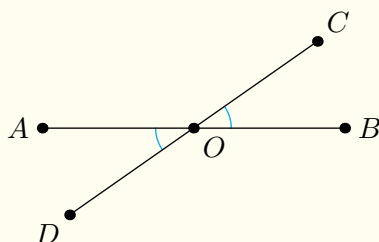


7.1 Vertical angles

Two opposite angles with a common vertex is called *vertical angles*. Vertical angles are of equal measure.



7.1 Vertical angles (explanation)



We have

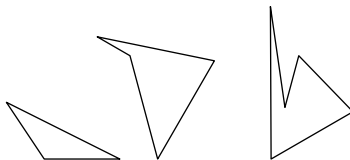
$$\angle BOC + \angle DOB = 180^\circ$$

$$\angle AOD + \angle DOB = 180^\circ$$

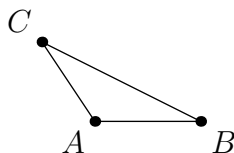
Hence, $\angle BOC = \angle AOD$. Similarly, $\angle COA = \angle DOB$.

Sides and vertices

When line segments form an enclosed shape, they form a *polygon*. The below figure shows, from left to right, a triangle (3-gon), a quadrilateral (4-gon) and a pentagon (5-gon).

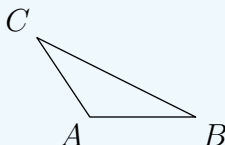


The segments of a polygon are called *edges* or *sides*. The respective intersection points of the segments are the *vertices* of the polygon. That is, the triangle below has vertices A , B and C and sides (edges) AB , BC and AC .



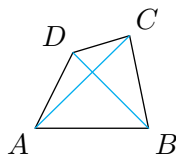
Note

Often we'll write a letter only to indicate a vertex of a polygon.



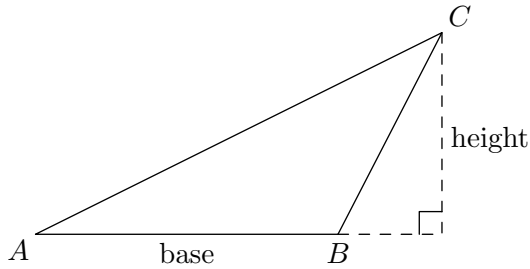
Diagonals

Segments between two vertices not belonging to the same side of a polygon is called a *diagonal*. The below figure shows the diagonals AC and BD .

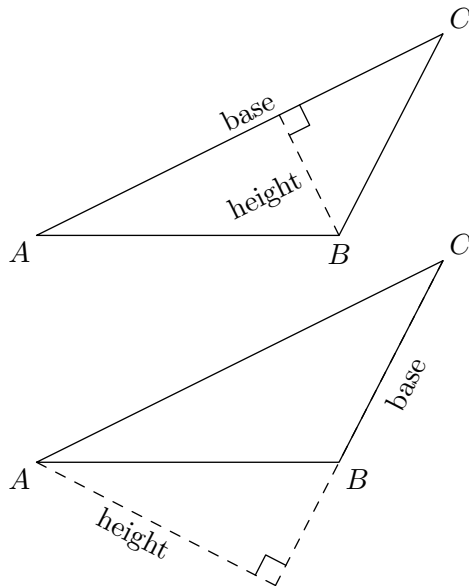


Altitudes and base lines

In [Section 7.4](#), the terms *base* and *height* (*altitude*) play an important role. To find the height of a triangle, we choose one of the sides to be the base. In the below figure, let's start with AB as the base. Then the height is the segment from AB (potentially, as is the case here, the extension of AB) to C , perpendicular to AB .



Since there are three sides which can be bases, a triangle has three heights.



Notice

The terms altitude and base also applies to other polygons.

7.2 Attributes of triangles and quadrilaterals

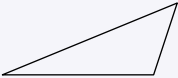
In addition to having a certain number of sides and vertices, polygons have other attributes, such as sides or angles of equal measure, or parallel sides. There are specific names of polygons with special attributes, and these names can be put into an overview where some "inherit"¹ attributes from others.

7.2 Triangles

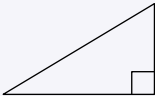
Triangle

Right triangle

Isosceles triangle → Equilateral triangle



Triangle
Have three sides and three vertices.



Right triangle
Have an angle of 90°.



Isosceles triangle
At least two sides are of equal length.
At least two angles are of equal measure.



Equilateral triangle
The sides are of equal length.
Each of the angles equals 60°.

Example

Since an equilateral triangle have three sides of equal length and three angles equal to 60°, it is also an isosceles triangle.

The language box

¹In [Rule 7.2](#) and [Rule 7.4](#) this is indicated by arrows.

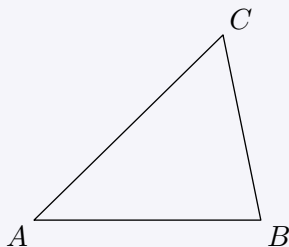
129

The longest side of a right triangle is called the *hypotenuse* and the shorter sides are called *legs*.

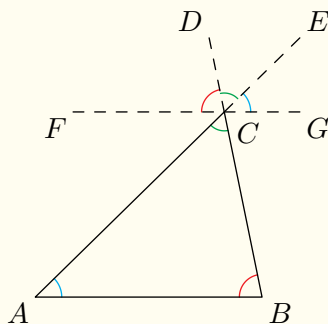
7.3 The sum of angles in a triangle

In a triangle, the sum of the angles equals 180° .

$$\angle A + \angle B + \angle C = 180^\circ$$



7.3 The sum of angles in a triangle (explanation)



We draw a segment FG passing through C and parallel to AB . Moreover, we place E and D on the extension of AC and BC , respectively. Then $\angle A = \angle GCE$ and $\angle B = \angle DCF$. $\angle ACB = \angle ECD$ because they are vertical angles. Now

$$\angle DCF + \angle ECD = \angle GCE = 180^\circ$$

Hence

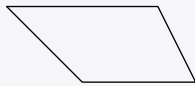
$$\angle CBA + \angle ACB + \angle BAC = 180^\circ$$

7.4 Quadrilaterals



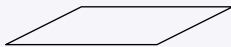
Quadrilateral

Have four sides and four vertices.



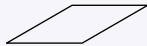
Trapezoid

Have at least one pair of parallel sides.



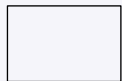
Parallelogram

Have two pairs of parallel sides.
Have two pairs of equal angles.



Rhombus

All sides are of equal length.



Rectangle

All angles equals 90° .



Square

Example

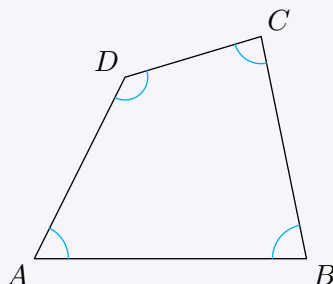
The square is both a rhombus and a rectangle, which means it "inherits" their attributes. From this it follows that, in a square,

- all sides are of equal length.
- all angles equals 90° .

7.5 The sum of angles in a quadrilateral

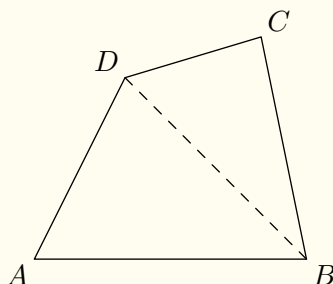
In a quadrilateral, the sum of the angles equals 360° .

$$\angle A + \angle B + \angle C + \angle D = 360^\circ$$



7.5 The sum of angles in a quadrilateral (explanation)

The total sum of angles of $\triangle ABD$ and $\triangle BCD$ equals the sum of the angles in $\square ABCD$. By [Rule 7.3](#), the sum of angles of triangles 180° , therefore the sum of the angles of $\square ABCD$ equals $2 \cdot 180^\circ = 360^\circ$.

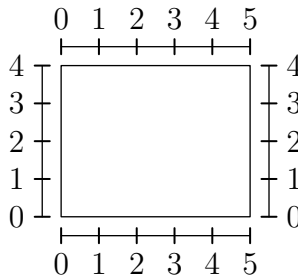


7.3 Perimeter

When we measure the length around an enclosed shape, we find its *perimeter*. Let's find the perimeter of this rectangle:



The rectangle has two sides of length 4 and two sides of length 5.



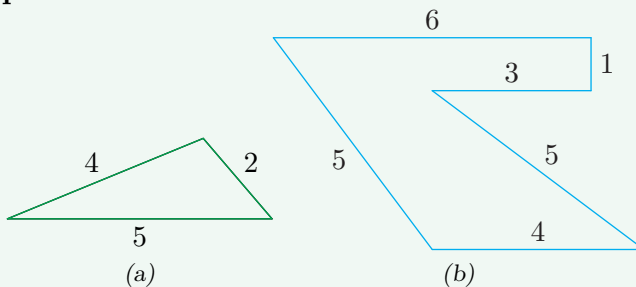
Hence

$$\begin{aligned} \text{The perimeter of the rectangle} &= 4 + 4 + 5 + 5 \\ &= 18 \end{aligned}$$

7.6 Perimeter

A perimeter is the length around a closed shape.

Example



In figure (a) the perimeter equals $5 + 2 + 4 = 11$.

In figure (b) the perimeter equals $4 + 5 + 3 + 1 + 6 + 5 = 24$.

7.4 Area

Our surroundings are full of *surfaces*, for example on a floor or a sheet. When measuring surfaces, we find their *area*. The concept of area is the following:

We imagine a square with sides of length 1. We call this the *unit square*.

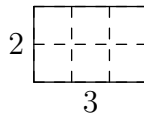


Then, regarding the surface for which we seek the area of, we ask:

”How many unit squares does this surface contain?”

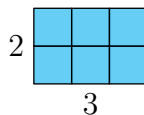
The area of a rectangle

Let's find the area of a rectangle with baseline 3 and altitude 2.



Simply by counting, we find that the rectangle contains 6 unit squares:

The area of the rectangle = 6

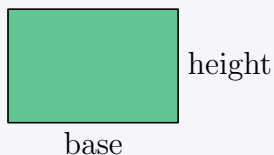


Looking back at [Section 6.4](#), we notice that

$$\begin{aligned}\text{The area of the rectangle} &= 3 \cdot 2 \\ &= 6\end{aligned}$$

7.7 The area of a rectangle

$$\text{Area} = \text{baseline} \cdot \text{altitude}$$

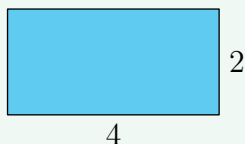


Width and length

In a rectangle, the baseline and the altitude are also referred to as (in random order) the *width* and the *length*.

Example 1

Find the area of the rectangle¹.

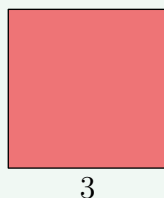


Answer

The area of the rectangle = $4 \cdot 2 = 8$

Example 2

Find the area of the square.



Answer

The area of the square = $3 \cdot 3 = 9$

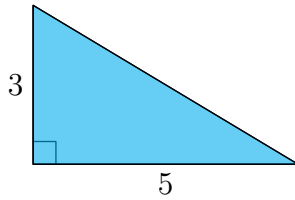
Note: The lengths used in one figure will not necessarily correspond to the lengths in another figure. That is, a side of length 1 in one figure can might as well be shorter than a side of length 1 in another figure.

The area of a triangle

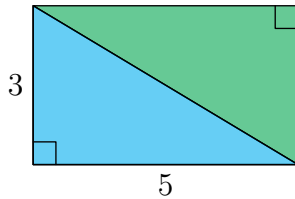
Concerning triangles, there are three different cases to study:

1) *The baseline and the altitude have a common end point*

Let's find the area of a right triangle with baseline 5 and altitude 3.



We can make a rectangle by copying our triangle and joining the hypotenuses:



By [Rule 7.7](#), the area of the rectangle equals $5 \cdot 3$. The area of one of the triangles makes up half the area of the rectangle, so

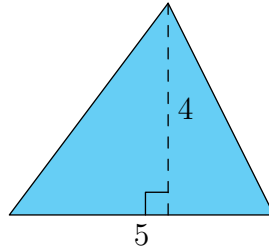
$$\text{The area of the blue triangle} = \frac{5 \cdot 3}{2}$$

Regarding the blue triangle we have

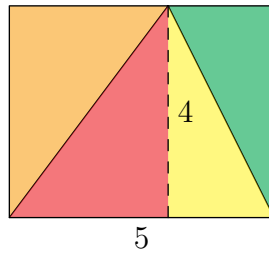
$$\frac{5 \cdot 3}{2} = \frac{\text{baseline} \cdot \text{height}}{2}$$

2) *The altitude is placed inside the triangle, but have no common end point with the baseline*

The below triangle has baseline 5 and altitude 4.



We make a rectangle containing the blue triangle (split into the red triangle and the yellow triangle):



Observe that

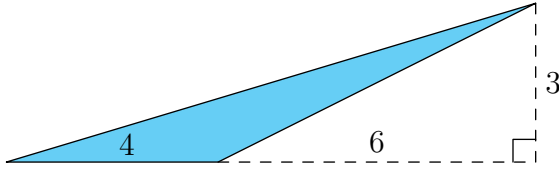
- the area of the red triangle makes up half the area of the rectangle formed by the red and the yellow triangle.
- the area of the yellow triangle makes up half the area of the rectangle formed by the yellow and the green triangle.

It now follows that the sum of the areas of the yellow triangle and the red triangle makes up half the area of the rectangle formed by the four colored triangles. The area of this rectangle equals $5 \cdot 4$, and since our original triangle (the blue) includes the red triangle and the orange triangle, we have

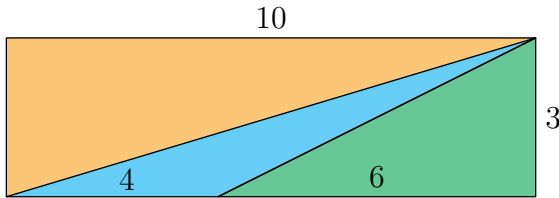
$$\text{The area of the blue triangle} = \frac{5 \cdot 4}{2} = \frac{\text{baseline} \cdot \text{height}}{2}$$

2) The altitude is placed outside the triangle

The below triangle has baseline 4 and altitude 3.



We now make a rectangle containing the blue triangle:



Now we introduce the following names:

The area of the rectangle = R

The area of the blue triangle = B

The area of the orange triangle = O

The area of the green triangle = G

We have

$$R = 3 \cdot 10 = 30$$

$$O = \frac{3 \cdot 10}{2} = 15$$

$$G = \frac{3 \cdot 6}{2} = 9$$

Moreover,

$$\begin{aligned} B &= R - O - G \\ &= 30 - 15 - 9 \\ &= 6 \end{aligned}$$

Observe that we can write

$$6 = \frac{4 \cdot 3}{2}$$

Regarding the blue triangle we recognize this as

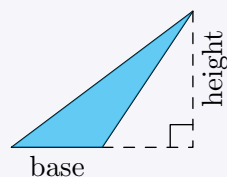
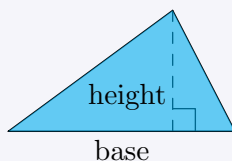
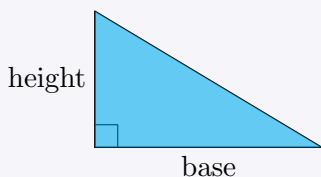
$$\frac{4 \cdot 3}{2} = \frac{\text{base} \cdot \text{height}}{2}$$

The three cases summarized

For a chosen baseline in a triangle, one of the cases discussed will always be valid. All cases resulted in the same expression for the area of the triangle.

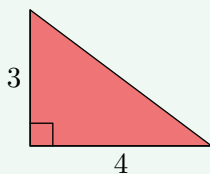
7.8 The area of a triangle

$$\text{Area} = \frac{\text{base} \cdot \text{height}}{2}$$



Example 1

Find the area of the triangle.

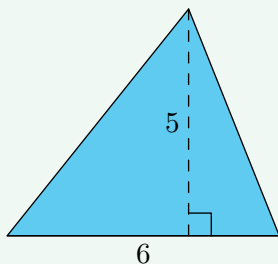


Answer

$$\begin{aligned}\text{The area of the triangle} &= \frac{4 \cdot 3}{2} \\ &= 6\end{aligned}$$

Example 2

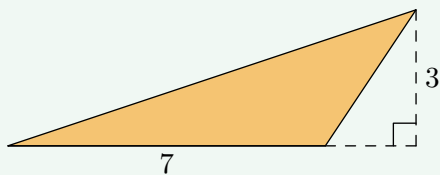
Find the area of the triangle.



Answer The area of the triangle = $\frac{6 \cdot 5}{2} = 15$

Example 3

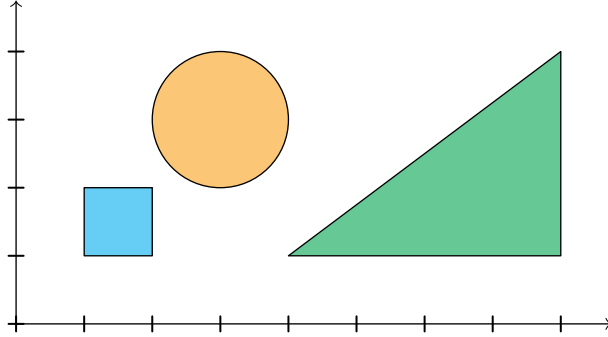
Find the area of the triangle.



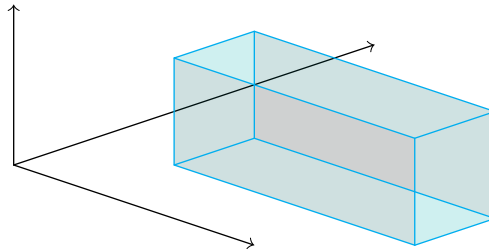
Answer The area of the triangle = $\frac{7 \cdot 3}{2} = \frac{21}{2}$

7.5 Tredimensional geometri

So far, we have studied two-dimensional shapes as triangles, quadrilaterals, circles etc. Every two-dimensional figure can be drawn in a coordinate system with two axes.



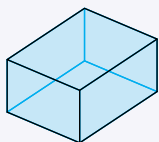
In order to draw **three-dimensional** shapes however, we need three axes:



While we have used the phrase *width/length* and *height* to describe a rectangle, we say that the box above has a width, a height *and* a length (depth).

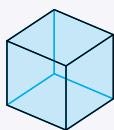
The area lying on the "outside" of a three-dimensional shape is called the **surface**. The surface of the box above is made up by 6 rectangles. Polygons that are parts of a surface are called **lateral surfaces**.

7.9 Three-dimensional figures



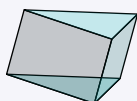
Right prism

Has two equal and four equal rectangles as lateral surfaces. All the surfaces in contact are perpendicular to each other.



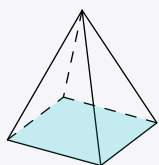
Kube

A right prism with squares as lateral surfaces.



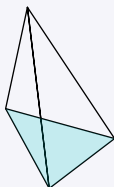
Triangular prism

Two of the lateral surfaces are congruent triangles parallel to each other. These triangles are connected by lateral surfaces that are rectangles.



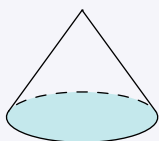
Rectangular pyramid

The surface is made up by one rectangle and four triangles.



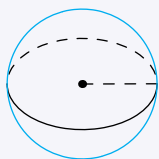
Triangular pyramid

The surface is made up by four triangles.



Cone

One part of the surface is a circle, the other part is a folded sector.



Sphere

All the points on the surface have equal distance to the center.

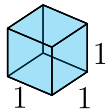
Tip

It is not that easy to imagine what a *folded sector* is, but try this:

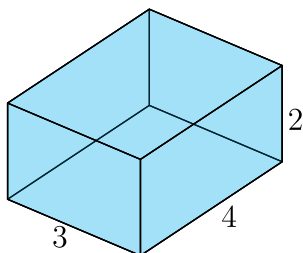
Draw a sector on a sheet. Cut the sector, and join the two edges of the sector. Then you have a cone without bottom surface.

7.6 Volum

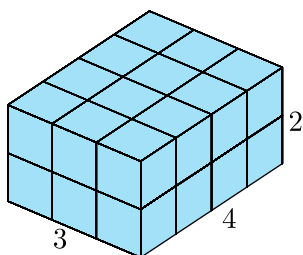
When we speak of how much space there is inside an object, we speak of it's **volume**. As a measure of volume, we imagine a cube with sides of length 1.



We shall call such a cube a 'unit cube'. Say we have a rectangular prism with width 3, length 4 and height 2.



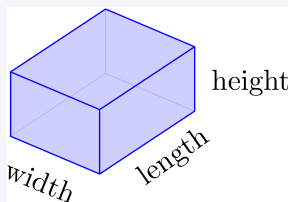
Inside it, there is room for exactly 24 unit cubes.



Other than simply counting unit cubes, we can find this by calculating

$$3 \cdot 4 \cdot 2 = 24$$

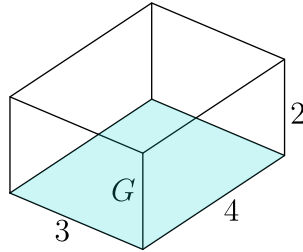
7.10 The volume of a rectangular prism I



$$\text{volume} = \text{width} \cdot \text{length} \cdot \text{height}$$

Base area

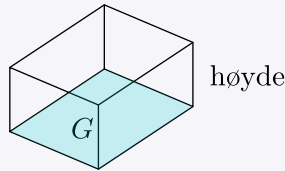
To calculate the volume of elementary shapes, it is useful to apply the term **base surface**. In the same way as with baselines¹, it is our choice of base surface that decides what the height is. Regarding the prism on the preceding page, it is natural to choose the horizontally aligned surface as the base. To indicate the base we shall use the letter G :



The base surface has area $3 \cdot 4 = 12$, and the corresponding height is 2. The volume of the prism equals the base surface area multiplied by the height:

$$\begin{aligned} V &= 3 \cdot 4 \cdot 2 \\ &= G \cdot 2 \\ &= 24 \end{aligned}$$

7.11 Volumee of a rectangular prism II



$$\text{volume} = G \cdot \text{height}$$

Grunnflata eller grunnflatearealet?

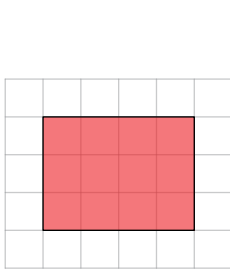
In the above text, we first let G denote the base surface, and later we let G denote the *base surface area*. In this book the term *base surface* is so strongly connected to the *base surface area* that we will let the same symbol denote them both, leaving the reader to differ them from the context of use.

¹See page ??.

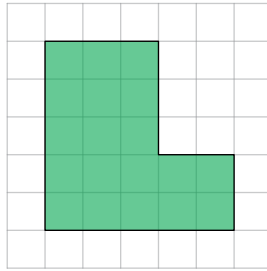
Exercises for Chapter 7

7.1.1

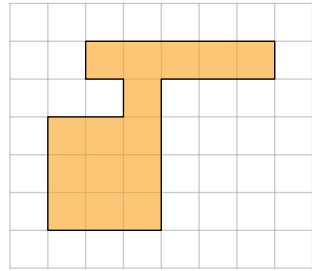
Find the perimeter of the colored figure.



(a)



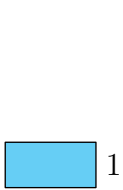
(b)



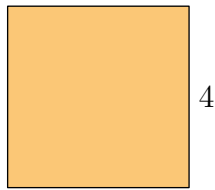
(c)

7.1.2

Calculate the area of the rectangles.



(a)



(b)



(c)

7.1.3

Find the width and height of the rectangle, given that

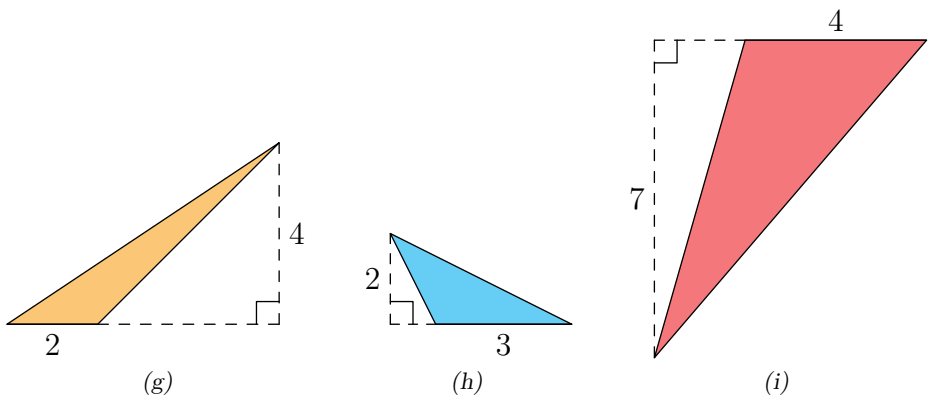
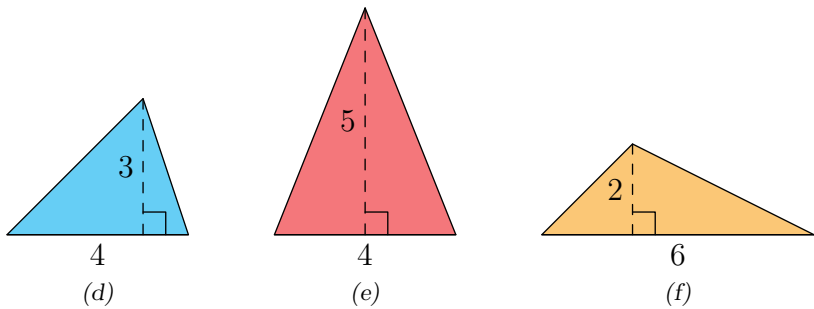
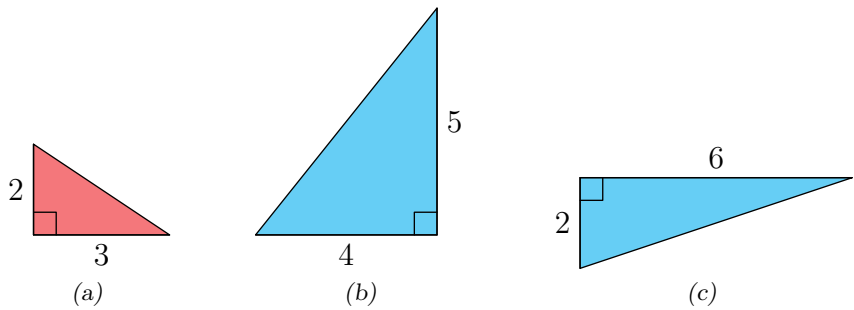
- a) the area is 16 and the perimeter is 20.
- b) the area is 12 and the perimeter is 14.
- c) the area is 18 and the perimeter is 18.

7.1.4

- a) Find the area of a square with perimeter 16.
- b) Give three examples of rectangles with perimeter 36. Provide the answer by stating the width, height and area of the rectangle.

7.1.5

Find the area of the triangle.

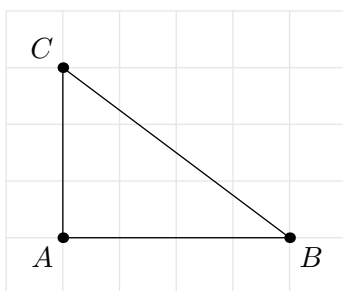


7.1.6

A prism has length 9, width 10 and height 8.

- a) Find the base area of the prism.
- b) Find the volume of the prism.

7.2.1

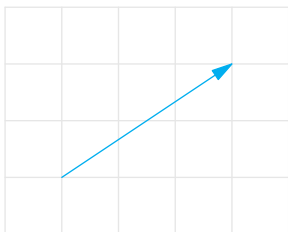


Shift the triangle with the vector shown.

a)



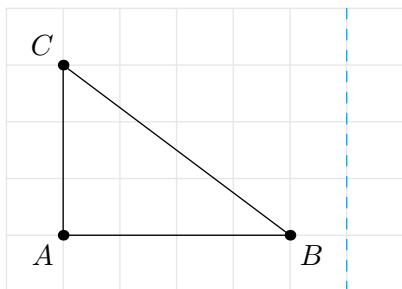
b)



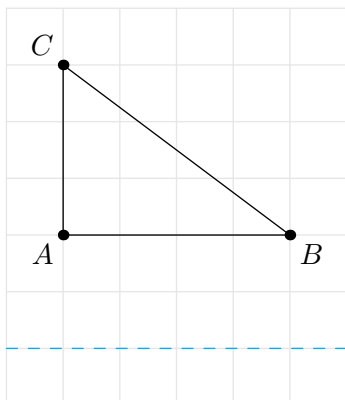
7.2.2

Mirror the triangle about the dashed line.

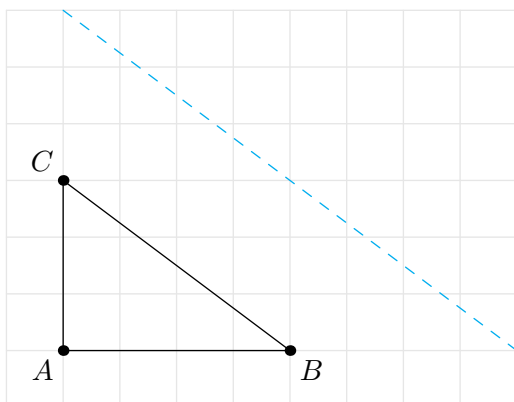
a)



b)



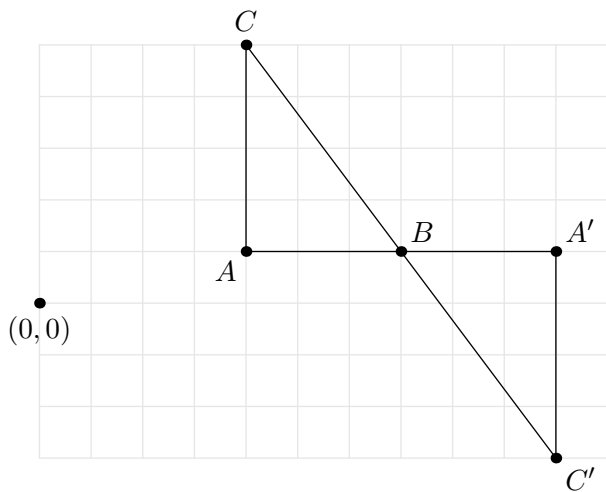
c)



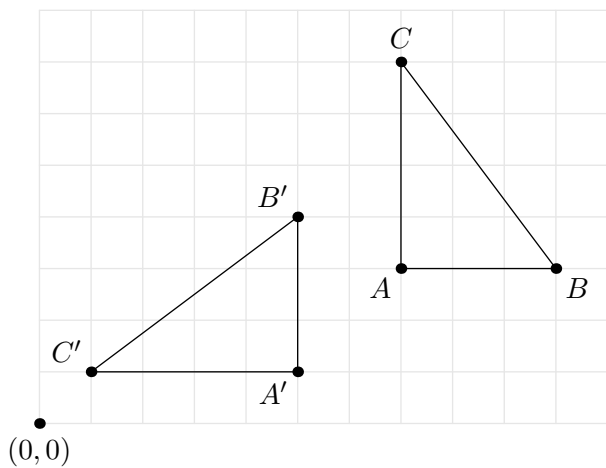
7.2.3

Find the angle and point of rotation.

a)



b)



Ponder 6

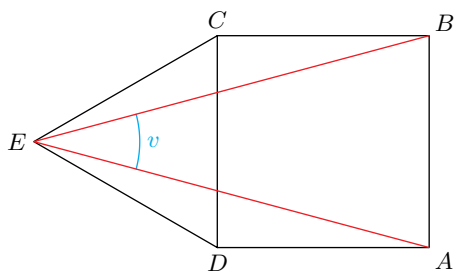
- a) Explain why the perimeter of a rectangle is always an even number.
- b) "If both the width and height of a rectangle is an odd number, there is no way the area and the perimeter of the rectangle can have the same value."

Explain why the proposition is correct/incorrect.

- c) What is the side length of the only square with area and perimeter of equal value?

Ponder 7

$\square ABCD$ is a square and $\triangle DEC$ is equilateral. Find the value of v .



Part II

Algebra and geometry

Chapter 8

Algebra

8.1 Introduction

Simply said, **algebra** is mathematics where letters represent numbers. This makes it easier working with *general* cases. For example, $3 \cdot 2 = 2 \cdot 3$ and $6 \cdot 7 = 7 \cdot 6$ but these are only two of the infinitely many examples of the commutative property of multiplication! One of the aims of algebra is giving *one* example that explains *all* cases, and since our digits (0-9) are inevitably connected to specific numbers, we apply letters to reach this target.

The value of the numbers represented by letters will often vary, in that case we call the letter-numbers *variables*. If letter-numbers on the other hand have a specific value, they are called *constants*.

In [Part I](#), we studied calculations through examples with specific numbers, however, most of these rules are *general*; they are valid for all numbers. On page 156-159, many of these rules are reproduced in a general form. A good way of getting acquainted with algebra is comparing the rules here presented by the way they are expressed in¹ [Part I](#).

8.1 Addition is commutative (6.1)

$a + b = b + a$

Example

$7 + 5 = 5 + 7$

8.2 Multiplication is commutative (6.2)

$a \cdot b = b \cdot a$

Example 1

$9 \cdot 8 = 8 \cdot 9$

Example 2

$8 \cdot a = a \cdot 8$

¹The number of the rules as found in [Part I](#) are written inside parentheses.

Multiplication involving letters

When multiplication involves letters, it is common to omit the symbol of multiplication. If a specific number and a letter are multiplied together, the specific number is written first. For example,

$$a \cdot b = ab$$

and

$$a \cdot 8 = 8a$$

We also write

$$1 \cdot a = a$$

In addition, it is common to omit the symbol of multiplication when an expression with parentheses is involved:

$$3 \cdot (a + b) = 3(a + b)$$

8.3 Fractions as rewriting of division (4.1)

$$a : b = \frac{a}{b}$$

Example

$$a : 2 = \frac{a}{2}$$

8.4 Fractions multiplied by fractions (4.8)

$$\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$$

Example 1

$$\frac{2}{11} \cdot \frac{13}{21} = \frac{2 \cdot 13}{11 \cdot 21} = \frac{26}{231}$$

Example 2

$$\frac{3}{b} \cdot \frac{a}{7} = \frac{3a}{7b}$$

8.5 Division by fractions (4.10)

$$\frac{a}{b} : \frac{c}{d} = \frac{a}{b} \cdot \frac{d}{c}$$

Example 1

$$\frac{1}{2} : \frac{5}{7} = \frac{1}{2} \cdot \frac{7}{5}$$

Example 2

$$\begin{aligned}\frac{a}{13} : \frac{b}{3} &= \frac{a}{13} \cdot \frac{3}{b} \\ &= \frac{3a}{13b}\end{aligned}$$

8.6 Distributive law (3.2)

$$(a + b)c = ac + bc$$

Example 1

$$(2 + a)b = 2b + ab$$

Example 2

$$a(5b - 3) = 5ab - 3a$$

8.7 Multiplication by negative numbers I (5.6)

$$a \cdot (-b) = -(a \cdot b)$$

Example 1

$$\begin{aligned}3 \cdot (-4) &= -(3 \cdot 4) \\ &= -12\end{aligned}$$

Example 2

$$\begin{aligned}(-a) \cdot 7 &= -(a \cdot 7) \\ &= -7a\end{aligned}$$

8.8 Multiplication by negative numbers II (5.7)

$$(-a) \cdot (-b) = a \cdot b$$

Example 1

$$\begin{aligned}(-2) \cdot (-8) &= 2 \cdot 8 \\ &= 16\end{aligned}$$

Example 2

$$(-a) \cdot (-15) = 15a$$

Extensions of the rules

One of the strengths of algebra is that we can express compact rules which are easily extended to apply for other cases. Let's, as an example, find another expression of

$$(a + b + c)d$$

[Rule 8.6](#) does not directly imply how to calculate between the expression inside the parentheses and d , but there is no wrongdoing in defining $a + b$ as k :

$$a + b = k$$

Then

$$(a + b + c)d = (k + c)d$$

Now, by [Rule 8.6](#), we have

$$(k + c)d = kd + cd$$

Inserting the expression for k , we have

$$kd + cd = (a + b)d + cd$$

By applying [Rule 8.6](#) once more we can write

$$(a + b)d + cd = ad + bc + cd$$

Then

$$(a + b + c)d = ad + bc + cd$$

Notice! This example is not meant to show how to handle expressions not directly covered by [Rule 8.1](#) - [Rule 8.8](#), but to emphasize why it's always sufficient to write rules with the least amount of terms, factors etc. Usually you apply extension of the rules without even thinking about it, and surely not in such meticulous manner as here provided.

8.2 Powers

base $\rightarrow 2^3 \leftarrow$ exponent

A power is composed by a *base* and an *exponent*. For example, 2^3 is a power with base 2 and exponent 3. An exponent which is a positive integer indicates the amount of instances of the base to be multiplied together, that is

$$2^3 = 2 \cdot 2 \cdot 2$$

8.9 Powers

a^n is a power with base a and exponent n .

If n is a natural number, a^n corresponds to n instances of a multiplied together.

Note: $a^1 = a$

Example 1

$$\begin{aligned} 5^3 &= 5 \cdot 5 \cdot 5 \\ &= 125 \end{aligned}$$

Example 2

$$c^4 = c \cdot c \cdot c \cdot c$$

Example 3

$$\begin{aligned} (-7)^2 &= (-7) \cdot (-7) \\ &= 49 \end{aligned}$$

The language box

Common ways of saying 2^3 are¹

- "2 (raised) to the power of 3"
- "2 to the third power"

In computer programming, the symbol $\wedge$ or the symbols $**$ is usually written between the base and the exponent.

Raising a number to the power of 2 is called "squaring" the number.

¹Attention! The examples illustrate a paradox in the English language; *power* is also a synonym for *exponent*.

Note

The next pages declares rules concerning powers with corresponding explanations. Even though one wish to have these explanations as general as possible, we choose to use, mostly, specific numbers as exponents . Using variables as exponents would lead to less reader-friendly expressions, and it is our claim that the general cases are well illustrated by the specific cases.

8.10 Multiplication by powers

$$a^m \cdot a^n = a^{m+n} \quad (8.1)$$

Example 1

$$\begin{aligned} 3^5 \cdot 3^2 &= 3^{5+2} \\ &= 3^7 \end{aligned}$$

Example 2

$$\begin{aligned} b^4 \cdot b^{11} &= b^{3+11} \\ &= b^{14} \end{aligned}$$

Example 3

$$\begin{aligned}
 a^5 \cdot a^{-7} &= a^{5+(-7)} \\
 &= a^{5-7} \\
 &= a^{-2}
 \end{aligned}$$

(See [Rule 8.13](#) regarding how powers with negative exponents can be interpreted.)

8.10 Multiplication by powers (explanation)

Let's study the case

$$a^2 \cdot a^3$$

We have

$$a^2 = 2 \cdot 2$$

$$a^3 = 2 \cdot 2 \cdot 2$$

Hence we can write

$$\begin{aligned}
 a^2 \cdot a^3 &= \overbrace{a \cdot a}^{a^2} \cdot \overbrace{a \cdot a \cdot a}^{a^3} \\
 &= a^5
 \end{aligned}$$

8.11 Division by powers

$$\frac{a^m}{a^n} = a^{m-n}$$

Example 1

$$\frac{3^5}{3^2} = 3^{5-2} = 3^3$$

Example 2

$$\begin{aligned}
 \frac{2^4 \cdot a^7}{a^6 \cdot 2^2} &= 2^{4-2} \cdot a^{7-6} \\
 &= 2^2 a \\
 &= 4a
 \end{aligned}$$

8.11 Division by powers (explanation)

Let's examine the fraction $\frac{a^5}{a^2}$. Expanding the powers, we get

$$\begin{aligned}\frac{a^5}{a^2} &= \frac{a \cdot a \cdot a \cdot a \cdot a}{a \cdot a} \\ &= \frac{\cancel{a} \cdot \cancel{a} \cdot a \cdot a \cdot a}{\cancel{a} \cdot \cancel{a}} \\ &= a \cdot a \cdot a \\ &= a^3\end{aligned}$$

The above calculations are equivalent to writing

$$\begin{aligned}\frac{a^5}{a^2} &= a^{5-2} \\ &= a^3\end{aligned}$$

8.12 The special case of $a^0 = 1$

Example 1 $1000^0 = 1$

Example 2 $(-b)^0 = 1$

8.12 The special case of a^0 (explanation)

A number divided by itself always equals 1, therefore

$$\frac{a^n}{a^n} = 1$$

From this, and [Rule 8.11](#), it follows that

$$\begin{aligned}1 &= \frac{a^n}{a^n} \\ &= a^{n-n} \\ &= a^0\end{aligned}$$

8.13 Powers with negative exponents

$$a^{-n} = \frac{1}{a^n}$$

Example 1

$$a^{-8} = \frac{1}{a^8}$$

Example 2

$$(-4)^{-3} = \frac{1}{(-4)^3} = -\frac{1}{64}$$

8.13 Powers with negative exponents (explanation)

By [Rule 8.12](#), we have $a^0 = 1$. Thus

$$\frac{1}{a^n} = \frac{a^0}{a^n}$$

By [Rule 8.11](#), we obtain

$$\begin{aligned}\frac{a^0}{a^n} &= a^{0-n} \\ &= a^{-n}\end{aligned}$$

8.14 Fractions as base

$$\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m} \quad (8.2)$$

Example 1

$$\left(\frac{3}{4}\right)^2 = \frac{3^2}{4^2} = \frac{9}{16}$$

Example 2

$$\left(\frac{a}{7}\right)^3 = \frac{a^3}{7^3} = \frac{a^3}{343}$$

8.14 Fractions as base (explanation)

Let's study

$$\left(\frac{a}{b}\right)^3$$

We have

$$\begin{aligned}\left(\frac{a}{b}\right)^3 &= \frac{a}{b} \cdot \frac{a}{b} \cdot \frac{a}{b} \\ &= \frac{a \cdot a \cdot a}{b \cdot b \cdot b} \\ &= \frac{a^3}{b^3}\end{aligned}$$

8.15 Factors as base

$$(ab)^m = a^m b^m \quad (8.3)$$

Example 1

$$\begin{aligned}(3a)^5 &= 3^5 a^5 \\ &= 243a^5\end{aligned}$$

Example 2

$$(ab)^4 = a^4 b^4$$

8.15 Factors as base (explanation)

Let's use $(a \cdot b)^3$ as an example. We have

$$\begin{aligned}(a \cdot b)^3 &= (a \cdot b) \cdot (a \cdot b) \cdot (a \cdot b) \\ &= a \cdot a \cdot a \cdot b \cdot b \cdot b \\ &= a^3 b^3\end{aligned}$$

8.16 Powers as base

$$(a^m)^n = a^{m \cdot n} \quad (8.4)$$

Example 1

$$\begin{aligned}(c^4)^5 &= c^{4 \cdot 5} \\ &= c^{20}\end{aligned}$$

Example 2

$$\begin{aligned}\left(3^{\frac{5}{4}}\right)^8 &= 3^{\frac{5}{4} \cdot 8} \\ &= 3^{10}\end{aligned}$$

8.16 Powers as base (explanation)

Let's use $(a^3)^4$ as an example. We have

$$(a^3)^4 = a^3 \cdot a^3 \cdot a^3 \cdot a^3$$

By [Rule 8.10](#), we get

$$\begin{aligned}a^3 \cdot a^3 \cdot a^3 \cdot a^3 &= a^{3+3+3+3} \\ &= a^{3 \cdot 4} \\ &= a^{12}\end{aligned}$$

8.17 n -root

$$a^{\frac{1}{n}} = \sqrt[n]{a}$$

The symbol $\sqrt{}$ is called the *radical sign*. In the case of an exponent equal to $\frac{1}{2}$, it is common to omit 2 from the radical:

$$a^{\frac{1}{2}} = \sqrt{a}$$

Example

By Rule 8.16, we have

$$\begin{aligned}(a^b)^{\frac{1}{b}} &= a^{b \cdot \frac{1}{b}} \\ &= a\end{aligned}$$

For example is

$$9^{\frac{1}{2}} = \sqrt{9} = 3, \text{ since } 3^2 = 9$$

$$125^{\frac{1}{3}} = \sqrt[3]{125} = 5, \text{ since } 5^3 = 125$$

$$16^{\frac{1}{4}} = \sqrt[4]{16} = 2, \text{ since } 2^4 = 16$$

The language box

$\sqrt{9}$ is called "the square (the 2nd) root of 9"

$\sqrt[3]{8}$ is called "the cube (the 3th) root of 8"

8.3 Irrational numbers

8.18 Irrational numbers

A number which is *not* a rational number, is an irrational number¹.

The value of an irrational number are decimal numbers with infinite digits in a non-repeating manner.

Example 1

$\sqrt{2}$ is an irrational number.

$$\sqrt{2} = 1.414213562373\dots$$

¹Strictly speaking, irrational numbers are all *real* numbers which are not rational numbers. But to explain what *real* numbers are, we have to mention *imaginary* numbers, and this we choose not to do in this book.

Exercises for Chapter 8

8.1.1

Exploit the relation between repeated addition and multiplication (see [Rule 6.3](#) and [Rule 5.5](#)) to simplify the expressions.

- a) $a + a + a$ b) $a + a + a + a$ c) $a + a + a + a + a + a + a$
d) $-b - b$ e) $-b - b - b - b - b$ f) $-k - k - k$

8.1.2

Simplify the expressions.

- a) $2a + b - a$ b) $-4a + 2b + 3a$ c) $7b - 3a + 2b$

8.1.3

Simplify the expressions.

- a) $4c + 2b - 5a - 3c$ b) $-9a - 3c + 3b + 3c$ c) $9b - 3a + 2b$

8.1.4

Use [Rule 3.2](#) to write the expression without parentheses.

- a) $7(a + 2)$ b) $9(b + 3)$ c) $8(b - 3c)$ d) $(-2)(3a + 5b)$
e) $(9a + 2)$ f) $(3b + 8)a$ g) $(b - 3c)(-a)$
h) $2(a + 3b + 4c)$ i) $9(3b - c + 7a)$ j) $(3b - c + 7a)(-2)$

8.1.5

Use [Rule 3.2](#) to factorize the expression.

- a) $2a + 2b$ b) $4ab + 5b$ c) $9bc - c$ d) $4ac - 2a$

8.1.6

Prove that

a) $(a + b)^2 = a^2 + 2ab + b^2$

b) $(a - b)^2 = a^2 - 2ab + b^2$

c) $(a + b)(a - b) = a^2 - b^2$

8.1.7 (GV21D1)

a) Simplify the expression.

$$\frac{a + a + a + a}{4a}$$

b) What value will the expression $\frac{y^2 - 2y}{y^2}$ attain if $x = 4$ and $y = -2$?

8.1.8 (E22)

Given the expression $(a + b)^2 = 16$. Decide whether the below alternatives makes the expression valid.

- $a = 2$ and $b = 2$
- $a = 8$ and $b = 4$
- $a = 8$ and $b = -4$

8.2.1

Write as a power.

a) $3 \cdot 3 \cdot 3 \cdot 3$ b) $5 \cdot 5$ c) $7 \cdot 7 \cdot 7 \cdot 7 \cdot 7 \cdot 7$

d) $a \cdot a \cdot a$ e) $b \cdot b$ f) $(-c)(-c)(-c)(-c)$

8.2.2

Find the value of the power.

a) 8^2 b) 2^5 c) 4^3 d) $(-2)^3$ e) $(-3)^5$ f) $(-4)^4$

8.2.3

Write the expression as a power.

a) $2^7 \cdot 2^9$ b) $3^4 \cdot 3^7$ c) $9 \cdot 9^5$ d) $6^8 \cdot 6^{-3}$ e) $5^3 \cdot 5^{-7}$

f) $10^8 \cdot 10^{-3} \cdot 10^6$ g) $a^9 \cdot a^7$ h) $k^5 \cdot k^2$ i) $x^5 \cdot x^{-2}$

k) $x^{-4} \cdot x^5$ l) $a^{-5} \cdot a \cdot a^4$ m) $a^3 \cdot b^5 \cdot a^2 \cdot b^{-8}$

8.2.4

Calculate..

a) $\sqrt{25}$ b) $\sqrt{100}$ c) $\sqrt{144}$

d) $\sqrt[3]{27}$ e) $\sqrt[3]{729}$ f) $\sqrt[5]{100000}$

Ponder 8

(1TH21D1)

Simplify the expression

$$\frac{9^{\frac{1}{2}} \cdot 3^{-1} + 9^0}{8^{\frac{3}{4}}}$$

Ponder 9

By adding the digits of a number, we find the **digit sum** of the number. For example is the digit sum of 14 equal to $1 + 4 = 5$, and the digit sum of 918 equals $9 + 1 + 8 = 18$. Prove that if the digit sum of a 3-digit integer is divisible by 3, then the number is also divisible by 3.

Note: Generalizing the 3-digit case is quite easy, thus proving that the rule is valid for an integer with any number of digits.

Comment (for the particularly interested)

Mathematics is *axiomatically* founded. This means we declare¹ some propositions to be true, and these are called *axioms* or *postulates*. For the subject of calculations we have 12 axioms², but in this book we have confined ourselves to explicitly mention the following 6:

Axioms

For the numbers a , b and c we have

$$a + (b + c) = (a + b) + c \quad (\text{A1})$$

$$a + b = b + a \quad (\text{A2})$$

$$a(bc) = (ab)c \quad (\text{A3})$$

$$ab = ba \quad (\text{A4})$$

$$a(b + c) = ab + ac \quad (\text{A5})$$

$$a \cdot \frac{1}{a} = 1 \quad (a \neq 0) \quad (\text{A6})$$

-
- (A1) Associative law for addition
 - (A2) Commutative law for addition
 - (A3) Associative law for multiplication
 - (A4) Commutative law for multiplication
 - (A5) Distributive law
 - (A6) Existence of a multiplicative identity

By applying axioms, we can derive more complex contexts which we call *theorems*. In this book we chose to let *rules* be the collective name for definitions, theorems and axioms. This is because all three, in practice, draws up guidelines (rules) inside the mathematical system in which we wish to operate.

¹Preferably, as few as possible.

²The number can slightly vary, depending on how the axioms are expressed.

In [Part I](#) we have tried to present the *motivation* behind the axioms, because (obviously) they are not randomly selected. The train of thoughts that leads us to them is the following:

1. Vi define positive numbers as representations of either an amount or a placement on a number line.
2. We define what addition, subtraction, multiplication and division entail for positive integers (and 0).
3. From the marks above, it's as good as self-evident that (A1) - (A6) is valid for all positive integers.
4. We define also fractions as representations of either an amount or a placement on a number line. What the elementary operations entail for fractions rests upon what is valid for the positive integers.
5. From the remarks above, we conclude that (A1) - (A6) is valid for all rational numbers.
6. We introduce negative numbers and an extended interpretation of addition and subtraction. This in turn leads to the interpretations of multiplication and division involving negative numbers.
7. (A1) - (A6) is still valid after the introduction of negative integers. Deriving that they are also valid for negative rational numbers is a formality (omitted in the book).
8. We can never write the value of an irrational number exact, but it can be approximated by a rational number¹. Therefore, all calculations involving irrational numbers is, in practice, calculations involving rational numbers, and in this way we can conclude that² (A1) - (A6) is also valid for irrational numbers.

A similar train of thoughts can be applied concerning the power-rules found in [Section 8.2](#).

¹For example, we can write $\sqrt{2} = 1.414213562373... \approx \frac{1414213562373}{1000000000000}$

²*Attention!* This explanation is good enough for the aim of this book but is a rather extreme simplification. Irrational numbers are a very complex subject, in fact, many books presenting advanced mathematics utilize several chapters to cover the subject in full depth .

Chapter 9

Equations

9.1 Introduction

Even though every mathematical expression involving $=$ is an *equation*, the word is, traditionally, closely linked to the presence of an *unknown* number.

Say we want to find the number which when added by 4 results in 7. The name of this unknown number is free to choose but commonly it's called x . Our equation can now be written as

$$x + 4 = 7$$

The x -value¹ which results in the same values on each side of the equal sign is the *solution* of the equation. It is nothing wrong done by simply observing what the value of x must be. Probably you have already realized that $x = 3$ is the solution of the equation, since

$$3 + 4 = 7$$

However, most equations are difficult to solve simply by observing, and it is therefore vice to take the advantage of more general methods. In reality, there is only one principle to follow:

We can always carry out a mathematical operation on one of the sides of the equal sign, as long as we carry out the operation on the other side too.

The mathematical operations presented in this book is the four elementary operations. Concerning these the principal sounds:

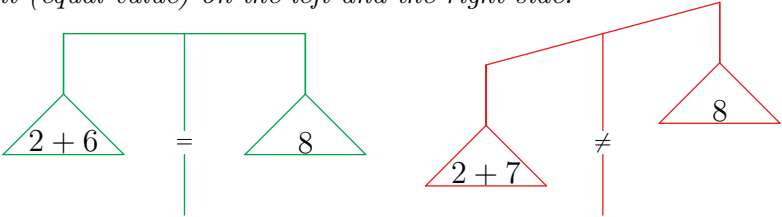
We can always add, subtract, multiply or divide by a number on one side of the equal sign, as long as we also do it on the other side.

The principle follows from the meaning of $=$. When two expressions are of equal value, their values are necessarily still equal as long as we carry out identical mathematical operations on them. Anyways, in the coming section we'll specify this principle for every single elementary operation. If you already feel things make sense, you can, without no great loss of insight, skip to [Section 9.3](#).

¹In other cases it can be several values.

9.2 Solving with the elementary operations

In the figures of this section we'll understand equations from what we call the *weight principle*. In that case, $=$ indicates¹ there is equal weight (equal value) on the left and the right side.

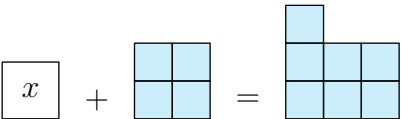


Addition and subtraction; moving terms

First example

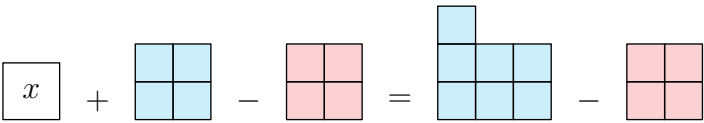
We have already found the solution of this equation, but let's now solve it in a different way²:

$$x + 4 = 7$$



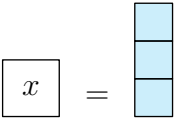
The value of x becomes clear if x is alone on one of the sides, and we can isolate x on the left side by removing 4. But if we are to remove 4 from the left side, we must also remove 4 from the right side, in order to preserve equal values on both sides.

$$x + 4 - 4 = 7 - 4$$



Since $4 - 4 = 0$ and $7 - 4 = 3$, we get

$$x = 3$$



¹ $\neq$ symbols "not equal".

²Note: In earlier figures, there have been a correspondence between the size of the boxes and the (absolute) value of the number they represent. This does not apply to the boxes representing x .

In a more abbreviated way this can be written as

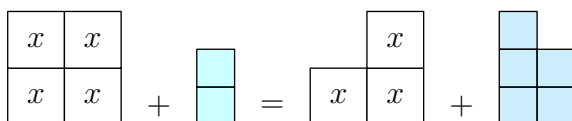
$$\begin{aligned}x + 4 &= 7 \\x &= 7 - 4 \\x &= 3\end{aligned}$$

Between the first and second line it is common to say that *4 has shifted side and therefore also sign (from + to -)*.

Second example

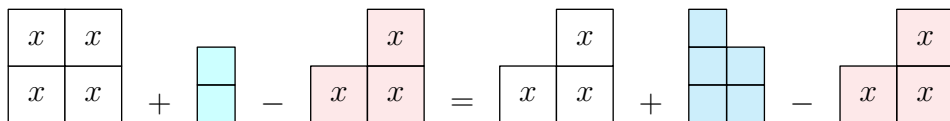
Let's move on to a somehow more complex equation¹:

$$4x - 2 = 3x + 5$$



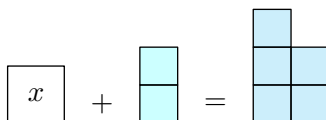
To get an expression with x exclusively on one side, we remove $3x$ on both sides:

$$4x - 2 - 3x = 3x + 5 - 3x$$



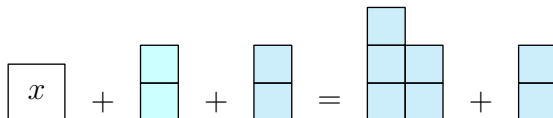
Now,

$$x - 2 = 5$$



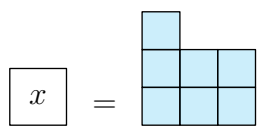
To isolate x we add 2 on the left side. Then we must also add 2 on the right side:

$$x - 2 + 2 = 5 + 2$$



¹Notice that the figure illustrates $4x + (-2)$ (see [Section 5.3](#)) on the left side. However, $4x + (-2)$ equals $4x - 2$ (see [Section 5.2](#)).

Hence

$$x = 7$$


The steps we have made can be summarized in this way:

$4x - 2 = 3x + 5$	1. figure
$4x - 3x - 2 = 3x - 3x + 5$	2. figure
$x - 2 = 5$	3. figure
$x - 2 + 2 = 5 + 2$	4. figure
$x = 7$	5. figure

In a more abbreviated way we can write

$$\begin{aligned}
 4x - 2 &= 3x + 5 \\
 4x - 3x &= 5 + 2 \\
 x &= 7
 \end{aligned}$$

9.1 Moving numbers across the equal sign

To solve an equation, we gather all x -terms and all known terms on respective sides of the equal sign. A term which shifts side, also shifts sign.

Example 1

Solve the equation

$$3x + 5 = 2x + 9$$

Answer

$$3x - 2x = 9 - 5$$

$$x = 4$$

Example 2

Solve the equation

$$-4x - 3 = -5x + 12$$

Answer

$$-4x + 5x = 12 + 3$$

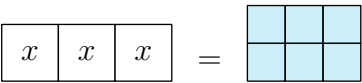
$$x = 15$$

Multiplication and division

Division

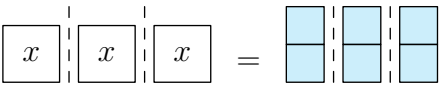
So far we have studied equations resulting in a single instance of x on one side of the equal sign. Often there are several instances of x , as, for example, in the equation

$$3x = 6$$



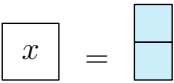
If we separate the left side into three equal groups, we get a single x in each group. And by separating the right side into three equal groups, all groups present are of equal value

$$\frac{3x}{3} = \frac{6}{3}$$



Therefore

$$x = 2$$



Let's summarize our calculations:

$$3x = 6 \qquad \qquad \qquad 1. \text{ figure}$$

$$\frac{3x}{3} = \frac{6}{3} \qquad \qquad \qquad 2. \text{ figure}$$

$$x = 2 \qquad \qquad \qquad 3. \text{ figure}$$

9.2 Division on both sides of an equation

We can divide both sides of an equation by the same number.

Example 1

Solve the equation

$$4x = 20$$

Answer

$$\begin{aligned}\frac{4x}{4} &= \frac{20}{4} \\ x &= 5\end{aligned}$$

Example 2

Solve the equation

$$2x + 6 = 3x - 2$$

Answer

$$2x - 3x = -2 - 6$$

$$-x = -8$$

$$\begin{aligned}\cancel{2}x &= \frac{-8}{\cancel{-1}} \\ x &= 8\end{aligned}$$

$$(-x = -1x)$$

Multiplication

Let's solve the equation

$$\frac{x}{3} = 4$$

$$\boxed{\frac{x}{3}} = \begin{array}{|c|c|} \hline & \\ \hline & \\ \hline \end{array}$$

We can get a unit x on the left side if we add two more instances of $\frac{x}{3}$. The equation informs that $\frac{x}{3}$ equals 4, this implies that for every instance of $\frac{x}{3}$ we add to the left side, we must add 4 to the right side, in order to keep the balance.

$$\frac{x}{3} + \frac{x}{3} + \frac{x}{3} = 4 + 4 + 4$$

$$\boxed{\frac{x}{3}} + \boxed{\frac{x}{3}} + \boxed{\frac{x}{3}} = \begin{array}{|c|c|} \hline & \\ \hline & \\ \hline \end{array} + \begin{array}{|c|c|} \hline & \\ \hline & \\ \hline \end{array} + \begin{array}{|c|c|} \hline & \\ \hline & \\ \hline \end{array}$$

Now we notice that $\frac{x}{3} + \frac{x}{3} + \frac{x}{3} = \frac{x}{3} \cdot 3$ and that $4 + 4 + 4 = 4 \cdot 3$:

$$\frac{x}{3} \cdot 3 = 4 \cdot 3$$

$$\begin{array}{|c|c|c|} \hline \frac{x}{3} & \frac{x}{3} & \frac{x}{3} \\ \hline \hline \hline \end{array} = \begin{array}{|c|c|c|c|} \hline & & & \\ \hline & & & \\ \hline & & & \\ \hline \end{array}$$

Since $\frac{x}{3} \cdot 3 = x$ and $4 \cdot 3 = 12$, we have

$$x = 12$$

$$\begin{array}{|c|} \hline x \\ \hline \end{array} = \begin{array}{|c|c|c|c|} \hline & & & \\ \hline & & & \\ \hline & & & \\ \hline \end{array}$$

Our steps can be summarized in the following way:

- | | |
|---|-----------|
| $\frac{x}{3} = 4$ | 1. figure |
| $\frac{x}{3} + \frac{x}{3} + \frac{x}{3} = 4 + 4 + 4$ | 2. figure |
| $\frac{x}{3} \cdot 3 = 4 \cdot 3$ | 3. figure |
| $x = 12$ | 4. figure |

In a more abbreviated form this can be written as

$$\begin{aligned} \frac{x}{3} &= 4 \\ \frac{x}{\cancel{3}} \cdot \cancel{3} &= 4 \cdot 3 \\ x &= 12 \end{aligned}$$

9.3 Multiplication on both sides of an equation

We can multiply both sides of an equation by the same number.

Example 1

Solve the equation

$$\frac{x}{5} = 2$$

Answer

$$\frac{x}{\cancel{5}} \cdot \cancel{5} = 2 \cdot 5$$

$$x = 10$$

Example 2

Solve the equation

$$\frac{7x}{10} - 5 = 13 + \frac{x}{10}$$

Answer

$$\frac{7x}{10} - \frac{x}{10} = 13 + 5$$

$$\frac{6x}{10} = 18$$

$$\frac{6x}{\cancel{10}} \cdot \cancel{10} = 18 \cdot 10$$

$$6x = 180$$

$$\frac{\cancel{6}x}{\cancel{6}} = \frac{180}{6}$$

$$x = 30$$

9.3 Solving with elementary operations summarized

9.4 Solving methods with elementary operations

We can always

- add or subtract both sides of an equation by the same number. This is equivalent to shifting a term from one side to the other, also shifting the terms sign.
- multiply or divide both sides of an equation by the same number.

Example 1

Solve the equation

$$3x - 4 = 6 + 2x$$

Answer

$$3x - 2x = 6 + 4$$

$$x = 10$$

Example 2

Solve the equation

$$9 - 7x = -8x + 3$$

Answer

$$8x - 7x = 3 - 9$$

$$x = -6$$

Example 3

Solve the equation

$$10x - 20 = 7x - 5$$

Answer

$$10x - 7x = 20 - 5$$

$$3x = 15$$

$$\frac{3x}{3} = \frac{15}{3}$$

$$x = 5$$

Example 4

Solve the equation

$$15 - 4x = x + 5$$

Answer

$$15 - 5 = x + 4x$$

$$10 = 5x$$

$$\frac{10}{5} = \frac{5x}{5}$$

$$2 = x$$

Note: In the other examples, we have chosen to collect the instances of x on the left side of the equation but we can just as well collect them on the right side. By doing that here, we have avoided calculations with negative numbers.

Example 5

Solve the equation

$$\frac{4x}{9} - 20 = 8 - \frac{3x}{9}$$

Answer

$$\frac{4x}{9} + \frac{3x}{9} = 20 + 8$$

$$\frac{7x}{9 \cdot 7} = \frac{28}{7}$$

$$\frac{x}{9} \cdot 9 = 4 \cdot 9$$

$$x = 36$$

Example 6

Solve the equation

$$\frac{1}{3}x + \frac{1}{6} = \frac{5}{12}x + 2$$

Answer

To avoid fractions, we multiply both sides by the common denominator 12:

$$\left(\frac{1}{3}x + \frac{1}{6}\right) 12 = \left(\frac{5}{12}x + 2\right) 12 \quad (9.1)$$

$$\frac{1}{3}x \cdot 12 + \frac{1}{6} \cdot 12 = \frac{5}{12}x \cdot 12 + 2 \cdot 12 \quad (*)$$

$$4x + 2 = 5x + 24 \quad (9.2)$$

$$4x - 5x = 24 - 2 \quad (9.3)$$

$$-x = 22 \quad (9.4)$$

$$\frac{\cancel{1}x}{\cancel{-1}} = \frac{22}{-1} \quad (9.5)$$

$$x = -22 \quad (9.6)$$

Tip

Some like to make the rule that "we can multiply or divide all terms by the same number". In that case, we could have jumped to the second line in the calculations of the example above.

Example 7

Solve the equation

$$3 - \frac{6}{x} = 2 + \frac{5}{2x}$$

Answer

We multiply both sides by the common denominator $2x$:

$$2x \left(3 - \frac{6}{x} \right) = 2x \left(2 + \frac{5}{2x} \right)$$

$$6x - 12 = 4x + 5$$

$$6x - 4x = 5 + 12$$

$$2x = 17$$

$$x = \frac{17}{2}$$

9.4 Power equations

Let's solve the equation

$$x^2 = 9$$

This is called a *power equation*. In general, power equations are difficult to solve applying the four elementary operations only. Applying power-rules, we raise both sides to the power of the inverse of the exponent associated with x :

$$\left(x^2\right)^{\frac{1}{2}} = 9^{\frac{1}{2}}$$

By [Rule 8.16](#), we have

$$\begin{aligned}x^{2 \cdot \frac{1}{2}} &= 9^{\frac{1}{2}} \\x &= 9^{\frac{1}{2}}\end{aligned}$$

Since $3^2 = 9$, we have $9^{\frac{1}{2}} = 3$. Now observe this:

The principle stated on page 178 says we can, like we just did, carry out a mathematical operation on both sides of an equation. However, sticking to this principle does not guarantee that all solutions are found.

Concerning our equation, we know that $x = 3$ is a solution. For the sake of it, we can confirm this by the calculation

$$3^2 = 3 \cdot 3 = 9$$

But we also have

$$(-3)^2 = (-3)(-3) = 9$$

Hence, -3 is also a solution of our original equation!

9.5 Power equations

An equation which can be written as

$$x^a = b$$

where a and b are constants, is a *power equation*.

The equation has a distinct solutions.

Example 1

Solve the equation

$$x^2 + 5 = 21$$

Answer

$$x^2 + 5 = 21$$

$$x^2 = 21 - 5$$

$$x^2 = 16$$

Since $4 \cdot 4 = 16$ and $(-4) \cdot (-4) = 16$, we have

$$x = 4 \quad \vee \quad x = -4$$

Example 2

Solve the equation

$$3x^2 + 1 = 7$$

Answer

$$3x^2 = 7 - 1$$

$$3x^2 = 6$$

$$\frac{3x^2}{3} = \frac{6}{3}$$

$$x^2 = 2$$

Hence,

$$x = \sqrt{2} \quad \vee \quad x = -\sqrt{2}$$

Note

Although the equation

$$x^a = b$$

has a solutions, they are not necessarily all *real*¹. Concerning this book, it means we settle with finding all rational or irrational numbers which solves the equation. For example,

$$x^3 = 8$$

has 3 solutions, but we settle with the solution $x = 2$.

¹As earlier mentioned, *real* and *imaginary* numbers lie outside the scope of this book.

Exercises for Chapter 9

9.2.1

Solve the equation.

a) $x + 8 = 18$ b) $x - 3 = 2$ c) $x - 8 = 1$

d) $x + 12 = 14$ e) $x - 1 = 2$ f) $x - 3 = 1$

g) $21 = x + 11$ h) $24 = x + 16$ i) $4 = x - 6$

9.2.2

Solve the equation.

a) $16x - 20 = 15x + 17$ b) $18x - 11 = 17x + 18$

c) $17x - 15 = 16x + 8$ d) $4x - 9 = 6 + 3x$

e) $12x - 6 = 11x + 2$ f) $2x + 10 = 3x - 1$

g) $5 + 8x = 9x - 18$ h) $15 + 2x = 3x - 4$

i) $9x + 8 = 10x - 2$ j) $17x + 9 = 18x - 19$

9.2.3

Solve the equation.

a) $3x = 12$ b) $10x = 50$ c) $7x = 63$ d) $2x = 30$

9.2.4

Solve the equation.

a) $\frac{x}{4} = 2$ b) $\frac{x}{9} = 8$ c) $\frac{x}{7} = 7$ d) $\frac{x}{15} = 10$

9.2.5

Solve the equation.

a) $18x - 27 = 9x + 36$ b) $7x - 27 = 4x + 3$

c) $15x - 16 = 7x + 32$ d) $13x - 42 = 7x + 12$

e) $4 + 9x = 13x - 32$ f) $7x + 8 = 11x - 24$

g) $5x + 4 = 8x - 11$ h) $7 + 10x = 14x - 9$

9.2.6

Given $\triangle ABC$ with $\angle C = 90^\circ$. Prove that

$$\angle A = 90^\circ - \angle B$$

9.2.7 (E22)

Solve the equation

$$3 \cdot 24 \cdot 9 = 4 \cdot 9 \cdot x$$

9.5.1 (1TV21D1)

Solve the equation set.

$$2x - y = 4$$

$$x - 2y = 5$$

Ponder 10

a) Prove that

$$0,2626\dots = \frac{26}{99}$$

Given

$$a = b \left(\frac{1}{10^c} + \frac{1}{10^{2c}} + \frac{1}{10^{3c}} + \dots \right)$$

where b is a number with c digits.

b) Prove that if $b = 26$, then $a = 0,2626\dots$.

c) Prove that

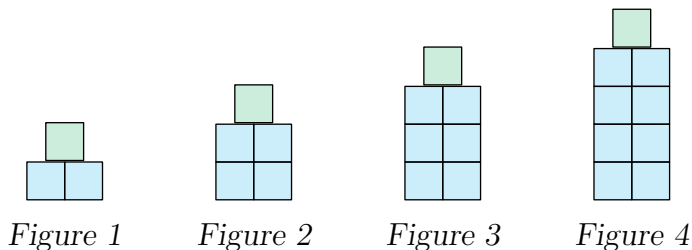
$$a = \frac{b}{10^c - 1}$$

Chapter 10

Functions

10.1 Introduction

Variables are values that change. A value which changes in compliance with a variable is called a *function*.



In the above figure, the amount of boxes follows a specific pattern. Mathematically we can describe the pattern like this:

$$\text{Number of boxes in Figure 1} = 2 \cdot 1 + 1 = 3$$

$$\text{Number of boxes in Figure 2} = 2 \cdot 2 + 1 = 5$$

$$\text{Number of boxes in Figure 3} = 2 \cdot 3 + 1 = 7$$

$$\text{Number of boxes in Figure 4} = 2 \cdot 4 + 1 = 9$$

Hence, for a figure of a random number x , we have

$$\text{Number of boxes in Figure } x = 2x + 1$$

The amount of boxes changes in compliance with the change of x , in this case we say that

"Number of boxes in Figure x " is a function of x .

$2x + 1$ is the expression of the function "Number of boxes in Figure x ".

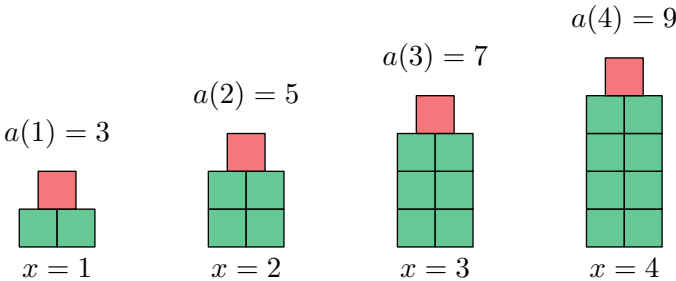
General expressions

If we were to continue working with the function just studied, writing "Number of boxes in *Figure x*" all the time would be very cumbersome. It is common to let letters indicate functions and to write the associated variable inside parentheses. Let's rename "Number of boxes in *Figure x*" to $a(x)$. Then

$$\text{Number of boxes in Figure } x = a(x) = 2x + 1$$

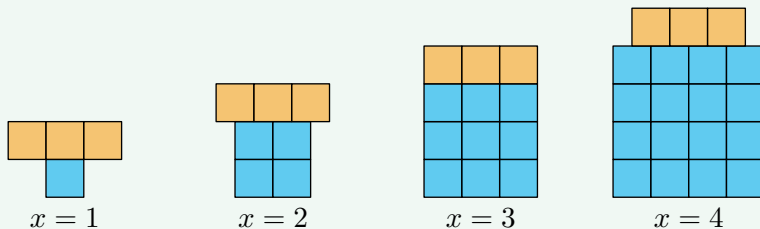
If we write $a(x)$, but substitute x by a specific number, we substitute x by this number in the expression of our function:

$$\begin{aligned} a(1) &= 2 \cdot 1 + 1 = 3 \\ a(2) &= 2 \cdot 2 + 1 = 5 \\ a(3) &= 2 \cdot 3 + 1 = 7 \\ a(4) &= 2 \cdot 4 + 1 = 9 \end{aligned}$$



Example

Let the number of boxes in the below pattern be given by $a(x)$.



- Find the expression of $a(x)$.
- How many boxes are there when $x = 10$?
- What is the value of x when $a(x) = 628$?

Answer

a) We notice that

- When $x = 1$, there are $1 \cdot 1 + 3 = 4$ boxes.
- When $x = 2$, there are $2 \cdot 2 + 3 = 7$ boxes.
- When $x = 3$, there are $3 \cdot 3 + 3 = 12$ boxes.
- When $x = 4$, there are $4 \cdot 4 + 3 = 17$ boxes.

Therefore

$$a(x) = x \cdot x + 3 = x^2 + 3$$

b)

$$a(10) = 10^2 + 3 = 100 + 3 = 103$$

When $x = 10$, there are 103 boxes.

c) We have the equation

$$x^2 + 3 = 628$$

$$x^2 = 625$$

Hence

$$x = 15 \quad \vee \quad x = -15$$

Since we seek a positive value of x , we have $x = 15$.

10.2 Linear functions and graphs

When a variable x and a function $f(x)$ are present, we have two values; the value of x and the associated value of $f(x)$. These pairs of values can be put into a coordinate system (see [Section ??](#)) to form the *graph* of $f(x)$.

Let's use the function

$$f(x) = 2x - 1$$

as an example. We have

$$f(0) = 2 \cdot 0 - 1 = -1$$

$$f(1) = 2 \cdot 1 - 1 = 1$$

$$f(2) = 2 \cdot 2 - 1 = 3$$

$$f(3) = 2 \cdot 3 - 1 = 5$$

These pairs of values can be put into a table:

x	0	1	2	3
$f(x)$	-1	1	3	5

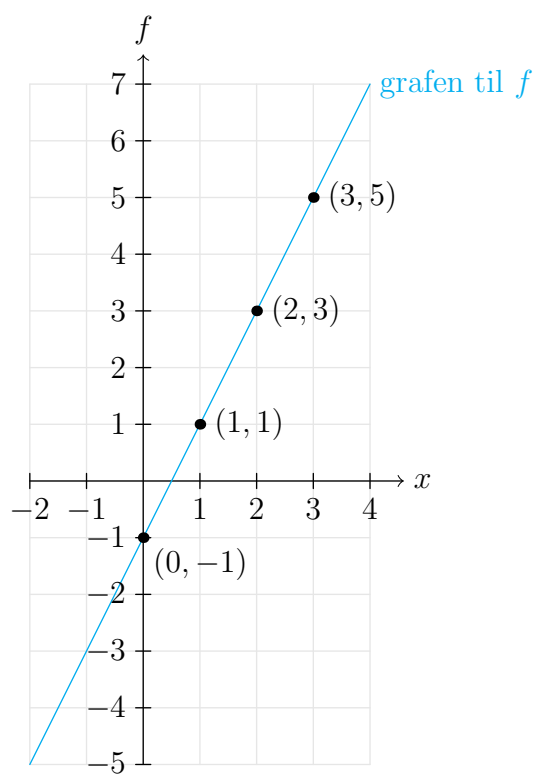
The above table yields the points

$$(0, -1) \quad (1, 1) \quad (2, 3) \quad (3, 5)$$

Now we place these points into a coordinate system (see the figure on page 200). Concerning functions, it is common to name the horizontal and the vertical axis the x -axis and the y -axis, respectively. Now the graph of $f(x)$ is the curve passing through all the infinite many points we can create by the x -values and their associated $f(x)$ -values. Our function is a *linear* function, which means its graph is a straight line. Hence, the graph is created by drawing the line going through the points we found.

As earlier mentioned, we can never draw a line, only a part of it. This also applies to graphs. In the figure on page 200 we have drawn the graph of $f(x)$ for x -values in the range -2 to 4 . That x is included in this *interval* we write as¹ $-2 \leq x \leq 4$ or $x \in [-2, 4]$.

¹Consult the list of symbols on page 4.



10.1 Linear functions

A function with the expression

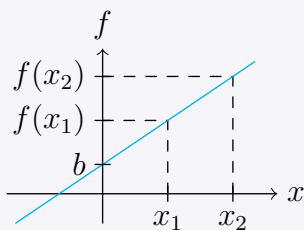
$$f(x) = ax + b$$

where a and b are constants, is a *linear* function with *slope* a and *intercept* b .

The graph of a linear function is a straight line passing through the point $(0, b)$.

For two distinct x -values, x_1 and x_2 , we have

$$a = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$



Example 1

Find the slope and the intercept of the functions.

$$f(x) = 2x + 1$$

$$g(x) = -3 + \frac{7}{2}$$

$$h(x) = \frac{1}{4}x - \frac{5}{6}$$

$$j(x) = 4 - \frac{1}{2}x$$

Answer

- $f(x)$ have slope 2 and intercept 1.
- $g(x)$ have slope -3 and intercept $\frac{7}{2}$.
- $h(x)$ have slope $\frac{1}{4}$ and intercept $-\frac{5}{6}$.
- $j(x)$ have slope $-\frac{1}{2}$ and intercept 4.

Example 2

Draw the graph of

$$f(x) = \frac{3}{4}x - 2$$

for $x \in [-5, 6]$.

Answer

To draw the graph of a linear function, we only need to know two points lying on it. The points are free to choose, therefore, in order to make calculations as simple as possible, we start off by finding the point where $x = 0$:

$$f(0) = \frac{3}{4} \cdot 0 - 2 = -2$$

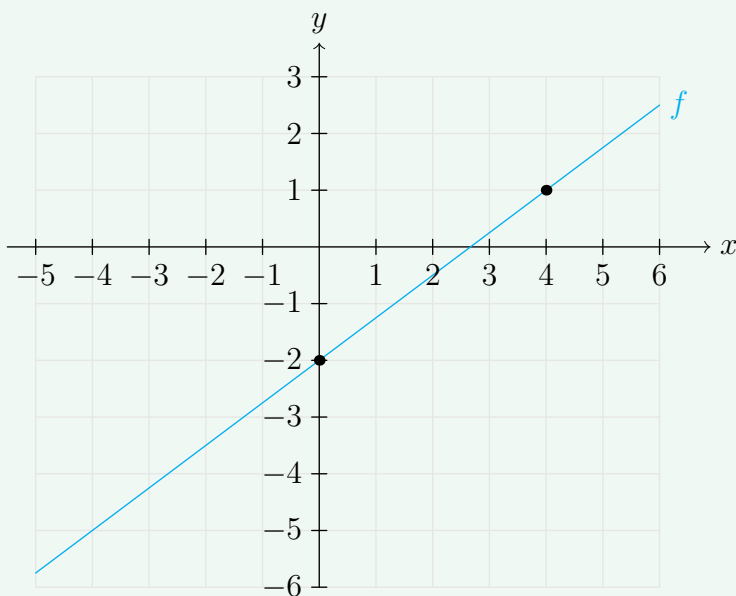
Further on, we choose $x = 4$, since this also results in easy calculations:

$$f(4) = \frac{3}{4} \cdot 4 - 2 = 1$$

Now we have all the information we need and for the record we put it into a table:

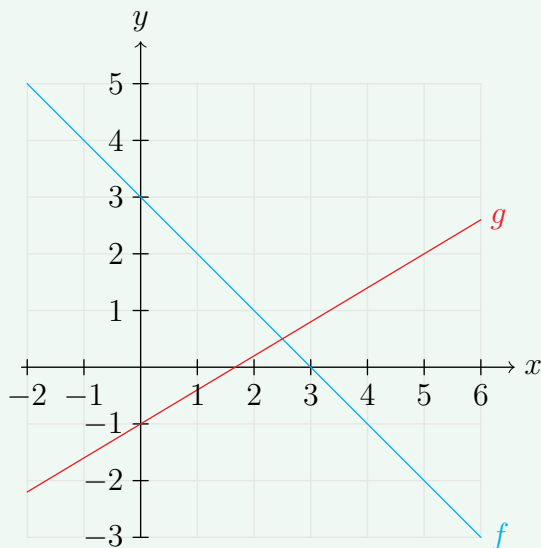
x	0	4
$f(x)$	-2	1

Now we place the points in a coordinate system and draw the line passing through them:



Example 3

Find the respective expressions of $f(x)$ and $g(x)$.



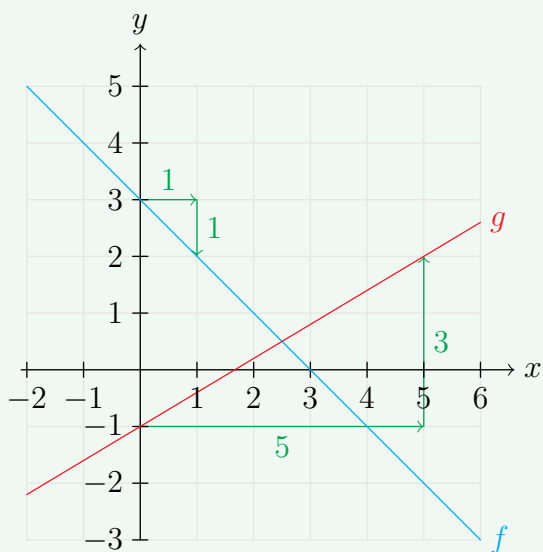
Answer

Firstly, we find the expression of $f(x)$. The point $(0, 3)$ lies on the graph of $f(x)$ (also see the figure on the next page). It then follows that $f(0) = 3$, and hence 3 is the intercept of $f(x)$. Moreover, we observe that $(1, 2)$ also lies on the graph of $f(x)$. The slope of $f(x)$ is then expressed by the fraction

$$\frac{2 - 3}{1 - 0} = -1$$

Therefore

$$f(x) = -x + 3$$



We now move our attention to finding the expression of $g(x)$. The point $(0, -1)$ lies on the graph of $g(x)$. It then follows that $f(0) = -1$, and hence -1 is the intercept of $g(x)$. Moreover, we observe that $(5, 2)$ also lies on the graph of $g(x)$. The slope $g(x)$ is then expressed by the fraction

$$\frac{2 - (-1)}{5 - 0} = \frac{3}{5}$$

Therefore

$$g(x) = \frac{3}{5}x + 1$$

10.1 Linear functions (explanation)

The expression of a

Given a linear function

$$f(x) = ax + b$$

For two distinct x -values, x_1 and x_2 , we have

$$f(x_1) = ax_1 + b \quad (10.1)$$

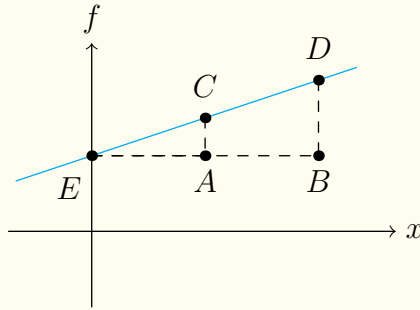
$$f(x_2) = ax_2 + b \quad (10.2)$$

Subtracting (10.1) from (10.2), we get

$$\begin{aligned} f(x_2) - f(x_1) &= ax_2 + b - (ax_1 + b) \\ f(x_2) - f(x_1) &= ax_2 - ax_1 \\ f(x_2) - f(x_1) &= a(x_2 - x_1) \\ \frac{f(x_2) - f(x_1)}{x_2 - x_1} &= a \end{aligned} \quad (10.3)$$

The graph of a linear function is a straight line

Given a linear function $f(x) = ax + b$ and two distinct x -values x_1 and x_2 . Let $A = (x_1, b)$, $B = (x_2, b)$, $C = (b, f(x_1))$, $D = (0, f(x_2))$ and $E = (0, b)$.



By (10.3), we obtain

$$\begin{aligned} \frac{f(x_1) - f(0)}{x_1 - 0} &= a \\ \frac{ax_1 + b - b}{x_1} &= a \\ \frac{ax_1}{x_1} &= a \end{aligned} \quad (10.4)$$

Similarly,

$$\frac{ax_2}{x_2} = a \quad (10.5)$$

Moreover,

$$AC = f(x_1) - b = ax_1$$

$$BD = f(x_2) - b = ax_2$$

$$EA = x_1$$

$$EB = x_2$$

From (10.4) and (10.5) it follows that

$$\frac{ax_1}{x_1} = \frac{ax_2}{x_2}$$

Hence

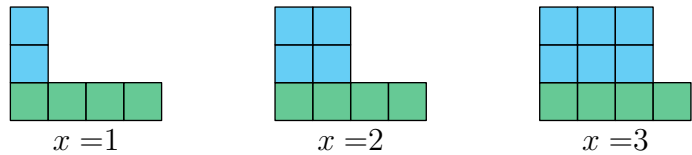
$$\frac{AC}{BD} = \frac{EA}{EB}$$

In addition, $\angle A = \angle B$, so $\triangle EAC$ and $\triangle EBD$ satisfy term iii from [Rule 11.12](#), and hence the triangles are similar. Consequently, C and D lies on the same line, which must be the graph of $f(x)$.

Exercises for Chapter 10

10.1.1

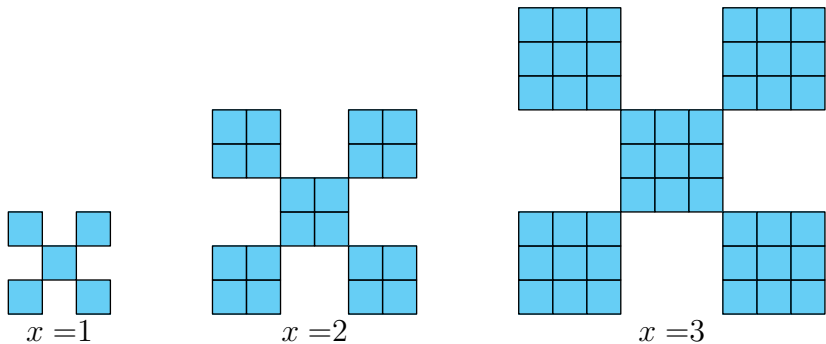
Let the number of boxes in the below figures be given by $f(x)$.



- a) Find an expression for $f(x)$.
- b) How many boxes are there when $x = 100$?
- c) What is x when $f(x) = 24$?

10.1.2

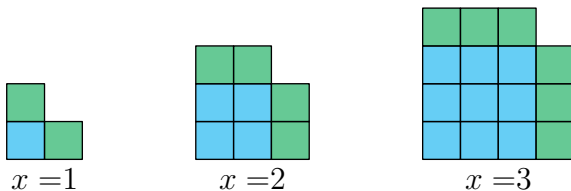
Let the number of boxes in the below figures be given by $a(x)$.



- a) Find an expression for $a(x)$.
- b) How many boxes are there when $x = 20$?
- c) What is x when $a(x) = 405$?

10.1.3

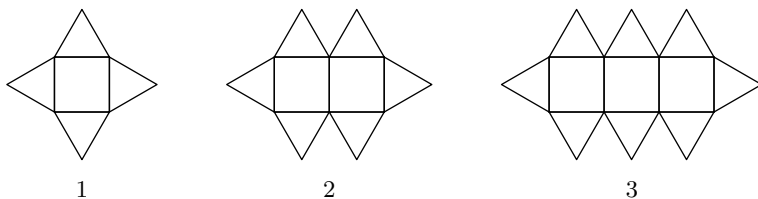
Let the number of boxes in the below figures be given by $b(x)$.



- a) Find an expression for $b(x)$.
- b) How many boxes are there when $x = 20$?
- c) What is x when $b(x) = 80$?

10.1.4 (EGV22D1)

The below figure shows the three first figures of a pattern. The figures are made up by triangles and squares.



How many triangles and how many squares will be present in figure number 10?

10.1.5

Let x be a positive integer.

- a) Make a function $p(x)$ yielding the value of the x th positive, even number.
- b) Make a function $o(x)$ yielding the value of the x th positive odd number.

10.2.1

Find the slope and the intercept of the function.

a) $f(x) = 5x + 10$ b) $g(x) = 3x - 12$

c) $h(x) = -\frac{1}{7}x - 9$ d) $i(x) = \frac{3}{2}x - \frac{1}{4}$

10.2.2

Draw the function on the interval $x \in [-5, 5]$:

a) $f(x) = 2x - 1$ b) $g(x) = -3x + 5$

10.3.1

Given the equation set

$$x - y = 5 \tag{I}$$

$$x + y = 9 \tag{II}$$

- a) Explain why the solutions of this set yield the interception point of the functions

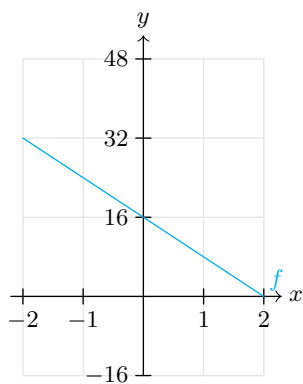
$$f(x) = x - 5$$

$$g(x) = 9 - x$$

- b) Solve the equation set.

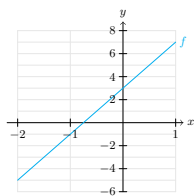
10.3.2

Find the expression of $f(x)$



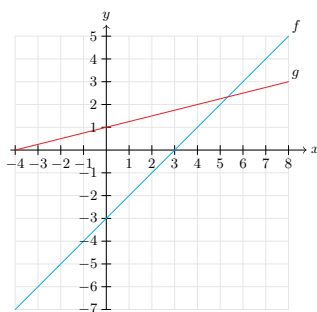
10.3.3

Find the expression of $f(x)$



10.3.4

Find the expression of $f(x)$ and $g(x)$.



Ponder 11

Ponder 12

Use the formulas from [Exercise 10.1.5](#) to prove that

- a) the sum/difference of two even numbers is an even number.
- b) sum/difference of two odd numbers is an even number.
- c) the sum/difference of an even number and an odd number is an odd number.

Ponder 13

- a) A linear function $f(x)$ has slope 3, and the point $(2, 1)$ lies on the graph of f . Find the expression of f .
- b) A linear function $f(x)$ has slope a , and the point (x_1, y_1) lies on the graph of f . Prove that

$$f(x) = a(x - x_1) + y_1$$

(This equation is called the **point-slope equation** .)

Ponder 14

Given the functions $f(x)$ and $g(x)$, where the graph of g is the line passing through $A = (a, f(a))$ and $B = (b, f(b))$. Prove that

$$f - g = f(x) - \frac{f(b) - f(a)}{b - a}(x - a) + f(a)$$

Chapter 11

Geometry

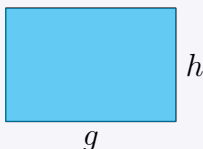
11.1 Formulas of area and perimeter

A *formula* is an equation where (usually) one variable is isolated on one side of the equal sign. In [Section 7.4](#) we have already looked at the formulas for the area of rectangles and triangles, but there using words instead of symbols. Here we shall reproduce these two formulas, followed by other classical formulas for area and perimeter.

11.1 The area of a rectangle (7.4)

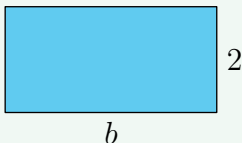
The area A of a rectangle with base g and height h is

$$A = gh$$



Example 1

Find the area of the rectangle.



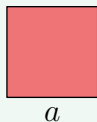
Answer

The area A of the rectangle is

$$A = b \cdot 2 = 2b$$

Example 2

Find the area of the square.



Answer

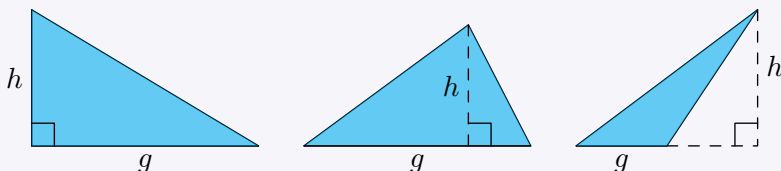
The area A of the square is

$$A = a \cdot a = a^2$$

11.2 The area of a triangle (7.4)

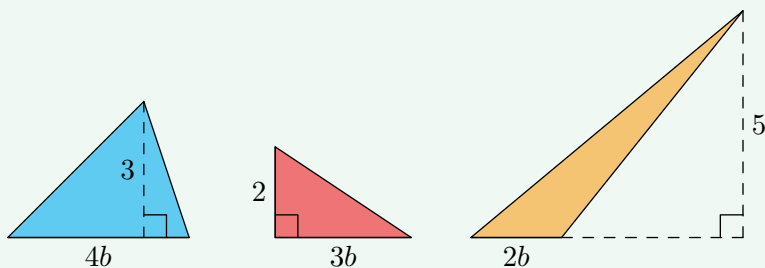
The area A of a triangle with base g and height h is

$$A = \frac{gh}{2}$$



Example

Which one of the triangles have the largest area?



Answer

Let A_1 , A_2 and A_3 denote the areas of, respectively, the triangle to the left, in the middle and to the right. Then

$$A_1 = \frac{4 \cdot 3}{2} = 6$$

$$A_2 = \frac{2 \cdot 3}{2} = 3$$

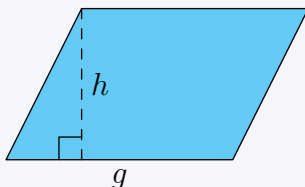
$$A_3 = \frac{2 \cdot 5}{2} = 5$$

Hence, it is the triangle to the left which has the largest area.

11.3 The area of a parallelogram

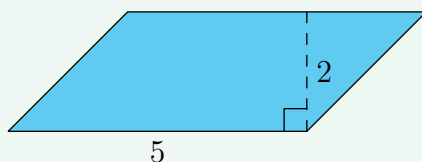
The area A of a parallelogram with base g and height h is

$$A = gh$$



Example

Find the area of the parallelogram



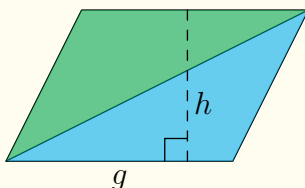
Answer

The area A of the parallelogram is

$$A = 5 \cdot 2 = 10$$

11.3 The area of a parallelogram (explanation)

From a parallelogram we can always, by drawing one of its diagonals, form two triangles which both have base g and height h .



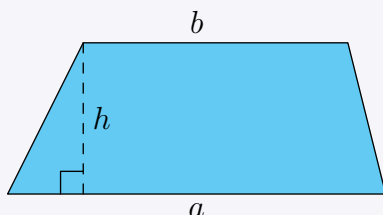
Hence, both triangles have an area equal to $\frac{gh}{2}$. Therefore, the area A of the parallelogram is

$$\begin{aligned} A &= \frac{gh}{2} + \frac{gh}{2} \\ &= g \cdot h \end{aligned}$$

11.4 The area of a trapezoid

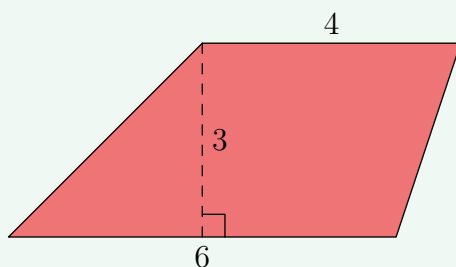
The area A of a trapezoid with parallel sides a and b and height h is

$$A = \frac{h(a + b)}{2}$$



Example

Find the area of the trapezoid.



Answer

The area A of the trapezoid is

$$\begin{aligned} A &= \frac{3(6 + 4)}{2} \\ &= \frac{3 \cdot 10}{2} \\ &= 15 \end{aligned}$$

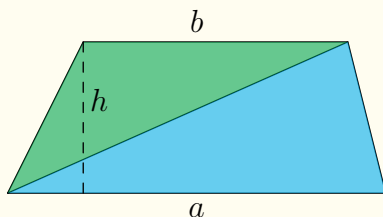
Note

In respect of a base and a height, the area formulas for a parallelogram and a rectangle are identical. Applying [Rule 11.4](#) on a parallelogram also results in an expression equal to gh . This follows from the fact that a parallelogram is just a special case of a trapezoid (and a rectangle is just a special case of a

parallelogram).

11.4 The area of a trapezoid (explanation)

In a trapezoid, we can, by drawing one of the diagonals, create two triangles:



In the above figure we have

$$\text{The area of the blue triangle} = \frac{ah}{2}$$

$$\text{The area of the green triangle} = \frac{bh}{2}$$

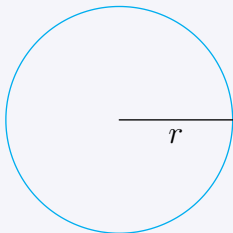
Therefore, the area A of the trapezoid is

$$\begin{aligned} A &= \frac{ah}{2} + \frac{bh}{2} \\ &= \frac{h(a+b)}{2} \end{aligned}$$

11.5 The perimeter of a circle (and the value of π)

The perimeter (the circumference) O of a circle with radius r is

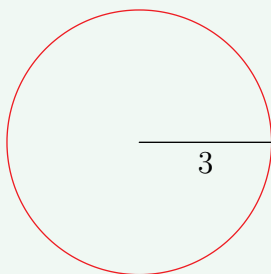
$$O = 2\pi r$$



$$\pi = 3.141592653589793....$$

Example 1

Find the circumference of the circle.



Answer

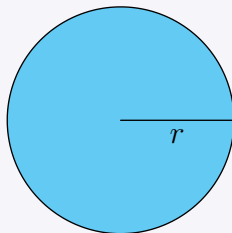
The circumference O is

$$\begin{aligned} O &= 2\pi \cdot 3 \\ &= 6\pi \end{aligned}$$

11.6 The area of a circle

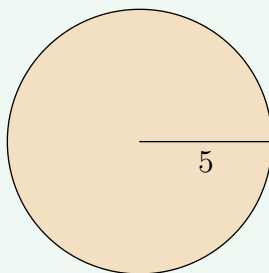
The area A of a circle with radius r is

$$A = \pi r^2$$



Example

Find the area of the circle.



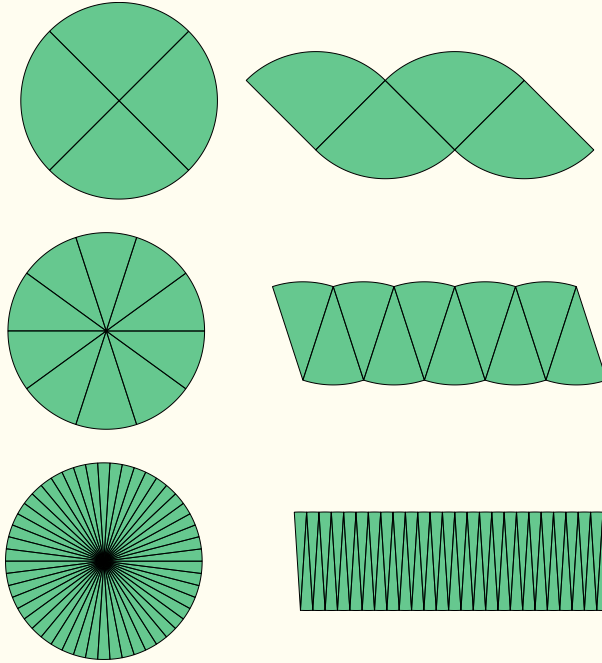
Answer

The area A of the circle is

$$A = \pi \cdot 5^2 = 25\pi$$

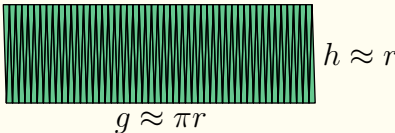
11.6 The area of a circle (explanation)

In the below figure, we have divided a circle into 4, 10 and 50 (equal-sized) sectors, and placed them consecutively.



In each case, the arcs make up the circumference of the circle. If the circle has radius r , the sum of the arcs equals $2\pi r$. And when there are equally many sectors turned upwards as downwards, the total length of the arcs equals πr on both the bottom and the top.

The more sectors the circle is divided into, the more the composition takes the form of a rectangle (in the figure below there are 100 sectors). The base g of this "rectangle" approximately equals πr , while the height h approximately equals r .



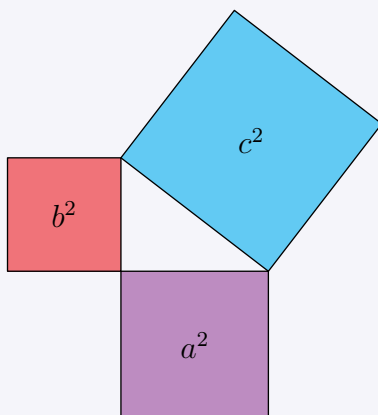
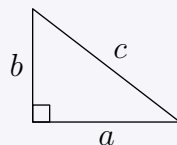
Hence, the area A of the "rectangle", that is, the circle, is

$$A \approx gh \approx \pi r \cdot r = \pi r^2$$

11.7 Pythagoras's theorem

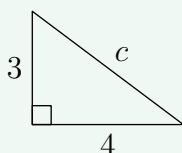
In a right triangle, the area of the square formed by the hypotenuse equals the sum of the areas of the squares formed by the legs.

$$a^2 + b^2 = c^2$$



Example 1

Find the length of c .



Answer

We know that

$$c^2 = a^2 + b^2$$

where a and b are the legs of the right triangle. Therefore

$$\begin{aligned} c^2 &= 4^2 + 3^2 \\ &= 16 + 9 \\ &= 25 \end{aligned}$$

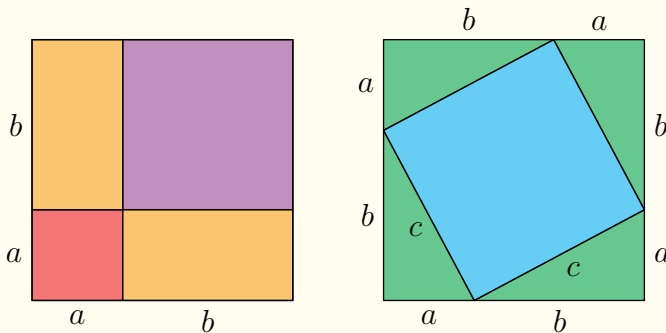
Hence,

$$c = 5 \quad \vee \quad c = -5$$

Since c is a length, $c = 5$.

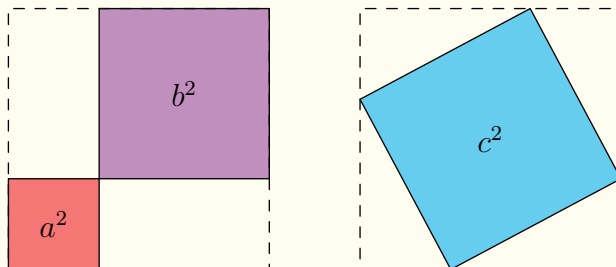
11.7 Pythagoras's theorem (explanation)

The below figure shows equal-sized squares divided into different shapes.



We observe the following:

1. The area of the red square is a^2 , the area of the purple square is b^2 and the area of the blue square is c^2 .
2. The area of an orange square is ab and the area of a green triangle is $\frac{ab}{2}$.
3. If we remove the two orange rectangles and the four green triangles, the remaining area to the left equals the remaining area to the right (by remark 2).



Hence

$$a^2 + b^2 = c^2 \quad (11.1)$$

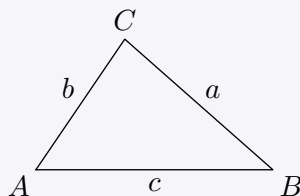
Given a triangle with sides of length a, b and c , of which c is the longest. As long as the triangle is right, we can always form two squares with sides of length $a + b$, as in the initial figure. Therefore, (11.1) is valid for alle right triangles.

11.2 Congruent and similar triangles

11.8 Unique construction of triangles

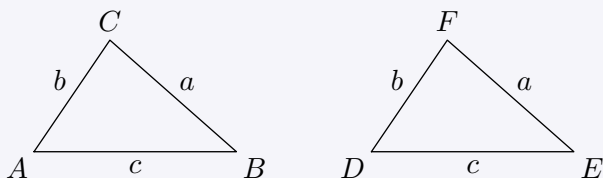
A triangle $\triangle ABC$, as shown in the below figure, can be uniquely constructed if one of the following terms are satisfied:

- i) $c, \angle A$ and $\angle B$ are known.
- ii) a, b and c are known.
- iii) b, c and $\angle A$ are known.



11.9 Congruent triangles

Two triangles of equal shape and size are congruent.

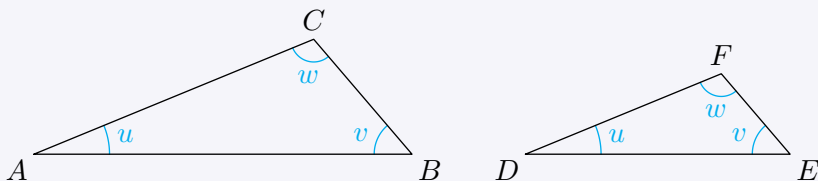


The congruence in the above figure is written

$$\triangle ABC \cong \triangle DEF$$

11.10 Similar triangles

Similar triangles constitute three pairs of equal angles.

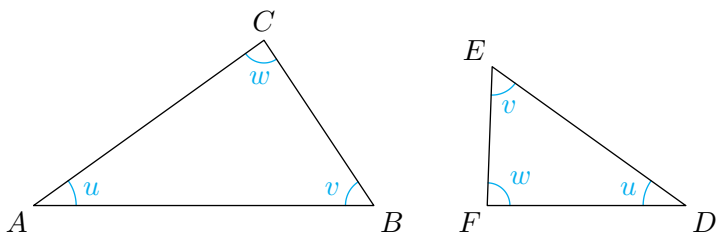


The similarity in the above figure is written

$$\triangle ABC \sim \triangle DEF$$

Corresponding sides

When studying similar triangles, *corresponding* sides plays an important role. Corresponding sides are sides in similar triangles adjacent to the same angle.



Regarding the similar triangles $\triangle ABC$ and $\triangle DEF$ we have

In $\triangle ABC$ is

- BC adjacent to u .
- AC adjacent to v
- AB adjacent to w .

I $\triangle DEF$ is

- FE adjacent to u .
- FD adjacent to v
- ED adjacent to w .

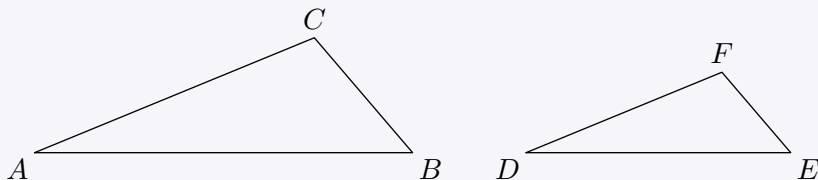
This means that these are corresponding sides:

- BC and FE
- AC and FD
- AB and ED

11.11 Ratios in similar triangles

If two triangles are similar, the ratios of corresponding sides are equal¹.

$$\frac{AB}{DE} = \frac{AC}{DF} = \frac{BC}{EF}$$



¹Here, we take it for granted that corresponding sides are apparent from the figure.

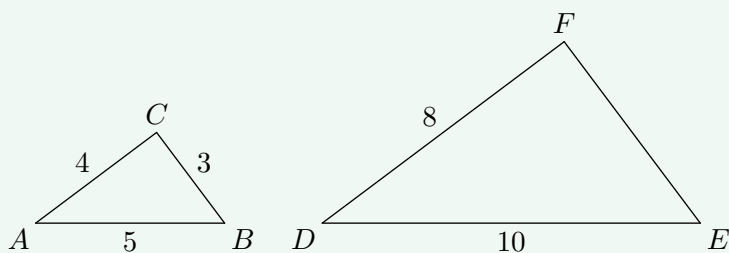
Notice

From [Rule 11.11](#) it follows that

$$\frac{AB}{BC} = \frac{DE}{EF} \quad , \quad \frac{AB}{AC} = \frac{DE}{DF} \quad , \quad \frac{BC}{AC} = \frac{EF}{DF}$$

Example

The triangles are similar. Find the length of EF .



Answer

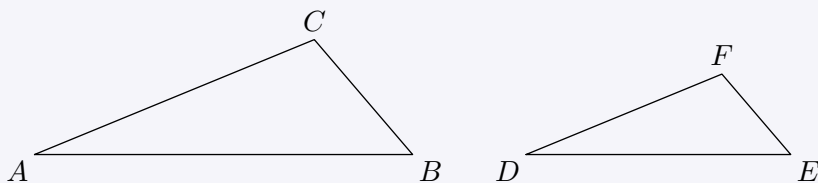
We observe that AB corresponds to DE , BC to EF and AC to DF . Therefore

$$\begin{aligned} \frac{DE}{AB} &= \frac{EF}{BC} \\ \frac{10}{5} &= \frac{EF}{3} \\ 2 \cdot 3 &= \frac{EF}{3} \cdot 3 \\ 6 &= EF \end{aligned}$$

11.12 Terms of similar triangles

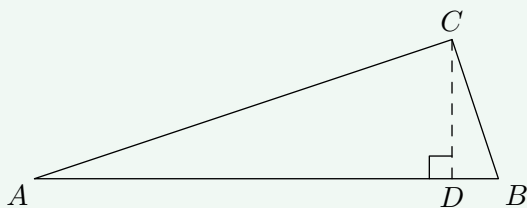
Two triangles $\triangle ABC$ and $\triangle DEF$ are similar if one of these terms are satisfied:

- i) They constitute two pairs of equal angles.
- ii) $\frac{AB}{DE} = \frac{AC}{DF} = \frac{BC}{EF}$
- iii) $\frac{AB}{DE} = \frac{AC}{DF}$ and $\angle A = \angle D$.



Example 1

$\angle ACB = 90^\circ$. Show that $\triangle ABC \sim \triangle ACD$.



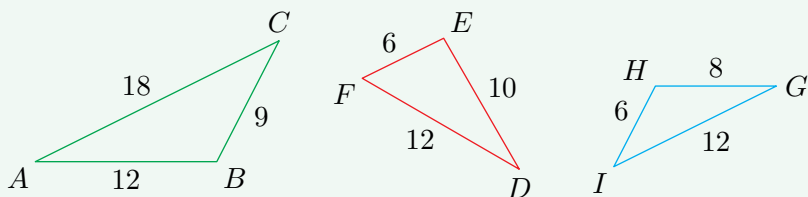
Answer

$\triangle ABC$ and $\triangle ACD$ are both right and they have $\angle DAC$ in common. Hence, the triangles satisfy term *i* from [Rule 11.12](#), and therefore they are similar.

Note: Similarly it can be shown that $\triangle ABC \sim \triangle CBD$.

Example 2

Examine whether the triangles are similar.



Answer

We have

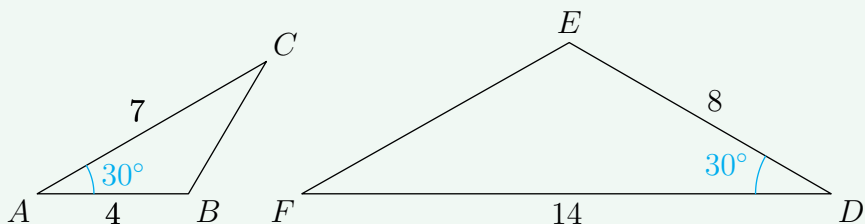
$$\frac{AC}{FD} = \frac{18}{12} = \frac{3}{2}, \quad \frac{BC}{FE} = \frac{9}{6} = \frac{3}{2}, \quad \frac{AB}{DE} = \frac{12}{10} = \frac{6}{5}$$

$$\frac{AC}{IG} = \frac{18}{12} = \frac{3}{2}, \quad \frac{BC}{IH} = \frac{9}{6} = \frac{3}{2}, \quad \frac{AC}{IG} = \frac{18}{12} = \frac{3}{2}$$

Hence, $\triangle ABC$ and $\triangle GHI$ satisfy term *ii* from [Rule 11.12](#), and therefore they are similar. (Hence, $\triangle GHI$ and $\triangle FED$ are not similar.)

Example 3

Examine whether the triangles are similar.



Answer

We have $\angle BAC = \angle EDF$. Also,

$$\frac{ED}{AB} = \frac{8}{4} = 2, \quad \frac{FD}{AC} = \frac{14}{7} = 2$$

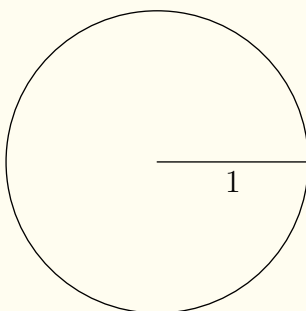
Hence, term *iii* from [Rule 11.12](#) is satisfied, and therefore the triangles are similar.

11.3 Explanations

11.5 The perimeter of a circle (and the value of π) (explanation)

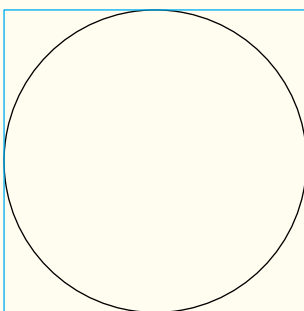
Here we shall use regular polygons along the path to our wanted result. In regular polygons, all sides are of equal length. Since all polygons here to be mentioned are regular, we'll mention them simply as polygons.

We'll start off by examining some approximations of the circumference O_1 of a circle with radius 1.



Upper and lower boundary

When seeking a value, it is a good habit to conclude how large or small you expect it to be. With this target, we enclose the circle by a square with sides of length 2:

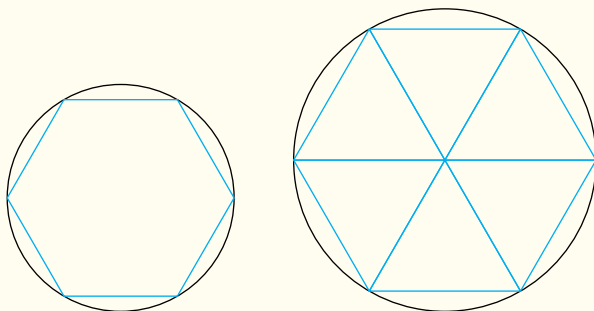


Clearly, the circumference of the circle is smaller than the perimeter of the square, therefore

$$\begin{aligned} O_1 &< 2 \cdot 4 \\ &< 8 \end{aligned}$$

Now we inscribe a 6-gon (hexagon). The hexagon can be divided into 6 equilateral triangles with, necessarily, sides of length 1. The circumference of the circle must be larger than the perimeter of the hexagon, so

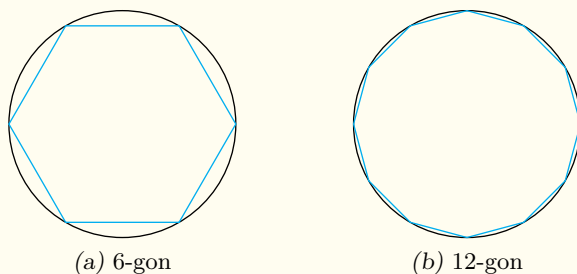
$$\begin{aligned} O_1 &> 6 \cdot 1 \\ &> 6 \end{aligned}$$



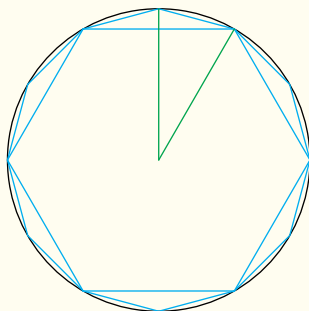
Now advancing to a more sophisticated hunt for the circumference, we know that we seek a value between 6 and 8.

Increasingly better approximations

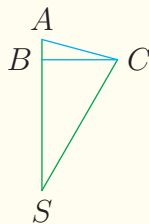
The idea of inscribing polygons carries on. We let the below figures work as a sufficient prove of the fact that the more sides of the polygon, the better estimate its perimeter makes of the circumference of the circle.



Since a 6-gon has sides of length 1, it is tempting to examine if this can help us find the side lengths of other polygons. By inscribing both a 6-gon and a 12-gon (and drawing a triangle) we have a figure like this:



(a) A 6-gon and a 12-gon together with a triangle formed by the circle center and one side of the 12-gon.



(b) The triangle from figure (a).

Let s_{12} and s_6 denote the side lengths of the 12-gon and the 6-gon, respectively. Moreover, we observe that both A and C lies on the circular arc and that both $\triangle ABC$ and $\triangle BSC$ are right-angled (explain to yourself why!). We have

$$SC = 1$$

$$BC = \frac{s_6}{2}$$

$$SB = \sqrt{SC^2 - BC^2}$$

$$BA = 1 - SB$$

$$AC = s_{12}$$

$$s_{12}^2 = BA^2 + BC^2$$

To find s_{12} , we need to know BA , and to find BA we need to know SB . Hence, we start off finding SB . Since $SC = 1$ and $BC = \frac{s_6}{2}$,

$$\begin{aligned} SB &= \sqrt{1 - \left(\frac{s_6}{2}\right)^2} \\ &= \sqrt{1 - \frac{s_6^2}{4}} \end{aligned}$$

Now we focus on finding s_{12} :

$$\begin{aligned} s_{12}^2 &= (1 - SB)^2 + \left(\frac{s_6}{2}\right)^2 \\ &= 1^2 - 2SB + SB^2 + \frac{s_6^2}{4} \end{aligned}$$

At first, it looks like the expression to the right cannot be simplified, but a small operation can change this. If -1 was a term present, we could have combined -1 and $\frac{s_6^2}{4}$ to become $-SB^2$. We obtain -1 by both adding and subtracting it on the right side of the equation:

$$\begin{aligned}
 s_{12}^2 &= 1 - 2SB + SB^2 + \frac{s_6^2}{4} - 1 + 1 \\
 &= 2 - 2SB + SB^2 - \left(1 - \frac{s_6^2}{4}\right) \\
 &= 2 - 2SB + SB^2 - SB^2 \\
 &= 2 - 2SB \\
 &= 2 - 2\sqrt{1 - \frac{s_6^2}{4}} \\
 &= 2 - \sqrt{4} \sqrt{1 - \frac{s_6^2}{4}} \\
 &= 2 - \sqrt{4 - s_6^2}
 \end{aligned}$$

Hence

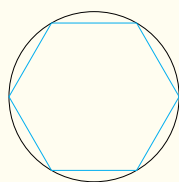
$$s_{12} = \sqrt{2 - \sqrt{4 - s_6^2}}$$

Even though we have derived a relation between the side lengths s_{12} and s_6 , this relation is valid for all pairs of side lengths where one is the side length of a polygon with twice as many sides as the other. Now let s_n and s_{2n} , respectively, denote the side lengths of a polygon and a polygon with twice as many sides. Then

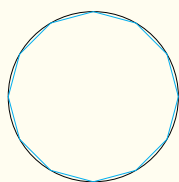
$$s_{2n} = \sqrt{2 - \sqrt{4 - s_n^2}} \quad (11.2)$$

The perimeter of a polygon inscribed in the circle is an estimate of the circumference. Applying (11.2), we can successively find the side length of a polygon with twice as many sides as the previous. The below table shows the side length and the associated estimate of the circumference up to a 96-gon:

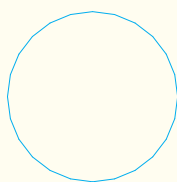
<i>Side length formula</i>	<i>Side length</i>	<i>Estimate, circumference</i>
	$s_6 = 1$	$6 \cdot s_6 = 6$
$s_{12} = \sqrt{2 - \sqrt{4 - s_6^2}}$	$s_{12} = 0.517\dots$	$12 \cdot s_{12} = 6.211\dots$
$s_{24} = \sqrt{2 - \sqrt{4 - s_{12}^2}}$	$s_{24} = 0.261\dots$	$24 \cdot s_{24} = 6.265\dots$
$s_{48} = \sqrt{2 - \sqrt{4 - s_{24}^2}}$	$s_{48} = 0.130\dots$	$48 \cdot s_{48} = 6.278\dots$
$s_{96} = \sqrt{2 - \sqrt{4 - s_{48}^2}}$	$s_{96} = 0.065\dots$	$96 \cdot s_{96} = 6.282\dots$



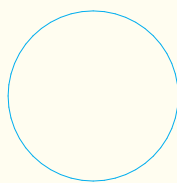
(a) 6-gon



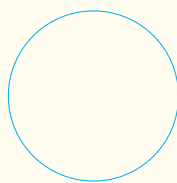
(b) 12-gon



(c) 24-gon



(d) 48-gon



(e) 96-gon

In fact, the mathematician [Archimedes](#) reached as far as the above calculation approximately 250 b.c.!

A computer has no problems performing calculations¹ on a polygon with extremely many sides. Calculating the perimeter of a 201 326 592-gon yields

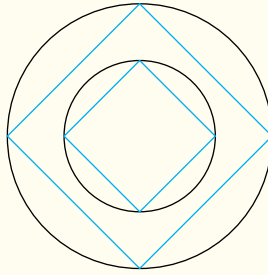
Circumference of a circle with radius 1 = 6.283185307179586...

(With the aid of more advanced mathematics it can be proved that the circumference of a circle with radius 1 is an irrational number, but that the digits shown above are correct, thereby the equal sign.)

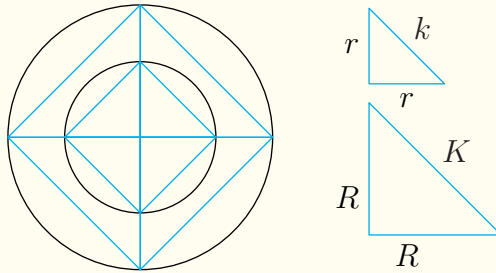
The formula and π

We shall now derive the famous formula for the circumference of any circle. Here as well, we take it for granted that the perimeter of an inscribed polygon yields an estimate of the circumference which gets more accurate the more sides the polygon has.

For the sake of simplicity, we shall use inscribed squares to illustrate the outline. We draw two circles of random size, but the one larger than the other, and inscribe a square in both. Let R and r denote the radius of the larger and the smaller circle, respectively. Also, let K and k denote the side length of the larger and the smaller square, respectively.



Both squares can be divided into four isosceles triangles:



Since these triangles are similar,

$$\frac{K}{R} = \frac{k}{r} \tag{11.3}$$

Let $\tilde{O} = 4K$ and $\tilde{o} = 4k$ denote the estimated circumferences of the larger and the smaller circle, respectively. Multiplying both

sides of (11.3) by 4 yields

$$\frac{4A}{R} = \frac{4a}{r} \quad (11.4)$$

$$\frac{\tilde{O}}{R} = \frac{\tilde{o}}{r} \quad (11.5)$$

Now we observe this:

If we were to inscribe polygons with 4, 100 or any number of sides, the polygons could still be divided into triangles obeying (11.3). And in the same way as we did in the above example, we could then rewrite (11.3) into (11.5).

Let's therefore imagine polygons with such a large number of sides that we accept their respective perimeters as equal to the respective circumferences of the circles. Letting O and o denote the circumferences of the larger and smaller circle respectively, we have

$$\frac{O}{R} = \frac{o}{r}$$

Since the circles are randomly chosen, we conclude that *all circles have the same ratio of the circumference to the radius*. An equivalent statement is that *all circles have the same ratio of the circumference to the diameter*.

The ratio of the circumference O to the diameter d in a circle is named π (pronounced "pi"):

$$\frac{O}{d} = \pi$$

The above equation yields the formula for the circumference of a circle with diameter d and radius r :

$$\begin{aligned} O &= \pi d \\ &= 2\pi r \end{aligned}$$

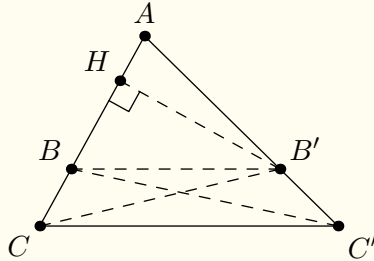
Earlier we found that the circumference of a circle with radius 1 (and diameter 2) equals 6.283185307179586... Hence

$$\begin{aligned} \pi &= \frac{6.283185307179586...}{2} \\ &= 3.141592653589793... \end{aligned}$$

¹For those interested in computer programming; the iteration algorithm must be altered in order to avoid instabilities when the number of sides are large.

11.11 Ratios in similar triangles (explanation)

Here, we shall write the area of a triangle $\triangle ABC$ as ABC .



In the figure above, we have $BB' \parallel CC'$. With BB' as base, HB' is the height of both $\triangle CBB'$ and $\triangle CBB'$. Therefore

$$CBB' = C'BB' \quad (11.6)$$

Moreover,

$$ABB' = AB \cdot HB'$$

$$CBB' = BC \cdot HB'$$

Hence

$$\frac{ABB'}{CBB'} = \frac{AB}{BC} \quad (11.7)$$

Similarly,

$$\frac{ABB'}{C'BB'} = \frac{AB'}{B'C'} \quad (11.8)$$

From (11.6), (11.7) and (11.8) it follows that

$$\frac{AB}{BC} = \frac{ABB'}{CBB'} = \frac{ABB'}{C'BB'} = \frac{AB'}{B'C'} \quad (11.9)$$

For the similar triangles $\triangle ACC'$ and $\triangle ABB'$,

$$\begin{aligned}\frac{AC}{AB} &= \frac{AB + BC}{AB} \\ &= 1 + \frac{BC}{AB}\end{aligned}$$

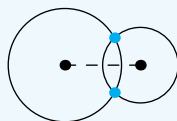
$$\begin{aligned}\frac{AC'}{AB'} &= \frac{AB' + B'C'}{AB'} \\ &= 1 + \frac{B'C'}{AB'}\end{aligned}$$

By (11.9), the ratio of corresponding sides in the two triangles are equal.

Note

In the following explanations of term *ii* and *iii* from [Rule 11.8](#) we assume this:

- Two circles intersect in maximum two points.
- Given a coordinate system placed in the center of one of the circles, such that the horizontal axis passes through both circle centers. If (a, b) is one of the intersection points, $(a, -b)$ is the other.

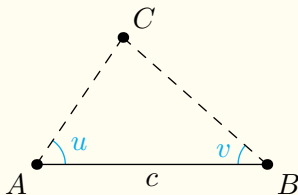


The remarks above are quite easy to prove, but since they are largely intuitively true, we hold them as granted. This implies that the triangle formed by the two centers and one of the intersection points is congruent to the triangle formed by the two centers and the other intersection point. By this, we can study attributes of triangles with the aid of semi-circles.

11.8 Unique construction of triangles (explanation)

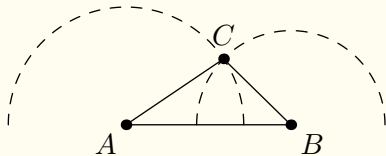
Term i

Given a length c and two angles u and v . We make a segment AB with length c . Then we draw two angle sides, such that $\angle A = u$ and $\angle B = v$. As long as these angle sides are not parallel, they must intersect in one, and one only, point (C in the figure). Together with A and B , this point will form a triangle uniquely determined by c , u and v .



Term ii

Given three lengths a , b and c . We make a segment AB with length c . Then we make two semi-circles with respective radii a and b and centers B and A . If a triangle $\triangle ABC$ is to have sides of length a , b and c , C must lie on both of the semi-circles. Since the semi-circles intersect in one point only, $\triangle ABC$ is uniquely determined by a , b and c .

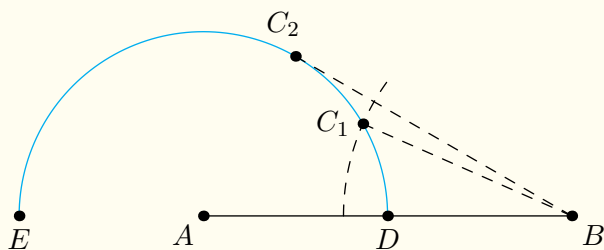


Term iii

Given two lengths b and c and an angle u . We begin as follows:

1. We make a segment AB with length c .
2. In A we draw a semi-circle with radius b .

By placing C randomly on the arc of the semi-circle, we get all instances of a triangle $\triangle ABC$ with sides of length $AB = c$ and $AC = b$. Specifically placing C on the arc of the semi-circle is equivalent to setting a specific value of $\angle A$. Now it remains to show that every placement of C implies a unique length of BC .



Let C_1 and C_2 denote two potential placements of C , where C_2 , along the semicircle, lies closer to E than C_1 . Now we draw a circular arc with radius BC_1 and center B . Since the dotted arc and the semi-circle only intersect in C_1 , other points will either lie inside or outside the dotted arc. Necessarily, C_2 lies outside the dotted arc, and therefore BC_2 is longer than BC_1 . From this we can conclude that the length of BC increases as C moves against E along the semi-circle. Therefore, specifying $\angle A = u$ yields a unique value of BC , and hence a unique triangle $\triangle ABC$ where $AC = b$, $c = AB$ and $\angle BAC = u$.

11.12 Terms of similar triangles (explanation)

Term i

Given two triangles $\triangle ABC$ and $\triangle DEF$. By [Rule 7.3](#),

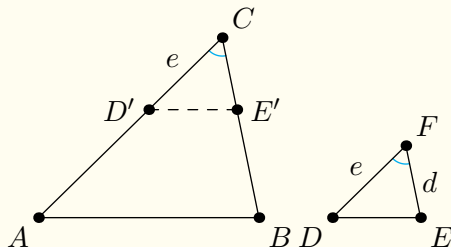
$$\angle A + \angle B + \angle C = \angle D + \angle E + \angle F$$

If $\angle A = \angle D$ and $\angle B = \angle E$, it follows that $\angle C = \angle F$.

Term ii

Given two triangles $\triangle ABC$ and $\triangle DEF$, where

$$\frac{AC}{DF} = \frac{BC}{EF} \quad , \quad \angle C = \angle F \quad (11.10)$$



Let $a = BC$, $b = AC$, $d = EF$ and $e = DF$. We place D' and E' on AC and BC , respectively, such that $D'C = e$ and $AB \parallel D'E'$. Then $\triangle ABC \sim \triangle D'E'C$, and hence

$$\frac{E'C}{BC} = \frac{D'C}{AC}$$

$$E'C = \frac{ae}{b}$$

By (11.10),

$$EF = \frac{ae}{b}$$

Hence $E'C = EF$. From term ii of [Rule 11.8](#) it now follows that $\triangle D'E'C \cong \triangle DEF$. This implies that $\triangle ABC \sim \triangle DEF$.

Term iii

Given two triangles $\triangle ABC$ and $\triangle DEF$, where

$$\frac{AB}{DE} = \frac{AC}{DF} = \frac{BC}{EF} \quad (11.11)$$

We place D' and E' on AC and BC , respectively, such that $D'C = e$ and $E'C = d$. From term i of [Rule 11.12](#) we have $\triangle ABC \sim \triangle D'E'C$. Therefore

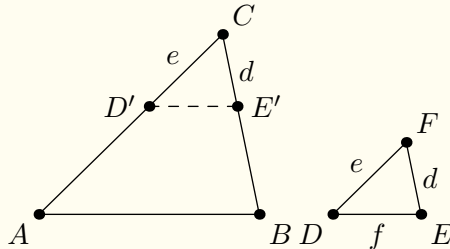
$$\frac{D'E'}{AB} = \frac{D'C}{AC}$$

$$D'E' = \frac{ae}{c}$$

By (11.11),

$$f = \frac{ae}{c}$$

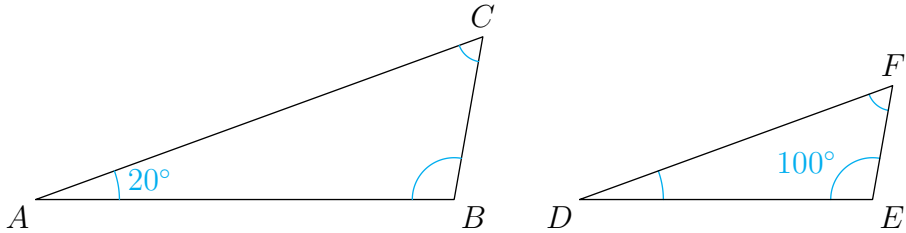
Hence, the side lengths of $\triangle D'E'C$ and $\triangle DEF$ are pairwise equal, and then, from term i of [Rule 11.8](#), they are congruent. This implies that $\triangle ABC \sim \triangle DEF$.



Exercises for Chapter 11

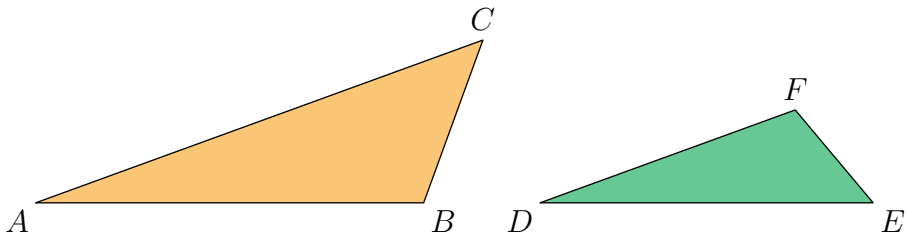
11.1.1

The triangles are similar. Find the value of $\angle ACB$.



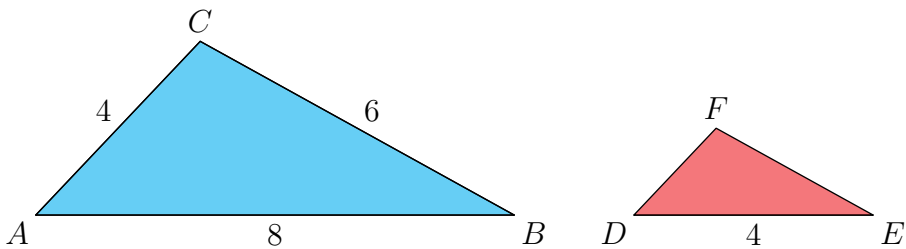
11.1.2

The triangles are similar. Find the three pairs of corresponding sides.



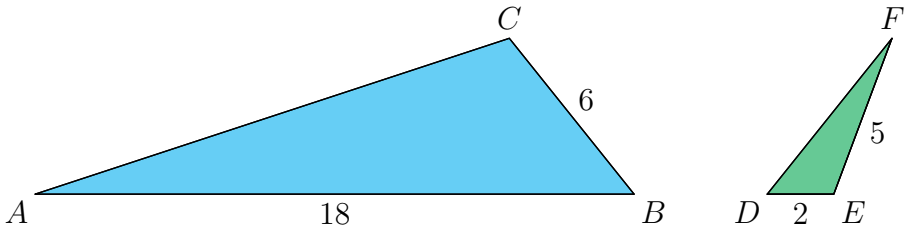
11.1.3

The triangles are similar. Find the length of EF and the length of DF .



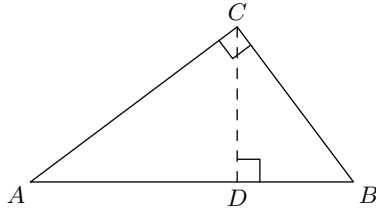
11.1.4

The triangles are similar. Find the length of AC and the length of DF .



11.1.5

Find all the similar triangles defined by A , B , C og D .



11.1.6

$\triangle ABC$ and $\triangle DEF$ are similar.

- What is the ratio of the area of $\triangle DEF$ to the area of $\triangle ABC$ if $h_1 = 2$ and $h_2 = 6$?
- Given a number a . What is the ratio of the area of $\triangle DEF$ to the area of $\triangle ABC$ if $h_2 = ah_1$?

11.1.7

En kjegle has radius 10 and height 4.

- a) Find the surface area of the cone.
- b) Find the volume of the cone.

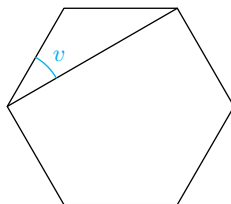
11.1.8

- a) Two spheres have radii 2 and 6, respectively. What is the ratio of the volume of the larger sphere to the volume of the smaller sphere?
- b) A sphere has radius r , and another sphere has radius ar , where $a > 1$. What is the ratio of the volume of the larger sphere to the volume of the smaller sphere?

Ponder 15

(GV21D1)

The polygon is a regular¹ hexagon. Find the value of v .

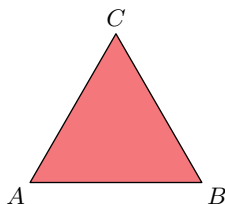


Ponder 16

Given an isosceles triangle $\triangle ABC$, with $AC = BC$. Prove that² the bisector of $\angle ACB$ is the perpendicular bisector of AB .

Ponder 17

$\triangle ABC$ is equilateral and have sides of length s .



- Prove that in a triangle with angles 30° , 60° , 90° , the hypotenuse is twice as long as the shortest leg.
- Prove that the height of $\triangle ABC$ is $\frac{\sqrt{3}}{2}s$.

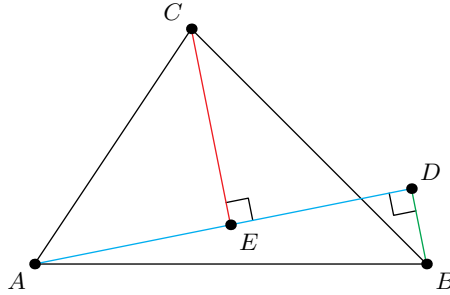
¹Regular polygons have sides of equal length.

²The definition of the bisector of an angle and the perpendicular bisector of a segment is found in [TM1](#).

Ponder 18

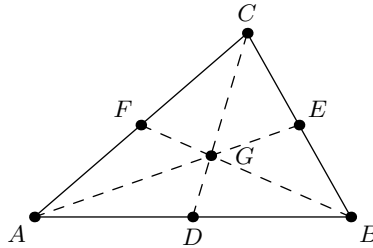
Prove that the double area of $\triangle ABC$ can be written as

$$AE \cdot BD + CE \cdot AD$$



Ponder 19

A **median** of a triangle is a segment joining a vertex to midpoint of the opposite side.



Given $\triangle ABC$ with medians AE , BF and CD .

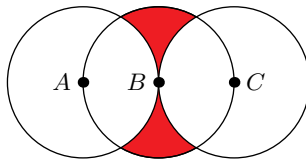
- Prove that AE , BF and CD intersect at exactly one point (G in the figure).
- Prove that

$$\frac{GC}{DG} = \frac{GB}{FG} = \frac{GA}{EG} = 2$$

Note: Part b) is probably easier than part a).

Ponder 20

The three circles have radius 2, and A , B and C lie on the same line. Find the area of the red surface.

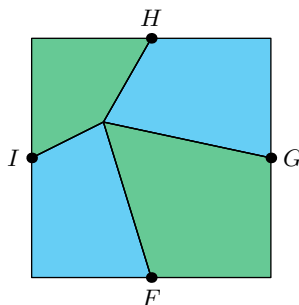


Hint: You could get use of the fact that a sector with angle v makes up $\frac{v}{360^\circ}$ of the area of a circle with equal radius.

Ponder 21

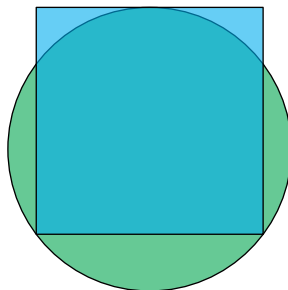
The colored surfaces makes up a square, and F , G , H and I are, respectively, the square's midpoints.

Prove that the area of the blue surface equals the area of the green surface.



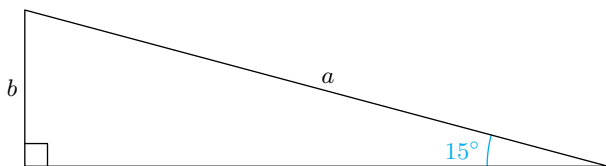
Ponder 22

The square has sides of length 4. Find the radius of the circle.



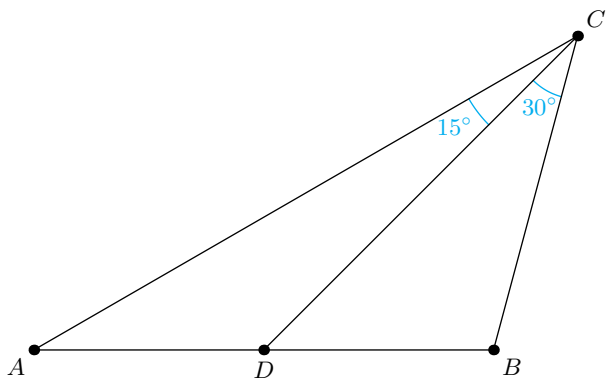
Ponder 23

- a) Prove that $\frac{a}{b} = \sqrt{2} + \sqrt{6}$.



Note: To solve this problem you may (but far from necessarily) need the *abc*-formula found in [TM1](#).

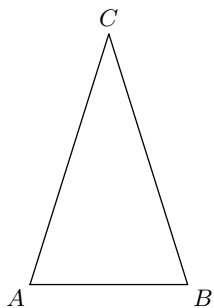
- b) $AD = BC$. Find the value of $\angle A$.



Ponder 24

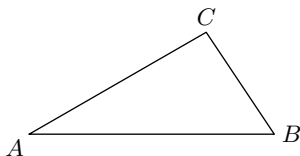
Note: This exercise deals with results that at first seems obvious, but turns out rather hard to prove.

- a) Prove that if $AC = BC$, then $\angle A = \angle B$.

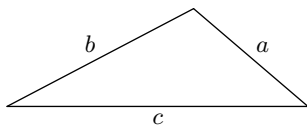


Note: Earlier on, we have declared that an isosceles triangle has two equal angles, but strictly we have to prove it.

- b) Prove that if $AC > BC$, then $\angle B > \angle C$.



- c) Given $\triangle ABC$, where AB is the longest side. Prove that when AB is the base, the height lies inside the triangle.
- d) In the figure below, c is the longest side of the triangle.



Prove that

$$c > a + b \quad , \quad b + c > a \quad , \quad a + c > b$$

Note: These three inequalities are called the **triangle inequality**.

Comment (for the particularly interested)

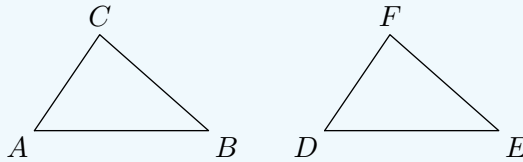
Also in geometry, axioms (see comment on page 174) lays the ground of the mathematical system we create, but the axiomatic structure of geometry is quite extensive and intricate. In addition, some theorems are such intuitively true that it, at least in a book like this, would be more confusing than clarifying to explain them all in detail.

However, it is worth noticing that [Rule 11.8](#) states three terms regarding the unique construction in a triangle, and [Rule 11.9](#) states a term regarding congruence. In more advanced texts on geometry, chances are that you will recognize the content of these rules as axioms and theorems of congruence:

Congruence

Two triangles $\triangle ABC$ and $\triangle DEF$ are congruent if one of the following terms are satisfied:

- i) $AB = DE$, $BC = EF$ and $\angle A = \angle D$.
- ii) $\angle A = \angle D$, $\angle B = \angle E$ and $AB = DE$.
- iii) $AB = DE$, $BC = EF$ and $AC = FD$.
- iv) $\angle A = \angle D$ and $\angle B = \angle E$ and, in addition, $AB = DE$ or $BC = EF$ or $AC = FD$.



-
- i) The Side-angle-side (SAS) axiom
 - ii) The Angle-side-angle (ASA) theorem
 - iii) The Side-side-side (SSS) theorem
 - iv) The Side-angle-angle (SAA) theorem

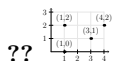
In the text-box on the previous side, term i) - iii) brings sufficient information regarding the unique construction of a triangle. However, in this book we have chosen to separate the concepts of congruence and unique construction. This is done under the presumption that most people will have a good intuition about congruent triangles, while having more difficulties stating terms of unique construction – and it is not necessarily easy to observe this directly from the terms of congruence.

Also, observe that iv) is just term ii) in a wider sense, but it cannot be used as a term of unique construction. Therefore, this term is not found in either [Rule 11.8](#) or [Rule 11.9](#).

Exercise answers

Chapter1

- 1.1.1 a) 22 b) 13 c) 36
 1.1.2 a) 17 b) 11 c) 29
 1.1.3 a) 1,7 b) 2,3
 1.1.4 a) 13,9 b) 32,8 c) 0,7 d) 2,4
 1.1.5 b)
 1.1.6 c)
 1.1.8 a)
 1.1.9 c)
 ?? a) 871 b) 178
 ?? $A = (1, 3)$, $B = (2, 4)$, $C = (6, 1)$, $D = (5, 5)$

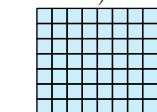


Chapter6

- 2.1.1 Note: Multiple possible answers. a) $4 = 1+3$ b) $5 = 2+3$ c) $6 = 2+4$
 d) $7 = 1+6$ e) $8 = 3+5$ f) $9 = 2+7$
 2.1.2 Note: Multiple possible answers. a) $5 = 2+2+1$ b) $6 = 1+3+2$
 c) $7 = 3+2+2$ d) $8 = 1+1+6$ e) $9 = 3+3+3$
 2.1.3
 1) a) 8 b) 7 c) 6 d) 5
 2) Fordi de er svarene i oppgave 1a), 1b) og 1c), og addisjon er kommutativ.
 2.1.4 a) 1) b) 1) c) 2)
 2.2.1 Note: Multiple possible answers. a) $2 = 7-5$ b) $3 = 10-7$ c) $4 = 5-1$
 d) $5 = 10-5$ e) $6 = 9-3$ f) $7 = 9-2$ g) $8 = 10-2$
 ?? a) 8 og 13 b) 42 og 19 c) 762 og 141 d) 15 og 9 e) 83
 og 25 f) 904 og 352
 2.2.2 a) 1) b) 1) c) 2)

- 2.3.1** a) $2+2+2+2=2\cdot 4=4+4$ b) $3+3+3+3+3+3=3\cdot 6=6+6+6$
 c) $4+4=4\cdot 2=2+2+2+2$ d) $5+5+5+5+5+5+5+5+5+5=5\cdot 10=10+10+10+10+10$
 e) $6+6+6+6+6=6\cdot 5=5+5+5+5+5$ f) $7+7+7+7=7\cdot 4=4+4+4+4+4+4+4$

- 2.3.2** a)  $4\cdot 5=20$ b)  $3\cdot 8=24$ c)  $2\cdot 9=18$ d)  $5\cdot 6=30$



- e) $7\cdot 8=56$

- ?? a) 5 og 9 b) 8 og 23 c) 56 og 42 d) 21 og 3 e) 216 og 9
 f) 3796 og 52

- Ponder 2.3.3** a) partall b) partall, 0 c) oddetall, 5

Chapter??

- 3.1.1** a) 34 b) 177 c) 100 d) 664 e) 2943

- 3.2.1** Note: Multiple possible answers. a) $5\cdot 20$ b) $3\cdot 10$ c) $2\cdot 20$
 d) $2\cdot 35$ e) $7\cdot 6$ f) $8\cdot 4$ g) $42\cdot 2$ h) $3\cdot 30$

- 3.2.2** a) $2\cdot 2\cdot 2\cdot 2\cdot 5$ b) $2\cdot 3\cdot 5$ c) $2\cdot 2\cdot 2\cdot 5$ d) $2\cdot 5\cdot 7$ e) $2\cdot 3\cdot 7$
 f) $2\cdot 2\cdot 2\cdot 2\cdot 2$ g) $2\cdot 2\cdot 3\cdot 7$ h) $2\cdot 3\cdot 3\cdot 5$

- 3.2.3** Note: Multiple possible answers. a) $5\cdot 20$, $10\cdot 10$, $5\cdot 2\cdot 20$ b) $3\cdot 10$, $6\cdot 5$,
 $3\cdot 2\cdot 5$ c) $2\cdot 20$, $4\cdot 10$, $2\cdot 2\cdot 10$ d) $2\cdot 35$, $14\cdot 5$, $2\cdot 7\cdot 5$ e) $7\cdot 6$,
 $7\cdot 2\cdot 3$, $21\cdot 2$ f) $8\cdot 4$, $16\cdot 2$, $8\cdot 2\cdot 2$ g) $42\cdot 2$, $21\cdot 4$, $7\cdot 6\cdot 2$ h) $3\cdot 30$,
 $9\cdot 10$, $3\cdot 3\cdot 10$

Ponder 3.2.5

- Ponder ??** a) See the solutions manual. b) 1715

Ponder 3.2.4 28

Ponder 2 See the solutions manual.

Chapter??

- 4.1.1** a) 6 b) 5 c) 2 d) 7 e) 9 f) 4

- 4.1.2** a) 0,5 b) 0,25 c) 0,2 d) 0,75 e) 0,4 f) 0,6
 g) 0,8 h) 1,5 i) 0,33... j) 2,5 k) 0,833... l) 1,4 m) 2,75
 n) 0,7

- ?? a) $\frac{2}{3}$ b) $\frac{5}{9}$ c) $\frac{3}{7}$ d) $\frac{7}{10}$

- 4.1.3** a) $\frac{3}{4}$ b) $\frac{2}{7}$ c) $\frac{2}{5}$

- 4.1.4** a) $\frac{14}{3}$ b) $\frac{13}{2}$ c) $\frac{11}{5}$

- 4.2.1** a) $\frac{20}{6}$ b) $\frac{9}{12}$ c) $\frac{12}{28}$ d) $\frac{45}{40}$ e) $\frac{54}{30}$ f) $\frac{77}{28}$

- 4.2.2** a) $\frac{9}{4}$ b) $\frac{9}{5}$ c) $\frac{2}{7}$

- ?? a) $\frac{4}{3} > \frac{6}{5}$ b) $\frac{20}{50} = \frac{2}{5}$ c) $\frac{7}{3} > \frac{9}{5}$

- 4.3.1** a) $\frac{10}{3}$ b) $\frac{14}{4}$ c) $\frac{11}{6}$ d) $\frac{10}{7}$ e) 1

Note: Én av brøkene kan forkortes.

4.3.3 a) $\frac{1}{3}$ b) $\frac{2}{4}$ c) $\frac{9}{6}$ d) 1 e) 0

Note: To av brøkene kan forkortes.

4.3.2 a) $\frac{22}{3}$ b) $\frac{8}{5}$ c) $\frac{17}{7}$

4.3.4 a) $\frac{6}{5}$ b) $\frac{5}{7}$ c) $\frac{14}{3}$

?? a) $\frac{17}{7}$ b) $\frac{29}{4}$

4.3.5 a) $\frac{9}{10}$ b) $\frac{73}{63}$ c) $\frac{101}{24}$ d) $\frac{73}{20}$ e) $\frac{5}{6}$

4.3.6 a) $\frac{1}{10}$ b) $\frac{29}{36}$ c) $\frac{71}{72}$ d) $\frac{11}{20}$ e) $\frac{5}{6}$

4.3.7 a) $\frac{5}{12}$ b) $\frac{157}{30}$ c) $\frac{229}{56}$

4.4.1 a) $\frac{20}{3}$ b) $\frac{40}{7}$ c) $\frac{54}{10}$ d) $\frac{80}{7}$ e) $\frac{21}{2}$ f) $\frac{28}{3}$ g) $\frac{35}{3}$

h) $\frac{30}{7}$ i) $\frac{5}{11}$ j) $\frac{72}{17}$

4.5.1 a) $\frac{4}{15}$ b) $\frac{5}{56}$ c) $\frac{9}{60}$ d) $\frac{8}{70}$ e) $\frac{3}{14}$ f) $\frac{9}{110}$ g) $\frac{1}{60}$

h) $\frac{9}{290}$ i) $\frac{8}{459}$ j) $\frac{4}{158}$

?? a) $\frac{2}{3}$ b) $\frac{2}{5}$ c) $\frac{4}{3}$ d) $\frac{6}{13}$

4.6.1 a) $\frac{20}{27}$ b) $\frac{7}{32}$ c) $\frac{18}{21}$ d) $\frac{60}{5}$ e) $\frac{21}{10}$ f) $\frac{10}{21}$ g) $\frac{16}{21}$

h) $\frac{80}{9}$ i) $\frac{36}{35}$ j) $\frac{35}{12}$

4.7.1 a) $\frac{15}{40}$ b) $\frac{153}{32}$ c) $\frac{46}{32}$ d) $\frac{21}{648}$ e) $\frac{203}{328}$

4.8.1 a) $\frac{4}{11}$ b) $\frac{35}{8}$ c) $\frac{1}{9}$ d) 4

4.8.2 a) $\frac{7}{4}$ b) $\frac{3}{7}$ c) $\frac{2}{3}$ d) $\frac{8}{7}$ e) $\frac{1}{2}$ f) $\frac{7}{2}$

4.8.3 a) 49 b) 54 c) 70 d) 16 e) 30 f) 12 g) 25

h) 14 i) 7 j) 63

4.9.3 a) $\frac{14}{15}$ b) $\frac{24}{45}$ c) $\frac{30}{21}$ d) $\frac{7}{20}$ e) $\frac{66}{15}$

4.9.4 a) $\frac{4}{9}$ b) $\frac{15}{4}$ c) $\frac{14}{5}$

Ponder 3 a) 2 b) 4 c) 5 d) 2 e) 4 f) 5 g) 3, 4

h) 2, 5 i) 3, 5 j) 4, 5

Ponder 4 a) 2 b) 4 c) 5 d) 4, 3 e) 5, 2 f) 5, 3

g) 5, 4

Ponder 5 a) $\frac{403}{6732}$ b) $\frac{269}{3150}$

Chapter 5

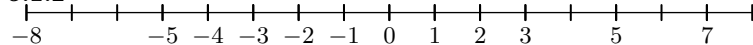
5.1.1 a) 9 har retning mot høyre og lengde 9. b) 4 har retning mot høyre

og lengde 4 c) -3 har retning mot venstre og lengde 3 d) 12 har ret-

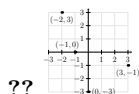
ning mot høyre og lengde 12 e) -11 har retning mot venstre og lengde 11

f) -25 har retning mot venstre og lengde 25

5.1.2



5.1.3 a) 9, 4 og 12. b) -3, -11 og -25.



??

5.2.1 a) 1 b) 8 c) 6 d) 0 e) 3 f) 1 g) 10

h) 5

- 5.2.2** a) -16 b) -8 c) -23 d) 0 e) -23 f) -17
g) -3 h) 0
- 5.2.3** a) 15 b) 17 c) 12 d) 14 e) -12 f) -19
g) -12 h) -13
- 5.2.4** a) 22 b) 22 c) -17 d) 14 e) 15 f) 13 g) -13
h) -12
- ?? a) $-\frac{5}{3}$ b) $\frac{1}{5}$ c) $-\frac{8}{11}$ d) $\frac{3}{7}$
- 5.2.5** a) -12 b) -50 c) -63 d) -24 e) -56 f) -27
g) -12 h) -40 i) 21 j) 25 k) 12 l) 72
- 5.2.6** a) -4 b) -6 c) -5 d) -4 e) -8 f) -9 g) -5
h) -5 i) 8 j) 9 k) 5

Chapter??

- 6.1.1** a) 96 b) 87 c) 899 d) 298
- 6.1.2** a) 103 b) 143 c) 1000 d) 1010
- 6.2.1** a) 61 b) 234 c) 128 d) 651
- 6.2.2** a) 59 b) 325 c) 523 d) 87
- ?? a) $57,9$ b) 32 c) 90
- ?? a) $314,6$ b) 210 c) 500
- ?? a) 4010 b) 4100 c) $53\,000$
- ?? a) 9 b) $3,252$ c) $0,9$
- ?? a) $4,146$ b) $0,021$ c) $0,07$
- ?? a) $0,00401$ b) $0,0041$ c) $0,053$
- 6.3.1** a) 36 b) 112 c) 380 d) 258 e) 763 f) 784
g) 1917
- 6.3.2** a) 348 b) 2573 c) 6916 d) 1162
- 6.3.3** a) $29\,736$ b) $58\,183$ c) $61\,243$ d) $48\,977$
- 6.3.4** a) 135 og $13,5$ b) $47\,424$ og $474,24$ c) $68\,672$ og $68,672$ d) $569\,366$ og $56,9366$
- 6.3.5** a) $411,5$ b) $66,57$ c) $38,08$
- 6.4.1** a) 49 b) 29 c) 23 d) 17 e) 12
- 6.4.2** a) 189 b) 56 c) 99 d) 19 e) 26 f) 97
- 6.5.1** a) $9,8 \cdot 10^4$ b) $1,67 \cdot 10^8$ c) $4,819 \cdot 10^3$ d) $2,1 \cdot 10^1$ e) $9,13227 \cdot 10^3$
f) $8,937 \cdot 10^2$ g) $1,80021 \cdot 10^4$ h) $3,024 \cdot 10^2$
- 6.5.2** a) $2,7 \cdot 10^{-2}$ b) $1,901 \cdot 10^{-4}$ c) $3,2 \cdot 10^{-1}$ d) $2,0032 \cdot 10^{-7}$
- Ponder 6.5.3** a) $9 \cdot 10^8 \cdot 7 \cdot 10^{-5}$ b) $6,3 \cdot 10^4$

Chapter??

- 7.1.1** a) 14 b) 20 c) 24
- 7.1.2** a) 12 b) 19 c) 15
- ?? a) 2 b) 16 c) 27

7.1.3 a) 2 og 8 b) 3 og 4 c) 3 og 6

7.1.4 a) 81 b) 1) Et åpenbart eksempel er kvadratet fra oppgave a), som har bredde og høyde 9, og areal 81. 2) Bredde 15 og høyde 3, areal 45. 3) Bredde 12 og høyde 6, areal 72. *Note:* Multiple possible answers.

7.1.5 a) 3 b) 10 c) 6 d) 6 e) 10 f) 6 g) 4
h) 3 i) 28

7.1.6 a) 90. *Note:* Grunnflatearealet kan også være 72 eller 80, avhengig av hvilken sideflate man velger ut som grunnflate.

7.2.1 See the solutions manual.

7.2.2 See the solutions manual.

Ponder 6 See the solutions manual.

Ponder 7 See the solutions manual.

Ponder 21 See the solutions manual.

Chapter??

8.1.1 a) $3a$ b) $4a$ c) $7a$ d) $-2b$ e) $-5b$ f) $-3k$

8.1.2 a) $a + b$ b) $a + 2b$ c) $9b - 3a$

8.1.3 a) $2b - 5a + c$ b) $3b - 9a$ c) $11b - 3a$

8.1.4 a) $7a+14$ b) $9b+27$ c) $8b-24c$ d) $-6a-10b$ e) $(9a+2)$
f) $(3b + 8)a$ g) $3ac - ab$ h) $2a + 6b + 8c$ i) $27b - 9c + 63a$
j) $2c - 6b - 14a$

8.1.5 a) $2(a + b)$ b) $b(4a + 5)$ c) $c(9b - 1)$ d) $2a(2c - 1)$

8.1.6 See the solutions manual.

8.1.7 a) $\frac{1}{4}$ b) $\frac{1}{2}$ når $x = 4$ og 2 når $x = -2$.

8.1.8 Uttrykkene gitt av a) og c) stemmer.

8.2.1 a) 3^4 b) 5^2 c) 7^6 d) a^3 e) b^2 f) $(-c)^4$ *Note:*
 $(-c)^4 = c^4$

8.2.2 a) 64 b) 32 c) 64 d) -8 e) -243 f) 256

8.2.3 a) 2^{16} b) 3^{11} c) 9^6 d) 6^5 e) 5^{-4} f) 10^{11} g) a^{16}
h) k^7 i) x^3 j) x k) 1 l) $a^5 \cdot b^{-3}$

8.2.4 a) 5 b) 10 c) 12 d) 3 e) 9 f) 10

Ponder ?? a) $\frac{2}{3}$ b) $\frac{1}{8}$ c) $\frac{1}{9}a^{-\frac{1}{2}}$ d) $\frac{7}{5}$

Ponder ?? a) $x - y$ b) $\frac{2(x+1)}{x-1}$

Ponder ?? a) $\frac{x+1}{x-3}$ b) $\frac{x+2}{x-2}$ c) $\frac{3(x+1)}{x-1}$ d) $\frac{1}{x-3}$

Ponder ?? a) $5 \cdot 10^9$ b) $2 \cdot 10^7$ c) $3,1 \cdot 10^{16}$ d) $1,9 \cdot 10^{-4}$

Ponder ?? See the solutions manual.

Ponder 9 See the solutions manual.

Ponder ?? $p = 3$, $q = 5$, $r = 2$

Chapter9

9.2.1 a) $x = 10$ b) $x = 5$ c) $x = 9$ d) $x = 2$ e) $x = 3$
f) $x = 4$ g) $x = 10$ h) $x = 8$ i) $x = 10$

9.2.2 a) $x = 37$ b) $x = 29$ c) $x = 23$ d) $x = 15$ e) $x = 8$
f) $x = 11$ g) $x = 23$ h) $x = 19$ i) $x = 10$ j) $x = 28$

9.2.3

a) $x = 4$ b) $x = 5$ c) $x = 9$ d) $x = 15$

9.2.4 a) $x = 8$ b) $x = 72$ c) $x = 49$ d) $x = 150$

9.2.5 a) $x = 7$ b) $x = 10$ c) $x = 6$ d) $x = 9$ e) $x = 9$
f) $x = 8$ g) $x = 5$ h) $x = 2$

?? a) $x = 10$ b) $x = 12$ c) $x = \frac{52}{3}$ d) $x = \frac{14}{15}$

?? a) $x = 6$ b) $x = \frac{1}{14}$ c) $x = \frac{5}{18}$ d) $x = \frac{76}{35}$

9.2.6 See the solutions manual.

9.2.7 $x = 18$

?? a) $x = 6, y = 19$ b) $x = 3, y = 7$ c) $x = 3, y = 0$ d) $x = \frac{63}{5}, y = 18$

Ponder ?? 36

Ponder ?? $x = 2$ eller $x = -3$.

Ponder ?? a) $x = 3$ b) $x = 2$ c) $x = 9$ d) $x = 11$

Ponder ?? a) $x = -\frac{21}{2}$ b) $x = \frac{90}{23}$ c) $x = 27$ d) $x = \frac{30}{11}$
e) $x = -6$ f) $x = \frac{71}{17}$

Ponder 10 See the solutions manual.

Chapter10

?? a) $f(2) = 6$ b) $g(9) = 52$ c) $h(5) = 39$ d) $i(3) = -2$
e) $j(10) = 100$ f) $k(5) = 54$ g) $l(10) = -100$

?? a) $a(x) = 4x + 5$ b) $a(20) = 85$ c) 21

?? a) $x = 10$ b) $x = 9$ c) $x = 7$ d) $x = 3$ e) $x = \pm 10$
f) $x = \pm 5$

10.1.1 a) $f(x) = 2x + 4$ b) 204 c) $x = 10$

10.1.2 a) $a(x) = 5x^2$ b) $x = 2000$ c) $x = 9$

10.1.3 a) $b(x) = x^2 + 2x$ b) 440 c) $x = 8$

10.1.4 22 trekanter og 10 firkanter.

10.1.5 La x være et positivt heltall. a) $p(n) = 2n$ b) $o(n) = 2n - 1$

10.2.1

a) Stigningstall 5 og konstantledd 10. b) Stigningstall 3 og konstantledd -12 .
c) Stigningstall $-\frac{1}{7}$ og konstantledd -9 . d) Stigningstall $\frac{3}{2}$ og konstantledd $-\frac{1}{4}$.

10.2.2 See the solutions manual.

10.3.1 a) (I) og (II) gir hver for seg en likning som beskriver en rett linje. Disse linjene kan også representeres ved f og g slik som de er definert. b) $x = 7$ og $y = 2$.

?? $x = 2$ og $y = 0$.

10.3.2 a) $f(x) = -8x + 16$ b) $f(x) = 4x + 3$

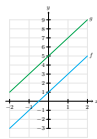
10.3.4 $f(x) = x - 3$ og $g(x) = \frac{1}{4}x + 1$

?? a) $y = -4x + 3$ b) $y = x - 5$ c) $y = -\frac{1}{5}x + \frac{5}{7}$ d) $y = x - 9$

Ponder ?? a) $y = 4x + 10$ b) $7x - 2$ c) $-7x + 10$ d) $-\frac{1}{3}x - 5$

Ponder ?? a) $f(x) = -2x + 3$, $g(x) = \frac{1}{2} - 2$ b) $\frac{5}{4}$

Ponder ?? $g(x) = 2x$



Ponder ?? a) b) f og g er parallelle. See the solutions manual.

Ponder ?? a) 3 b) $\frac{3}{2}$ c) 3

Ponder ?? $\frac{5}{2}$

Ponder ?? 4

Ponder ?? $f(x) = 2x^2 - 5x + 4$

Ponder 13 a) $f(x) = 3x - 5$ b) See the solutions manual.

Ponder ?? a) See the solutions manual. b) $x_1 = -2$, $x_2 = 4$ c) 3
for begge. d) $A = (-3, 7)$, $B = (5, 7)$. Horisontalavstanden er 4 for begge.
e) De har lik y -verdi. f) $x = 47$

Ponder 12 See the solutions manual.

Ponder 14 See the solutions manual.

Chapter??

?? $\frac{62}{135}$

?? a) 3 b) $9b$ c) 10

?? a) 8π b) 16π

?? a) 10 b) 9 c) $2k$ d) $10a$

?? a) 6π og 24π b) 9π og 144π c) 4 d) 16

?? a) a b) a^2

11.1.7 a) 100π b) 400π

11.1.8 a) $\frac{32\pi}{3}$ og 288 b) a^3

?? a) 15 b) 4 c) 7 d) $\sqrt{97}$

?? $c = \sqrt{5}a$

?? a) Ja. b) Nei.

?? a^3

11.1.1 60°

11.1.2 AC og DE , BC og EF , AB og DF .

11.1.3 $EF = 3$ og $DF = 2$.

11.1.4 $AC = 15$ og $DF = 6$.

?? To vinkler som er parvis like store. See the solutions manual..

?? a) To vinkler som er parvis like store. See the solutions manual.. b) CD
 og AC , BC og EC , AB og ED . c) 20.

11.1.5 $\triangle ABC \sim \triangle ACD \sim \triangle CBD$.

?? a) $AC^2 = BC^2 + AB^2$ b) $BC^2 = AC^2 - AB^2$

c) $AB^2 = AC^2 - BC^2$

?? a) 10 b) 17 c) 12 d) 15

?? a) Rettvinklet. b) Ikke rettvinklet.

Ponder ?? $x = 3$

Ponder ?? 20

Ponder ?? 15 eller $3\sqrt{7}$

Ponder 17 a) See the solutions manual. b) See the solutions manual.



Ponder ?? a)

b)

c)

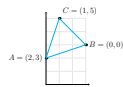
Ponder ?? $3\sqrt{10}$

Ponder ?? 30

Ponder ?? See the solutions manual.

Ponder 11.1.6 See the solutions manual.

Ponder ?? See the solutions manual.



Ponder ?? a) $f(x) = 3x + 2$, $g(x) = \frac{1}{3} + 2$. b) (navnene på
 punktene er vilkårlig valgt). c) $\triangle ABC$ er likebeint ($AB = AC$). d) 4
 e) See the solutions manual.

Ponder ?? a) 26 b) Linja er parallell med AB . Høyden vil være uendret.

Ponder ?? See the solutions manual.

Ponder ?? 8, 15 og 17

Ponder ?? $180^\circ(n - 2)$

Ponder ?? See the solutions manual.

Ponder ?? See the solutions manual.

Ponder 15 30°

Ponder ?? See the solutions manual.

Ponder ?? 10

Ponder 18 See the solutions manual.

Ponder 16 See the solutions manual.

Ponder 20 See the solutions manual.

Ponder 22 $\frac{5}{2}$

Ponder 21 See the solutions manual.

Ponder ?? See the solutions manual.

Ponder 19 See the solutions manual.

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Note: The text, at least a very similar one, about Pythagoras's theorem on page 224 was first printed in Skage Hansen's book Tempelgeometri (2020).

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